

Quantitative Volatility Corpus
Volume I: Single-Name Foundations

Quant Corpus Consolidation Committee

Contents

Chapter 1

Gamma P&L and the Volatility Bet

The value of a European option is a convex function of the underlying price. Convexity makes the position more than a linear exposure: its profit and loss over a price move carries a quadratic term, and holding that quadratic term costs time decay. The balance between the two turns an option into a bet on realized volatility. This chapter derives the balance exactly.

1.1 The Black–Scholes frame

Assumption 1.1 (Standing assumption). Rates and dividends are zero. The forward price F is used generally; every collapse $F = S$ is flagged at its point of use.

The market variables are the spot price S , the forward F , the strike K , and the time to expiry T . The strike log-moneyness is

$$m = \ln(K/F). \quad (1.1)$$

A European call with implied volatility σ has the Black–Scholes price

$$C(K, T) = F N(d_1) - K N(d_2), \quad d_1 = -\frac{m}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (1.2)$$

where N is the standard normal CDF and φ the standard normal pdf. Under Assumption ??, $F = S$, and (??) reads $C = S N(d_1) - K N(d_2)$.

Assumption 1.2 (Fixed implied volatility). Throughout this chapter σ is the fixed Black–Scholes input for a single option: it moves with neither spot nor time. Its dependence on m , on the forward level, and on T , and the P&L that its movement generates, are deferred to Chapters 2–3.

Definition 1.3 (Delta, gamma, theta).

$$\Delta = \frac{\partial C}{\partial S}, \quad \Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\partial \Delta}{\partial S}, \quad \theta = -\frac{\partial C}{\partial T}. \quad (1.3)$$

Δ is the sensitivity of the option price to the underlying, Γ the sensitivity of Δ itself, and θ the drift of the price as calendar time advances (T decreasing). Since (??) expresses C through F alone, the S -derivatives carry the chain rule through $F(S)$: $\Delta = (\partial C / \partial F)(dF/dS)$, and correspondingly for Γ . Under Assumption ??, $dF/dS = 1$ ($F = S$, flagged), so the S -partials coincide with the F -partials.

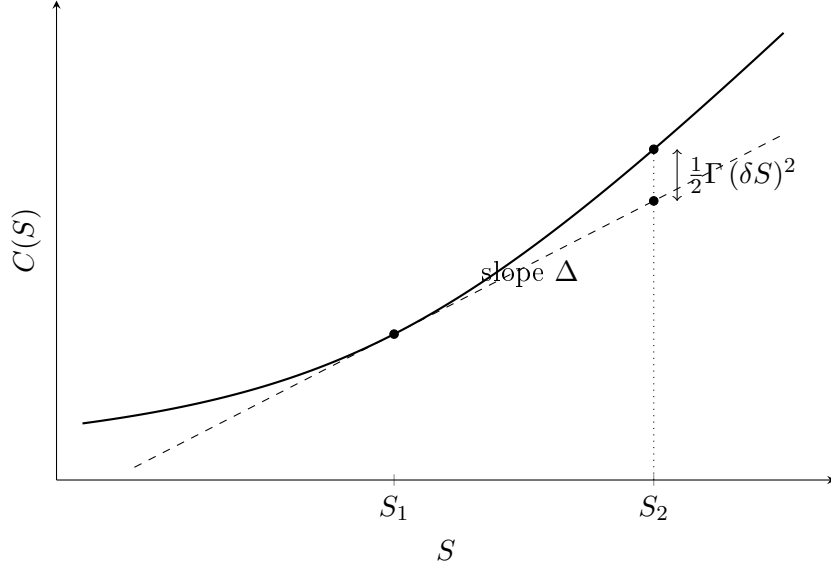


Figure 1.1: Convexity of the call price (schematic). The dashed tangent at S_1 is the delta approximation; over the move $\delta S = S_2 - S_1$ the option gains the quadratic gap above the tangent. For a long position the gap is favorable in both directions of the move.

Lemma 1.4 (Closed forms at zero rates). *Under Assumption ?? ($F = S$),*

$$\Delta = N(d_1), \quad \Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{T}}, \quad \theta = -\frac{S\varphi(d_1)\sigma}{2\sqrt{T}}. \quad (1.4)$$

Proof. Direct differentiation of (??); the terms in $\partial d_1/\partial S$ and $\partial d_2/\partial S$ cancel by the identity $F\varphi(d_1) = K\varphi(d_2)$. The identity is one line: $d_1^2 - d_2^2 = (d_1 + d_2)(d_1 - d_2) = -2m$, so $\varphi(d_1)/\varphi(d_2) = e^m = K/F$. Under Assumption ?? ($F = S$) it reads $S\varphi(d_1) = K\varphi(d_2)$. $\square$

Since $\varphi > 0$, $\Gamma > 0$: the call price is strictly convex in S , and $\theta < 0$: a long option loses value as time passes. These two signs are the entire subject of the chapter. Figure ?? shows the geometry: the tangent line at the current spot is the delta approximation, and the gap between curve and tangent grows quadratically with the size of the move.

1.2 The P&L of a price move

At fixed time, a move of the underlying from S_1 to $S_2 = S_1 + \delta S$ changes the option price by

$$C(S_2) - C(S_1) = \int_{S_1}^{S_2} \Delta(u) du, \quad (1.5)$$

the cumulative effect of delta along the path.

Theorem 1.5 (Gamma P&L formula). *For a move δS at fixed time,*

$$\boxed{\text{P\&L} = \Delta \cdot \delta S + \frac{1}{2} \Gamma (\delta S)^2} \quad (1.6)$$

with error $O((\delta S)^3)$; Δ and Γ are evaluated at S_1 .

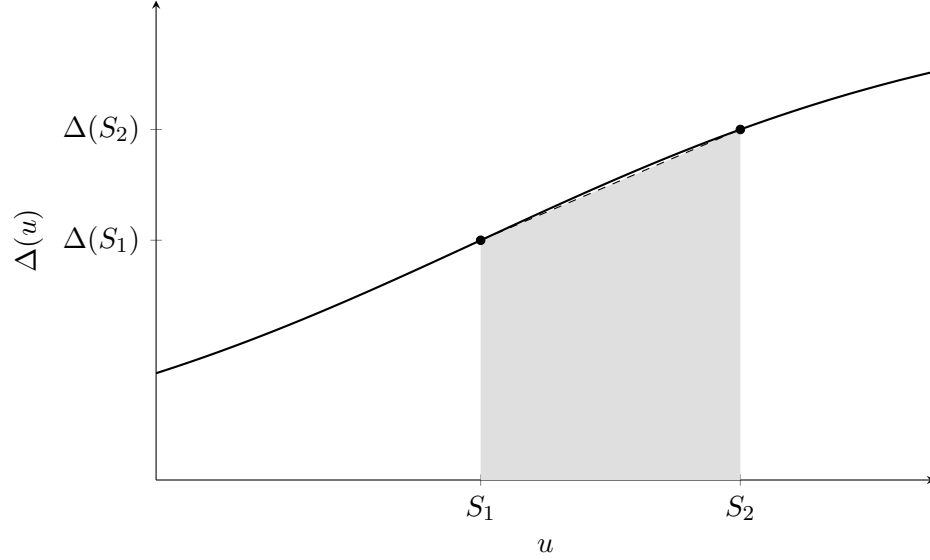


Figure 1.2: The trapezoidal reading of the gamma P&L formula (schematic). The exact P&L (??) is the area under the delta curve between S_1 and S_2 ; the shaded trapezoid, with base δS and average height $\bar{\Delta}$, approximates it and equals $\Delta \cdot \delta S + \frac{1}{2}\Gamma(\delta S)^2$. The error is the sliver between the chord and the curve.

Proof. Second-order Taylor expansion of C in S at fixed time, with Definition ??.

□

The first term is the delta P&L, linear in the move; the second is the gamma P&L, quadratic in the move and positive for a long position in either direction. Two further readings of (??) give the same expression and locate its accuracy.

Proposition 1.6 (Average delta and the trapezoidal rule). *Under the linearization $\Delta(S_2) \approx \Delta(S_1) + \Gamma \delta S$, the right-hand side of (??) equals*

- (i) *the average-delta P&L $\bar{\Delta} \cdot \delta S$, where $\bar{\Delta} = \frac{1}{2} [\Delta(S_1) + \Delta(S_2)]$;*
- (ii) *the trapezoidal-rule approximation of the exact integral (??).*

Proof. The trapezoidal rule applied to (??) is exactly $\bar{\Delta} \cdot \delta S$, so (i) and (ii) are the same quantity. Substituting the linearization,

$$\bar{\Delta} \cdot \delta S = \frac{\Delta + (\Delta + \Gamma \delta S)}{2} \delta S = \left(\Delta + \frac{\Gamma \delta S}{2} \right) \delta S = \Delta \cdot \delta S + \frac{1}{2} \Gamma (\delta S)^2. \quad (1.7)$$

□

1.3 A worked example

Example 1.7 (At-the-money call, zero rates). An at-the-money call under Assumption ?? ($F = S$, so $K = S$ gives $m = 0$):

- spot $S_1 = \$100$, strike $K = \$100$;
- time to expiry $T = 0.25$ (three months);

Component	Formula	Calculation	Value (\$)
Repricing at constant σ	$C(102) - C(100)$	$7.0908 - 5.9785$	1.1122
Delta P&L	$\Delta \cdot \delta S$	0.5299×2	1.0598
Gamma P&L	$\frac{1}{2} \Gamma (\delta S)^2$	$\frac{1}{2} \times 0.0265 \times 4$	0.0530
Approximation (??)	delta + gamma	$1.0598 + 0.0530$	1.1128
Error		$1.1128 - 1.1122$	0.0006
Error (relative)		$0.0006/1.1122$	0.05%

Table 1.1: P&L decomposition for the \$2 move of Example ?? . All values are computed under Assumption ?? and at fixed σ (Assumption ??): the first row is the single-option repricing at unchanged implied volatility, not the book P&L. The surface P&L generated by σ moving with spot is excluded here and enters in Chapters 2–3.

- implied volatility $\sigma = 30\%$;
- move $\delta S = \$2$, to $S_2 = \$102$.

Evaluating (??) and (??) ($d_1 = 0.075$, $d_2 = -0.075$ at S_1):

$$C(100) = 5.9785, \quad C(102) = 7.0908, \quad \Delta(100) = 0.5299, \quad \Gamma(100) = 0.0265. \quad (1.8)$$

Table ?? decomposes the P&L of the move.

The delta-only estimate understates the repricing by the gamma term: convexity works for the long position. The quadratic term, \$0.0530, is already material on a 2% move, and the two-term formula reproduces the constant- σ repricing to 0.05%. That figure belongs to the 2% move, not to the formula: the error term of (??) is $O((\delta S)^3)$, so the relative error grows like $(\delta S/S)^2$, and on a large or gapping move the estimate degrades on two counts — the cubic Taylor error and, chiefly, the excluded repricing of σ itself.

1.4 Theta and the breakeven move

Time decay is quoted per trading day, with the convention of 252 trading days per year:

$$\theta_d = \frac{\theta}{252}. \quad (1.9)$$

Consider a long option, delta-hedged: the hedge removes the first-order term of (??), and under Assumption ?? the hedge carries no financing cost. Over one trading day with underlying move δS , the position's P&L is, to second order in δS and at fixed σ (Assumption ??, which suppresses the day's $\delta\sigma$ terms),

$$\text{P\&L}_{\text{day}} = \frac{1}{2} \Gamma (\delta S)^2 + \theta_d. \quad (1.10)$$

The gamma term is a sure gain; the theta term is a sure loss. Their balance point is the breakeven move.

Theorem 1.8 (Theta–gamma breakeven). *The daily P&L (??) vanishes exactly when $|\delta S|$ equals the breakeven move*

$$\delta S_{\text{BE}} = \sqrt{\frac{2|\theta_d|}{\Gamma}}, \quad \text{equivalently} \quad |\theta_d| = \frac{1}{2} \Gamma (\delta S_{\text{BE}})^2, \quad (1.11)$$

Daily move (\$)	Gamma P&L (\$)	Theta P&L (\$)	Total P&L (\$)
0.50	0.0033	−0.0474	−0.0440
1.00	0.0133	−0.0474	−0.0341
1.89	0.0474	−0.0474	0.0000
2.00	0.0530	−0.0474	0.0057
3.00	0.1193	−0.0474	0.0720

Table 1.2: Daily P&L of the delta-hedged position of Example ?? across move sizes, from (??) with $\Gamma = 0.0265$, $\theta_d = -0.0474$. The breakeven row is $\delta S_{BE} = \$1.89$. All values are computed under Assumption ?? and at fixed σ (Assumption ??).

and the daily P&L takes the form

$$\boxed{\text{P\&L}_{\text{day}} = \frac{1}{2} \Gamma [(\delta S)^2 - (\delta S_{BE})^2]} . \quad (1.12)$$

Proof. Set (??) to zero and solve for $|\delta S|$, using $\theta_d < 0$ for the long option; (??) is (??) with $\theta_d = -\frac{1}{2}\Gamma(\delta S_{BE})^2$ substituted. $\square$

Equation (??) contains the whole trade: the day is profitable precisely when the underlying moves more than the breakeven, in either direction.

Corollary 1.9 (Closed form of the breakeven). *Under Assumption ?? ($F = S$), the ratio of the closed forms (??) gives $|\theta|/\Gamma = S^2\sigma^2/2$ exactly, so*

$$\delta S_{BE} = \sqrt{\frac{2|\theta|}{252\Gamma}} = \frac{\sigma S}{\sqrt{252}}, \quad (1.13)$$

independent of strike and of time to expiry: the factor $\varphi(d_1)$ cancels in the ratio.

For Example ??, $\theta = -11.9347$ per year, so $\theta_d = -0.0474$ per trading day, and

$$\delta S_{BE} = \frac{\sigma S}{\sqrt{252}} = \frac{0.30 \times 100}{\sqrt{252}} = \$1.89. \quad (1.14)$$

Table ?? and Figure ?? evaluate (??) across daily move sizes.

1.5 Options as volatility bets

Corollary ?? identifies the breakeven move: $\sigma S/\sqrt{252}$ is the one-standard-deviation daily move of the underlying under its implied volatility, since for the diffusion $dS = \sigma S dW$ the standard deviation of δS over one trading day is $\sigma S\sqrt{1/252}$ to first order. The delta-hedged long option is therefore a bet that the underlying's realized daily moves exceed the one-standard-deviation move priced by the implied volatility. Aggregated over the option's life, the bet is on realized versus implied variance.

Proposition 1.10 (Expected P&L of the volatility bet). *Denote by σ_{real} the realized volatility of the underlying over the option's life, treated as deterministic (equivalently, every expectation conditions on the realized variance), and by σ_{impl} the implied volatility at which the option was bought. With Γ and S held at their initial values, the expected P&L of the delta-hedged position over the option's life is*

$$\mathbb{E}[\text{P\&L}] \approx \frac{1}{2} \Gamma S^2 (\sigma_{\text{real}}^2 - \sigma_{\text{impl}}^2) T. \quad (1.15)$$

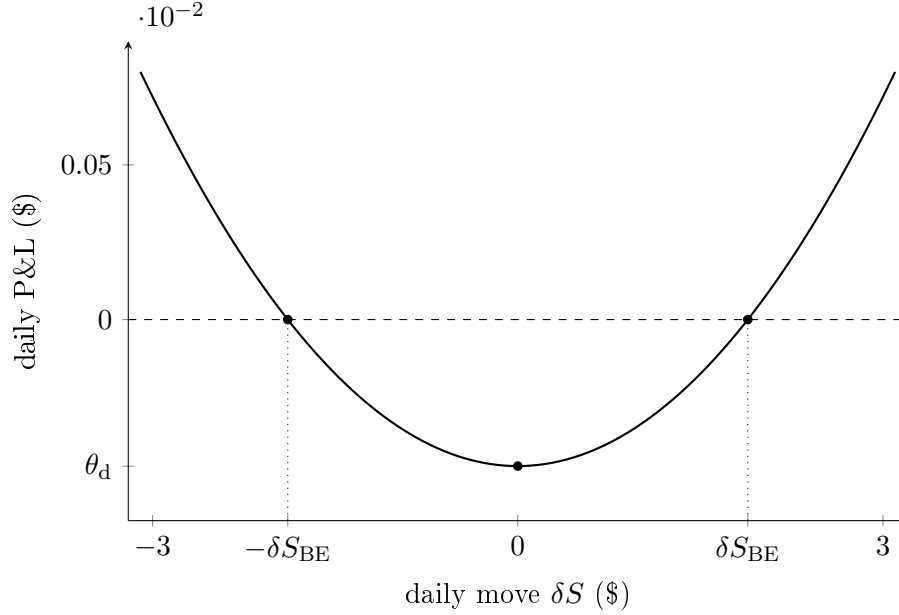


Figure 1.3: Daily P&L (??) of the delta-hedged position of Example ?? as a function of the day's move. The parabola crosses zero at $\pm\delta S_{\text{BE}} = \pm\1.89 ; days inside the breakeven band lose theta faster than gamma earns, days outside earn more. The intercept at $\delta S = 0$ is the full day's decay θ_d .

Proof. Summing the daily P&L (??) over the $252T$ trading days of the option's life, with $(\delta S_{\text{BE}})^2 = \sigma_{\text{impl}}^2 S^2 / 252$ from Corollary ?? and $\mathbb{E}[(\delta S)^2] = \sigma_{\text{real}}^2 S^2 / 252$ per day,

$$\mathbb{E}[\text{P\&L}] \approx 252T \cdot \frac{1}{2} \Gamma \frac{S^2}{252} (\sigma_{\text{real}}^2 - \sigma_{\text{impl}}^2) = \frac{1}{2} \Gamma S^2 (\sigma_{\text{real}}^2 - \sigma_{\text{impl}}^2) T. \quad (1.16)$$

□

The approximation freezes the dollar gamma ΓS^2 — and with it the breakeven δS_{BE} , which depends on Γ , S , and σ_{impl} through Corollary ?? — at initial values; along an actual path both vary with spot and with the passage of time. The exact statement carries the running weight.

Proposition 1.11 (Pathwise P&L of the hedged position). *Let the underlying follow $dS_t = \sigma_t S_t dW_t$ with σ_t the instantaneous realized volatility, and let the long option be delta-hedged continuously at the constant implied volatility σ_{impl} . Then, on every path,*

$$\text{P\&L} = \frac{1}{2} \int_0^T \Gamma_t S_t^2 (\sigma_t^2 - \sigma_{\text{impl}}^2) dt, \quad (1.17)$$

where Γ_t is the Black–Scholes gamma at σ_{impl} evaluated along the path.

Proof. Itô's formula on the Black–Scholes value $C(t, S_t; \sigma_{\text{impl}})$ gives $dC = \partial_t C dt + \Delta dS_t + \frac{1}{2} \Gamma_t S_t^2 \sigma_t^2 dt$; the Black–Scholes equation at zero rates gives $\partial_t C = -\frac{1}{2} \Gamma_t S_t^2 \sigma_{\text{impl}}^2$. The increment of the hedged position, $dC - \Delta dS_t$, is therefore $\frac{1}{2} \Gamma_t S_t^2 (\sigma_t^2 - \sigma_{\text{impl}}^2) dt$; integrating over $[0, T]$ gives (??). (N. El Karoui, M. Jeanblanc-Picqué, and S. E. Shreve, “Robustness of the Black and Scholes formula,” *Mathematical Finance* 8(2), 1998, 93–126.) □

Since $\Gamma_t > 0$, the sign of the P&L is guaranteed pathwise whenever $\sigma_t^2 - \sigma_{\text{impl}}^2$ keeps one sign over the option's life. No such guarantee attaches to the aggregate spread $\sigma_{\text{real}}^2 - \sigma_{\text{impl}}^2$: once the dollar-gamma weight in (??) is unfrozen, realized variance is collected only where the position has gamma, and the expected P&L can take the opposite sign to the spread — as when volatility spikes while spot sits far from the strike and Γ_t is negligible. Only the frozen form (??) carries the sign of the spread, and over a three-month life the freeze is a leading-order distortion, not a small correction: a gap away from the strike early in the life removes the gamma through which the remaining realized variance would have been collected. Two further omissions of (??): the expectation conceals wide dispersion across paths, so a position right on average — realized above implied — still loses on paths that concentrate the variance where its gamma is not; and the formula is gross of hedging frictions, the discrete-hedge error and the transaction costs of rebalancing. A long option position is long realized variance against implied, weighted by where its gamma lives; a short position is the opposite bet.

Everything above prices and differentiates a single option at a single fixed σ . On a real surface the implied volatility is a function $\sigma(m, \ell_F, T)$ of strike log-moneyness, forward level, and expiry, and its movement generates P&L terms of its own; that surface, and the sensitivities routed through it, occupy the rest of this volume.

Chapter 2

The Implied and Local Volatility Surface

The implied-volatility surface is not a collection of independent quotes: it is a single coherent object constrained, at every strike and every expiry, by the requirement that no static arbitrage be admitted. This chapter builds a parsimonious representation of that object, parameterising the smile through its *curvature*—its second derivative in log-moneyness—rather than directly, so that calibration collapses to a linear regression on compactly supported basis functions, the Dupire local volatility follows in closed form from the same coefficients, and the no-arbitrage conditions become low-degree polynomial inequalities on those coefficients. The construction opens the single-name admissibility ladder (wing and convexity bounds) that carries forward into the multi-name volume.

Standing assumption. The risk-free rate and the dividend yield are zero, so the forward and spot coincide, $F = S$; the collapse is flagged at each point of use. The forward F nonetheless remains the primary variable, so that reintroducing non-zero carry amounts only to restoring $F = S e^{(r-q)T}$ in the moneyness definition.

2.1 The Volatility Decomposition and Its Curvature Representation

2.1.1 The ATM/Smile Decomposition

The Black–Scholes implied volatility splits into an at-the-money *level* and a *smile shape*,

$$\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T), \quad \sigma_{\text{smile}}(0, T) = 0, \quad (2.1)$$

where the strike log-moneyness is $m = \ln(K/F)$ and the forward log-level is $\ell_F = \ln(F/F_{\text{ref}})$, with F_{ref} a fixed reference forward (the surface anchor). The ATM component $\sigma_{\text{ATM}}(\ell_F, T)$ carries all dependence on where the forward sits; the smile component $\sigma_{\text{smile}}(m, T)$ carries the shape across strikes at a fixed forward. The normalisation $\sigma_{\text{smile}}(0, T) = 0$ is what makes σ_{ATM} the genuine at-the-money volatility and makes the split unique: setting $m = 0$ in (??) gives

$$\sigma(0, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) \quad (2.2)$$

exactly. The whole of the present chapter constructs $\sigma_{\text{smile}}(m, T)$; the ATM level $\sigma_{\text{ATM}}(\ell_F, T)$ and its slope in ℓ_F (the vol.slope) are the objects that transmit surface dynamics to the routed sensitivities of Section ?? and to the multi-name book. This chapter does *not* calibrate that ℓ_F -dependence: it fits a single smile at the prevailing forward, so vol.slope and $\partial_{\ell_F}^2 \sigma_{\text{ATM}}$ are exogenous inputs owned by the surface-dynamics chapter, defaulting to zero (the C4 $\sigma_{\text{ATM}} = \text{const}$ case) until supplied—a default flagged at every spot-routed use below.

Remark 2.1 (Two slopes, never to be confused). Two distinct first-order objects live on the surface. The *smile skew* $s(m) = \frac{\partial \sigma_{\text{smile}}}{\partial m}$ is the slope of the shape *across strikes* at a fixed forward. The *vol.slope* $:= \frac{\partial \sigma_{\text{ATM}}}{\partial \ell_F}$ is the slope of the single ATM point *as the forward itself moves*. For equity indices $\text{vol.slope} < 0$. These are separate derivatives of separate arguments of (??); the routed Greeks of Section ?? keep them apart by construction.

2.1.2 From Curvature to Smile

Fix the expiry T and the forward. Define the smile *skew* and *curvature*,

$$s(m) = \frac{d\sigma_{\text{smile}}}{dm}(m), \quad c(m) = \frac{d^2\sigma_{\text{smile}}}{dm^2}(m).$$

With the anchoring data $\sigma_{\text{smile}}(0, T) = 0$ (normalisation) and $s(0) = s_0$ (the ATM smile skew), two integrations recover the shape from its curvature:

$$s(m) = s_0 + \int_0^m c(u) du, \quad (2.3)$$

$$\sigma_{\text{smile}}(m, T) = s_0 m + \int_0^m \int_0^u c(v) dv du. \quad (2.4)$$

This is the Green's-function solution of $\sigma_{\text{smile}}'' = c(m)$ under the boundary data $(\sigma_{\text{smile}}(0, T), s(0)) = (0, s_0)$. The ATM level $\sigma_{\text{ATM}}(\ell_F, T)$ enters (??) additively and is invisible to (??); this is precisely why calibrating the shape is a linear problem.

Example 2.2 (Constant curvature). If $c(m) = c_0$ for all m , then $\sigma_{\text{smile}}(m, T) = s_0 m + \frac{1}{2}c_0 m^2$, a parabolic smile with $\sigma_{\text{smile}}(0, T) = 0$. This quadratic smile is revisited as a fully worked local-volatility example in Section ??.

2.1.3 Kernel Representation of the Curvature

The curvature is a superposition of n compactly supported Epanechnikov kernels,

$$c(m) = \sum_{i=1}^n c_i \frac{1}{h_i} \kappa\left(\frac{m - \mu_i}{h_i}\right), \quad \kappa(u) = \begin{cases} \frac{3}{4}(1 - u^2), & |u| \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad (2.5)$$

where c_i are curvature coefficients, μ_i the kernel centres, and $h_i > 0$ the bandwidths. Because $\int_{-1}^1 \kappa = 1$, each c_i is the *total skew change* contributed by kernel i ; the total skew swing from $m = -\infty$ to $m = +\infty$ is $\sum_i c_i$. The choice of the Epanechnikov kernel is not incidental: it vanishes at its endpoints, $\kappa(\pm 1) = 0$, which (as Section ?? shows) makes the reconstructed smile globally C^2 and uniquely yields closed-form polynomial basis functions.

Remark 2.3 (The flexible-ruler picture). A flexible ruler laid along the moneyness axis carries the picture. Two conditions pin it at the origin: its height (here 0, by the normalisation) and its slope s_0 . Each curvature kernel c_i is a localised bump placed under the ruler, bending it within the support $[\mu_i - h_i, \mu_i + h_i]$ and leaving it straight elsewhere. Beyond the last bump on each side, the ruler continues as a straight line—it has no further reason to bend. The subsections that follow make this picture precise, and the wing constraints of Section ?? are exactly the requirement that these straight tails be flat rather than sloped.

Function-space structure of the kernel basis

Let $I_i(m)$ denote the double integral of the i -th kernel from the origin (defined precisely in (??) below). The model space

$$\mathcal{V} = \text{span}\{m, I_1(m), \dots, I_n(m)\}$$

of smile shapes decomposes canonically as

$$\mathcal{V} = \mathcal{V}_0 \oplus \mathcal{V}_c, \quad \mathcal{V}_0 = \ker \partial_m^2 \cap \mathcal{V} = \text{span}\{m\}, \quad \mathcal{V}_c = \text{span}\{I_1, \dots, I_n\}.$$

Here $\mathcal{V}_0$ carries the linear (zero-curvature) part—the ruler with no bumps, a straight line of slope s_0 through the origin—while each basis function $I_i \in \mathcal{V}_c$ records the deformation produced by one bump. The Green’s-function structure (??) gives every I_i two roles: a *local* role (inside its support it contributes genuine curvature) and a *propagating* role (beyond its support it becomes affine, collapsing into $\mathcal{V}_0$ and contributing only a permanent change of slope).

Let

$$\alpha_i = P_\kappa(-\mu_i/h_i) \tag{2.6}$$

denote the fraction of kernel i lying to the left of the origin, where P_κ is the kernel’s cumulative (single-integral) function, defined in (??). Then the single integral satisfies $J_i(m) \rightarrow 1 - \alpha_i$ as $m \rightarrow +\infty$ and $J_i(m) \rightarrow -\alpha_i$ as $m \rightarrow -\infty$, so the skew change c_i partitions into $c_i(1 - \alpha_i)$ propagating rightward and $-c_i\alpha_i$ propagating leftward. Because both J_i and I_i integrate from $m = 0$, one has $J_i(0) = I_i(0) = 0$ for *every* kernel, so $\sigma_{\text{smile}}(0, T) = 0$ and $s(0) = s_0$ regardless of kernel placement: the anchor at the origin is immovable. This is exactly the contract normalisation $\sigma_{\text{smile}}(0, T) = 0$ holding by construction.

(i) *Central kernel* ($\mu_i = 0$, hence $\alpha_i = \frac{1}{2}$). The propagation is symmetric: $+c_i/2$ rightward, $-c_i/2$ leftward, and by the symmetry of κ about its centre $I_i(-m) = I_i(m)$. The central kernel adds a symmetric dome (or trough, if $c_i < 0$) about the money.

(ii) *Off-centre and outermost kernels* ($\alpha_i \neq \frac{1}{2}$). The skew offset propagates asymmetrically, tilting one wing more than the other—in the ruler analogy, a bump away from the pin bends the longer lever arm more. Beyond the last kernel support, σ_{smile} lies entirely in $\mathcal{V}_0$: the smile is affine, with right-wing slope $s_0 + \sum_i c_i(1 - \alpha_i)$ and left-wing slope $s_0 - \sum_i c_i\alpha_i$ (each divided by $\sqrt{T}$ after the time scaling of Section ??). Imposing the wing constraints of Section ?? forces both slopes to vanish, flattening the smile to a constant beyond the grid on either side.

Remark 2.4 (Grid design). The grid need not be symmetric: a steep put skew calls for denser kernel placement on the put side than the call side, with one central kernel at $\mu_i = 0$. Non-overlapping or minimally overlapping supports give each c_i control of a distinct moneyness region and keep the design matrix Φ of Section ?? well conditioned. Since the two wing constraints consume two degrees of freedom among the c_i , a *buffer kernel* on each side beyond the last liquid strike prevents the constraint solver from distorting the fit in the data-rich region.

2.1.4 Integral Basis Functions

Define the cumulative integrals of the i -th kernel from the origin,

$$J_i(m) = \int_0^m \frac{1}{h_i} \kappa\left(\frac{u - \mu_i}{h_i}\right) du, \tag{2.7}$$

$$I_i(m) = \int_0^m J_i(u) du. \tag{2.8}$$

The function J_i is a C^1 sigmoid transitioning across the support $[\mu_i - h_i, \mu_i + h_i]$, and I_i is its antiderivative; both vanish at $m = 0$. (The kernel cumulative J_i , *always* carrying a kernel subscript, is distinct from the bare observation count J of Section ??; the two never collide in an unindexed form.) Their closed forms are collected in Section ?. Unlike spline interpolation, which imposes junction conditions at knots, the piecewise-polynomial structure here *arises* from double integration of the compactly supported κ ; the C^2 junctions at $\mu_i \pm h_i$ are automatic consequences of $\kappa(\pm 1) = 0$, not imposed constraints.

2.1.5 The Linear Model and the Bra-Ket Form

Substituting (??) into (??) and restoring the ATM level gives the full implied volatility as a single inner product. Writing the intercept parameter σ_0 for the calibrated ATM level,

$$\sigma(m, \ell_F, T) = \sigma_0 + s_0 m + \sum_{i=1}^n c_i I_i(m) = \langle \phi(m) | \beta \rangle, \quad (2.9)$$

with the feature ket and parameter ket

$$|\phi(m)\rangle = \begin{pmatrix} 1 \\ m \\ I_1(m) \\ \vdots \\ I_n(m) \end{pmatrix} \in \mathbb{R}^{n+2}, \quad |\beta\rangle = \begin{pmatrix} \sigma_0 \\ s_0 \\ c_1 \\ \vdots \\ c_n \end{pmatrix} \in \mathbb{R}^{n+2}.$$

Because $I_i(0) = 0$ for every kernel (the *origin-adjusted* convention of Section ??), the intercept is the ATM level itself, $\sigma_0 = \sigma_{\text{ATM}}(\ell_F, T)$, and the slope is the ATM smile skew, $s_0 = s(0)$; the smile part $\sigma_{\text{smile}}(m, T) = s_0 m + \sum_i c_i I_i(m)$ then satisfies $\sigma_{\text{smile}}(0, T) = 0$ exactly, as required by (??).

The calibration is performed at the prevailing forward, with ℓ_F frozen at its current market value, so the right-hand side of (??) carries no explicit ℓ_F argument and σ_0 estimates $\sigma_{\text{ATM}}(\ell_F, T)$ at that single point; the ℓ_F -dependence itself remains the exogenous input of Section ??, consumed by the routed Greeks of Section ?.

Remark 2.5 (Bra-ket convention). Dirac notation is used with the standard Euclidean inner product on $\mathbb{R}^{n+2}$: $|\phi(m)\rangle$ is the feature ket (column), $\langle \phi(m)|$ its transpose (row), and $\langle \phi(m) | \beta \rangle$ their dot product. The parameter ket $|\beta\rangle$ is the state of the calibrated market; $\langle \phi(m)|$ is the measurement functional that reads off the implied volatility at moneyness m . This bra-ket machinery is reserved throughout both volumes for the feature/parameter linear algebra; generic inner products elsewhere are written $x^\top y$.

Remark 2.6 (Derivative kets). Differentiation in m acts on the feature ket while $|\beta\rangle$ stays fixed:

$$|\phi'(m)\rangle = \begin{pmatrix} 0 \\ 1 \\ J_1(m) \\ \vdots \\ J_n(m) \end{pmatrix}, \quad |\phi''(m)\rangle = \begin{pmatrix} 0 \\ 0 \\ \kappa_1(m)/h_1 \\ \vdots \\ \kappa_n(m)/h_n \end{pmatrix}, \quad \kappa_i(m) = \kappa\left(\frac{m - \mu_i}{h_i}\right),$$

so that $s(m) = \langle \phi'(m) | \beta \rangle$ and $c(m) = \langle \phi''(m) | \beta \rangle$. The Dupire denominator of Section ?? depends on the 2-jet $(\sigma_{\text{smile}}, \partial_m \sigma_{\text{smile}}, \partial_m^2 \sigma_{\text{smile}}) = (\langle \phi | \beta \rangle, \langle \phi' | \beta \rangle, \langle \phi'' | \beta \rangle)$, the numerator on the 1-jet in T . Differentiation is thus an operator on the ket space that preserves linear dependence on $|\beta\rangle$: the formalism is structural, not merely notational.

Remark 2.7 (Origin-adjusted versus cumulative convention). If one instead builds the basis from the bare kernel cumulative P_κ without subtracting its value at the origin (the *cumulative* convention), kernels whose support straddles or lies left of $m = 0$ contribute non-zero offsets $J_i(0), I_i(0)$. The intercept σ_0 is then a bare regression coefficient, *not* the ATM level; the ATM level is recovered as $\sigma_{\text{ATM}}(\ell_F, T) = \sigma(0, \ell_F, T) = \sigma_0 + \sum_i c_i I_i(0)$ and the ATM skew as $s(0) = s_0 + \sum_i c_i J_i(0)$. The two conventions differ only by components in $\text{span}\{1, m\}$, so ordinary least squares compensates automatically and the calibrated curve $\sigma(m, \ell_F, T)$ —hence the decomposition $\sigma_{\text{smile}}(m, T) = \sigma(m, \ell_F, T) - \sigma_{\text{ATM}}(\ell_F, T)$ —is identical either way. The origin-adjusted convention is canonical here precisely because it makes the contract normalisation $\sigma_{\text{smile}}(0, T) = 0$ hold at the level of the intercept parameter.

Remark 2.8 (Regularisation implication). Since $\mathcal{V}_0$ carries no curvature and needs no smoothing, a natural ridge penalty acts only on the curvature coefficients: $\lambda \sum_{i=1}^n c_i^2$, leaving σ_0 and s_0 unpenalised. The rationale is again the ruler—the pins at the origin are set by the data; one wishes only to prevent the bumps from becoming unreasonably large.

Example 2.9 (Three-kernel equity smile). Take an intercept $\sigma_0 = 0.20$ and slope $s_0 = -0.16$ (both in the cumulative convention of Remark ??), with three kernels at a single expiry:

Kernel	μ_i	w_i	c_i
Put wing	$-1.5 \sigma_{\text{ref}} \sqrt{T}$	0.8	+0.60
ATM	0	0.8	+0.20
Call wing	$+1.0 \sigma_{\text{ref}} \sqrt{T}$	0.8	+0.10

With $\sigma_{\text{ref}} = 0.20$ and $T = 0.25$ the centres are $\mu_1 = -0.15$, $\mu_2 = 0$, $\mu_3 = +0.10$, all bandwidths $h_i = 0.08$, and the left-of-origin fractions are $\alpha_1 = 1$, $\alpha_2 = \frac{1}{2}$, $\alpha_3 = 0$. The put-side kernel bends the skew from $s_0 = -0.16$ up to $s_0 + c_1 = +0.44$ across its support, flattening the put wing.

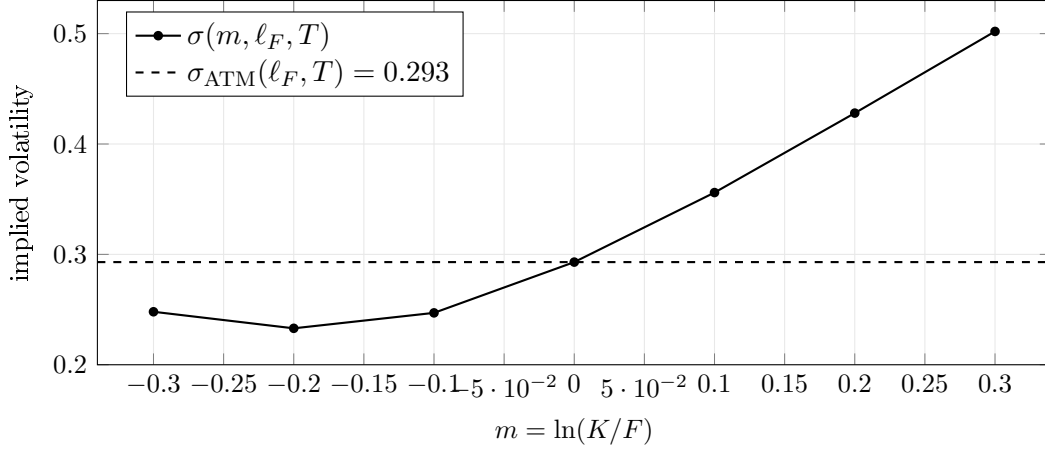
Reading off the decomposition. In the cumulative convention $J_1(0) = 1$, $I_1(0) = 0.15$, $J_2(0) = 0.5$, $I_2(0) = 0.015$, $J_3(0) = I_3(0) = 0$, so σ_0 is *not* the ATM level. The ATM component is recovered as

$$\sigma_{\text{ATM}}(\ell_F, T) = \sigma(0, \ell_F, T) = \sigma_0 + \sum_i c_i I_i(0) = 0.20 + 0.60(0.15) + 0.20(0.015) + 0.10(0) = 0.293,$$

and the ATM smile skew is $s(0) = s_0 + \sum_i c_i J_i(0) = -0.16 + 0.60 + 0.10 = 0.54$. The smile component is then $\sigma_{\text{smile}}(m, T) = \sigma(m, \ell_F, T) - 0.293$, which vanishes at $m = 0$ by construction. Representative values of the full implied vol:

m	-0.30	-0.20	-0.10	0	+0.10	+0.20	+0.30
$\sigma(m, \ell_F, T)$	0.248	0.233	0.247	0.293	0.356	0.428	0.502
$\sigma_{\text{smile}}(m, T)$	-0.045	-0.060	-0.046	0	+0.063	+0.135	+0.209

The smile minimum lies near $m = -0.20$, reflecting the interplay between the negative slope s_0 and the positive curvature contributions; σ_{smile} vanishes at the money, as (??) demands.



2.2 Time Dependence

2.2.1 Square-Root Scaling of the Feature Ket

Assumption 2.10 (Leading-order skew scaling). To leading order the ATM smile skew scales as $1/\sqrt{T}$. This is exact for surfaces that are functions of the standardised moneyness $m/(\sigma\sqrt{T})$ (the sticky-delta self-similar case). It is *not* exact for the general diffusive stochastic-volatility class: Heston, for instance, has an ATM skew that tends to a finite constant as $T \rightarrow 0$ and decays like $1/T$ at long maturity. In rough-volatility models the empirical scaling is closer to $T^{H-1/2}$ with Hurst exponent $H \approx 0.1$ (Fukasawa 2011; Bayer, Friz & Gatheral 2016), which the framework accommodates by replacing $\sqrt{\tilde{T}} = \tilde{T}^{1/2}$ with $\tilde{T}^{1/2-H}$ in the feature scaling below. Dividing the moneyness features by $\tilde{T}^{1/2-H}$ makes the skew scale as $\tilde{T}^{-(1/2-H)} = \tilde{T}^{H-1/2}$, as required (since $1/2 - H > 0$ for $H \approx 0.1$, this is a milder division than $1/\sqrt{\tilde{T}}$, giving a *flatter* short-dated skew—the opposite of $\tilde{T}^{H+1/2}$, which would steepen it), while leaving the regression linear.

Under Assumption ?? every moneyness-dependent component of the feature ket is divided by $\sqrt{\tilde{T}}$:

$$|\phi(m, T)\rangle = \begin{pmatrix} 1 \\ m/\sqrt{\tilde{T}} \\ I_1(m)/\sqrt{\tilde{T}} \\ \vdots \\ I_n(m)/\sqrt{\tilde{T}} \end{pmatrix}, \quad \sigma(m, \ell_F, T) = \langle \phi(m, T) | \beta \rangle. \quad (2.10)$$

The intercept row is left untouched: the ATM level carries its own term structure through $\sigma_{\text{ATM}}(\ell_F, T)$, developed in Section ?. The kernel centres and bandwidths scale with $\sqrt{\tilde{T}}$,

$$\mu_i(T) = w_i^{\text{ctr}} \sigma_{\text{ref}} \sqrt{\tilde{T}}, \quad h_i(T) = w_i \sigma_{\text{ref}} \sqrt{\tilde{T}}, \quad (2.11)$$

with w_i^{ctr}, w_i fixed constants. The reference volatility σ_{ref} is held *fixed* during calibration—chosen close to recent ATM vol but never set to the ATM parameter itself—so that the design matrix depends only on observable moneyness and time, and the model stays linear in $|\beta\rangle$.

2.2.2 Modified Time to Maturity

To prevent singular behaviour as $T \rightarrow 0$ and excessive flattening for large T , define the modified time $\tilde{T}$ by

$$T' = \frac{T_d + z}{252 + z}, \quad \tilde{T} = \frac{T'}{1 + T'/T_{\max}} + T_{\min}, \quad (2.12)$$

where T_d is time to expiry in *trading days* and $T' \approx T_d/252$ converts to years (with shift z for smoothing). Everywhere else in this chapter T denotes time to expiry in years; (??) is the sole appearance of trading-day units. The constants $T_{\min}, T_{\max}, z$ floor and cap $\tilde{T}$.

Proposition 2.11 (Bounds on $\tilde{T}$). *For $T \geq 0$, $z > 0$, $T_{\min} > 0$, $T_{\max} > 0$,*

$$T_{\min} < \tilde{T} < T_{\max} + T_{\min}.$$

Proof. The map $g(x) = x/(1 + x/T_{\max})$ sends $[0, \infty)$ onto $[0, T_{\max})$ strictly monotonically, with supremum $T_{\max}$ never attained. Since $T' \geq z/(252 + z) > 0$ and $T' < \infty$ for finite T , $g(T') \in (0, T_{\max})$; adding $T_{\min}$ gives the claim. $\square$

Remark 2.12 (Reference constants).

Parameter	Value	Rationale
$T_{\min}$	$2/252 \approx 0.008$	floor at 2 trading days
$T_{\max}$	5.0	cap at longest liquid maturity
z	2	smooths the $T = 0$ singularity

2.2.3 Time-Dependent Parameters

Each component of $|\beta\rangle$ —including the ATM level—is expanded over time-decay kernels,

$$\beta_i(T) = \sum_j b_{i,j} \psi(T; \tau_j, p), \quad \psi(T; \tau, p) = 2^{-(T/\tau)^p} = \exp(-(T/\tau)^p \ln 2).$$

At $T = \tau$ one has $\psi = \frac{1}{2}$ regardless of p , justifying the name half-life; here τ is a locally scoped decay parameter, not a second time variable. If each of the $n + 2$ parameters is expanded over L decay kernels, the block-diagonal matrix $\Lambda(T) \in \mathbb{R}^{(n+2) \times N_b}$, with $N_b = (n + 2)L$, collects the ψ -values and

$$|\beta(T)\rangle = \Lambda(T) |\mathbf{b}\rangle, \quad \sigma(m, \ell_F, T) = \langle \tilde{\phi}(m, T) | \mathbf{b} \rangle, \quad |\tilde{\phi}(m, T)\rangle = \Lambda(T)^\top |\phi(m, T)\rangle. \quad (2.13)$$

The model remains linear in the extended parameter ket $|\mathbf{b}\rangle$. The T -derivative of ψ , needed for the local-volatility numerator, is

$$\psi'(T; \tau, p) = -\frac{p \ln 2}{\tau} \left(\frac{T}{\tau} \right)^{p-1} \psi(T; \tau, p). \quad (2.14)$$

2.3 Calibration

2.3.1 Ordinary Least Squares

Given J market observations $\{(\hat{m}_j, \hat{\sigma}_j)\}_{j=1}^J$, form the design matrix $\Phi \in \mathbb{R}^{J \times (n+2)}$ whose j -th row is $\langle \phi(\hat{m}_j) |$, and the observation ket $|\hat{\sigma}\rangle = (\hat{\sigma}_1, \dots, \hat{\sigma}_J)^\top$. Since $\sigma(\hat{m}_j) = \langle \phi(\hat{m}_j) | \beta \rangle$, the model-implied vector is $\Phi |\beta\rangle$ and the OLS estimator is

$$|\hat{\beta}\rangle = (\Phi^\top \Phi)^{-1} \Phi^\top |\hat{\sigma}\rangle, \quad (2.15)$$

the orthogonal projection of $|\hat{\sigma}\rangle$ onto $\text{Im}(\Phi)$.

Proposition 2.13 (Regularity). *The estimator (??) is well defined if and only if Φ has full column rank $n + 2$. Sufficient: $J \geq n + 2$ and at least one observed moneyness inside each kernel support.*

Remark 2.14 (Conditioning). When supports overlap heavily or data is sparse, $\Phi^\top \Phi$ can be ill-conditioned; ridge regression $|\hat{\beta}\rangle = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top |\hat{\sigma}\rangle$ with small $\lambda > 0$ restores stability. Conversely, when the kernels have disjoint (or nearly disjoint) supports and each support holds at least one observation, the curvature columns of Φ have disjoint non-affine parts, $\Phi^\top W \Phi$ is nearly block-diagonal, and the problem is well conditioned.

2.3.2 Weighted and Constrained Regression

Observations are weighted by inverse bid–ask spread or by vega:

$$|\hat{\beta}\rangle = (\Phi^\top W \Phi + \lambda I)^{-1} \Phi^\top W |\hat{\sigma}\rangle, \quad W = \text{diag}(\varpi_1, \dots, \varpi_J), \quad \varpi_j \propto \nu_j^2 / \text{spread}_j^2, \quad (2.16)$$

where the locally scoped weight symbol ϖ_j is distinct both from the kernel width constants w_i and from the contract-reserved ω_M (Margrabe, R2), and ν_j is the Black–Scholes vega of quote j . To enforce the arbitrage constraints of Section ??, replace unconstrained OLS by constrained least squares,

$$\min_{|\beta\rangle} \| |\hat{\sigma}\rangle - \Phi |\beta\rangle \|_W^2 \quad \text{subject to} \quad g_\ell(|\beta\rangle) \geq 0, \quad \ell = 1, \dots, L,$$

where the g_ℓ are the admissibility constraints of Section ?? evaluated on a moneyness grid. These are *not* uniformly linear in $|\beta\rangle$, in keeping with the chapter’s opening claim that the no-arbitrage conditions are low-degree *polynomial* inequalities. Only some are affine: positivity (A1) $\sigma > 0$, the calendar condition (A3) $\mathcal{N} > 0$ (equivalently $\sigma + 2T \partial_T \sigma > 0$, affine in $|\beta\rangle$ once σ is), and the flat-wing constraints (??)–(??). The butterfly condition (A2) $D > 0$ is genuinely quadratic-to-quartic in $|\beta\rangle$ —it contains σc , s^2 and $\sigma^2 T^2 s^2$ with σ, s, c all linear in $|\beta\rangle$ —and even the sufficient surrogate (S1) $c > (\sigma T/4)s^2$ is quadratic. The programme is solved either in its true polynomially-constrained form (SOCP/SDP or sequential quadratic programming), or by successive linearisation of D about the current fit—a linear surrogate that is only *sufficient* and may reject some admissible surfaces—followed by a *mandatory* final pass verifying $D > 0$ and $\mathcal{N} > 0$ exactly on the grid.

2.3.3 Convexity Bias Between Mid-Price and Mid-Vol

Let $p(\sigma)$ be the Black–Scholes put price as a function of implied volatility, and set $p_{\text{mid}} = \frac{1}{2}(p(\sigma_{\text{bid}}) + p(\sigma_{\text{ask}}))$.

Proposition 2.15 (Convexity bias for OTM options). *Suppose $d_1 d_2 > 0$ for every $\sigma \in [\sigma_{\text{bid}}, \sigma_{\text{ask}}]$ (under the standing zero rates $F = S$ this is $|m| > \sigma_{\text{ask}}^2 T/2$, i.e. the whole quoted band is sufficiently out-of-the-money). Then the Volga $\partial^2 p / \partial \sigma^2 = \nu d_1 d_2 / \sigma = F \sqrt{T} \varphi(d_1) d_1 d_2 / \sigma > 0$ throughout the band, $p(\sigma)$ is convex there, and the implied vol σ_{mid} of p_{mid} satisfies*

$$\sigma_{\text{mid}} \geq \frac{1}{2}(\sigma_{\text{bid}} + \sigma_{\text{ask}}).$$

Proof. Write $\bar{\sigma} = \frac{1}{2}(\sigma_{\text{bid}} + \sigma_{\text{ask}})$. Convexity of p on the band gives, by Jensen, $p_{\text{mid}} = \frac{1}{2}(p(\sigma_{\text{bid}}) + p(\sigma_{\text{ask}})) \geq p(\bar{\sigma})$. Since p is strictly increasing in σ (vega > 0), its inverse is increasing, whence $\sigma_{\text{mid}} = p^{-1}(p_{\text{mid}}) \geq p^{-1}(p(\bar{\sigma})) = \bar{\sigma}$. The sign hypothesis is required across the *whole* band because $d_1 d_2$ changes sign at $|m| = \sigma^2 T/2$; a wide quoted spread straddling that point breaks the convexity and the conclusion. $\square$

Remark 2.16. Near the money $d_1 d_2 < 0$, the Volga is negative, $p(\sigma)$ is concave, and the inequality reverses. The effect is small for tight spreads but material for illiquid deep-OTM options; it biases naive mid-vol fits and is one reason to weight by vega rather than to average vols.

Example 2.17 (Convexity bias). Under the standing zero-rate assumption ($F = S$), take $S = 100$, $K = 85$ (an OTM put), $T = 0.25$, $\sigma_{\text{bid}} = 0.25$, $\sigma_{\text{ask}} = 0.35$. Then $p(0.25) \approx 0.52$, $p(0.35) \approx 1.54$, so $p_{\text{mid}} \approx 1.03$, whose implied vol is $\sigma_{\text{mid}} \approx 0.307$ —above the arithmetic mid 0.300.

2.4 Sensitivities and Total Greeks

2.4.1 Vega and Volga Decomposed per Kernel

For the operative time-scaled model (??), $\sigma(m, \ell_F, T) = \langle \phi(m, T) | \beta \rangle$, the first-order sensitivity of an option price C to each parameter factorises through the Black–Scholes vega $\nu = \partial C / \partial \sigma$:

$$\frac{\partial C}{\partial \beta_i} = \nu \cdot \phi_i(m, T), \quad (2.17)$$

where $\phi_i(m, T)$ is the i -th *time-scaled* feature component, carrying for the moneyness rows the $1/\sqrt{T}$ factor of (??) (using the un-scaled $\phi_i(m)$ mis-scales per-kernel risk attribution by $\sqrt{T}$). The second-order sensitivity factorises through the Volga:

$$\frac{\partial^2 C}{\partial \beta_i \partial \beta_j} = \text{Volga} \cdot \phi_i(m, T) \phi_j(m, T), \quad \text{Volga} = \frac{\partial^2 C}{\partial \sigma^2} = \nu \frac{d_1 d_2}{\sigma}. \quad (2.18)$$

Remark 2.18 (Curvature orthogonality is not feature orthogonality). For non-overlapping kernels the *curvature* functions satisfy $\kappa_i(m) \kappa_j(m) = 0$ for $i \neq j$, so curvature contributions are independent and risk attribution at the level of $c(m)$ is diagonal. The *feature* functions do not inherit this: beyond its support $I_i(m)$ is affine (not zero), so $I_i(m) I_j(m) \neq 0$ in general and the Hessian (??) is not diagonal in the β indices for a single instrument. Diagonality holds in curvature space, not feature space.

2.4.2 Routing Through the Forward: Total Greeks

The implied-vol model supplies the derivatives that convert flat Black–Scholes Greeks into smile-adjusted (total) Greeks. Here the decomposition (??) matters decisively: as spot moves, *both* the ATM level (through ℓ_F) and the smile shape (through m) respond. Carrying the chain rule through $F(S)$,

$$\ell_F = \ln(F/F_{\text{ref}}), \quad \frac{\partial \ell_F}{\partial S} = \frac{1}{F} \frac{dF}{dS}; \quad m = \ln(K/F), \quad \frac{\partial m}{\partial S} = -\frac{1}{F} \frac{dF}{dS}.$$

Under the standing zero-rate assumption $F = S$, $dF/dS = 1$, so $\partial \ell_F / \partial S = 1/S$ and $\partial m / \partial S = -1/S$. Hence the total first and second spot-derivatives of implied vol are

$$\frac{d\sigma}{dS} = \frac{\partial \sigma_{\text{ATM}}}{\partial \ell_F} \frac{\partial \ell_F}{\partial S} + \frac{\partial \sigma_{\text{smile}}}{\partial m} \frac{\partial m}{\partial S} = \frac{\text{vol.slope} - s(m)}{S}, \quad (2.19)$$

$$\frac{d^2 \sigma}{dS^2} = \frac{(\partial_{\ell_F}^2 \sigma_{\text{ATM}} - \text{vol.slope}) + (c(m) + s(m))}{S^2}. \quad (2.20)$$

The routed spot-derivatives on the left are *total* (d/dS), since they run through both arguments $m(S)$ and $\ell_F(S)$; the components $\partial\sigma_{\text{ATM}}/\partial\ell_F$ and $\partial\sigma_{\text{smile}}/\partial m$ on the right remain genuine partials. These feed the total (smile-adjusted) gamma,

$$\Gamma_{\text{adj}} = \Gamma + 2 \text{Vanna} \cdot \frac{d\sigma}{dS} + \text{Volga} \cdot \left(\frac{d\sigma}{dS} \right)^2 + \nu \cdot \frac{d^2\sigma}{dS^2}. \quad (2.21)$$

Equation (??) is the second total derivative d^2/dS^2 of $C(S, \sigma(S))$: expanding $dC/dS = \partial_S C + \partial_\sigma C \cdot d\sigma/dS$ once more, and using $C \in C^2$ so that mixed partials commute ($\partial_S \partial_\sigma C = \partial_\sigma \partial_S C = \text{Vanna}$), gives the four terms, with $\Gamma = \partial_S^2 C$, $\text{Vanna} = \partial_S \partial_\sigma C$, $\text{Volga} = \partial_\sigma^2 C$, $\nu = \partial_\sigma C$ the frozen-vol Black–Scholes Greeks.

Remark 2.19 (The $\sigma_{\text{ATM}} = \text{const}$ collapse, flagged). The C4 convention sets $\sigma_{\text{ATM}} = \text{const}$, i.e. $\text{vol.slope} = 0$ and $\partial_{\ell_F}^2 \sigma_{\text{ATM}} = 0$. Under that convention (??)–(??) reduce to $d\sigma/dS = -s(m)/S$ and $d^2\sigma/dS^2 = (c(m) + s(m))/S^2$ —the pure-smile terms only, discarding exactly the vol.slope contribution of Remark ???. The full forms (??)–(??) are the ones that must enter any book carrying genuine ATM dynamics, and it is through vol.slope that the surface term structure enters the multi-name shadow Greeks of the second volume.

2.5 Local Volatility

2.5.1 Dupire’s Formula and the Black–Scholes Partial

Under zero rates and dividends, Dupire’s formula (Dupire 1994) reads

$$\sigma_{\text{loc}}^2(K, T) = \frac{\partial_T C}{\frac{1}{2} K^2 \partial_K^2 C}. \quad (2.22)$$

The derivatives are those of the Black–Scholes call $C = F N(d_1) - K N(d_2)$, in which $d_1 = -m/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2$ and $d_2 = d_1 - \sigma\sqrt{T}$; $\sigma = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ depends on K only through $m = \ln(K/F)$ (the ATM component is constant in K at fixed forward).

Temporal derivative. Holding K fixed—and, under the standing zero-rate assumption $F = S$, the forward is constant across maturities so m is fixed as well—

$$\partial_T C = \frac{F \varphi(d_1)}{\sqrt{T}} \left(\frac{\sigma}{2} + T \frac{\partial \sigma}{\partial T} \right). \quad (2.23)$$

Spatial second derivative. With $\sigma = \sigma(m(K), T)$ the chain rule gives

$$\frac{d^2 C}{dK^2} = \frac{\partial^2 C}{\partial K^2} \Big|_\sigma + 2 \frac{\partial^2 C}{\partial K \partial \sigma} \frac{\partial \sigma}{\partial K} + \frac{\partial^2 C}{\partial \sigma^2} \left(\frac{\partial \sigma}{\partial K} \right)^2 + \frac{\partial C}{\partial \sigma} \frac{\partial^2 \sigma}{\partial K^2}, \quad (2.24)$$

where the frozen-vol Black–Scholes partials are

$$\frac{\partial^2 C}{\partial K^2} \Big|_\sigma = \frac{\varphi(d_2)}{K \sigma \sqrt{T}}, \quad \frac{\partial^2 C}{\partial K \partial \sigma} = \frac{\varphi(d_2) d_1}{\sigma}, \quad \frac{\partial^2 C}{\partial \sigma^2} = F \sqrt{T} \varphi(d_1) \frac{d_1 d_2}{\sigma}. \quad (2.25)$$

These are verified in Section ??; the key identity is $F \varphi(d_1) = K \varphi(d_2)$, exact in the forward measure since $d_1^2 - d_2^2 = -2m$ and $K/F = e^m$.

2.5.2 The Local-Volatility Master Formula

Converting K -derivatives to m -derivatives via $\partial\sigma/\partial K = (1/K) \partial\sigma/\partial m$ and $K^2 \partial_K^2 \sigma = \partial_m^2 \sigma - \partial_m \sigma$, writing $s = \partial_m \sigma = s(m)$ and $c = \partial_m^2 \sigma = c(m)$ for the smile skew and curvature, and factoring out $F \varphi(d_1)/(2\sigma\sqrt{T})$, the denominator becomes

$$\frac{1}{2} K^2 \partial_K^2 C = \frac{F \varphi(d_1)}{2\sigma\sqrt{T}} D(m, T), \quad (2.26)$$

with

$$D(m, T) = 1 - \frac{2m}{\sigma} \frac{\partial\sigma}{\partial m} + \left(\frac{m^2}{\sigma^2} - \frac{\sigma^2 T^2}{4} \right) \left(\frac{\partial\sigma}{\partial m} \right)^2 + \sigma T \frac{\partial^2 \sigma}{\partial m^2}. \quad (2.27)$$

The clean form (??) rests on one cancellation: the substitution $K^2 \partial_K^2 \sigma = \partial_m^2 \sigma - \partial_m \sigma$ injects a term $-\sigma T s$, while the expansion $2d_1\sqrt{T} = -2m/\sigma + \sigma T$ injects an equal and opposite $+\sigma T s$. Combining (??) and (??),

$$\boxed{\sigma_{\text{loc}}^2(m, T) = \sigma^2 \frac{\mathcal{N}(m, T)}{D(m, T)}}, \quad \mathcal{N}(m, T) = 1 + \frac{2T}{\sigma} \frac{\partial\sigma}{\partial T}, \quad (2.28)$$

and D as in (??). The numerator $\mathcal{N}$ carries the calendar-spread condition; the denominator D carries the butterfly condition (Section ??).

2.5.3 Application to the Kernel Model

From the time-scaled model (??),

$$\frac{\partial\sigma}{\partial m} = s(m) = \frac{1}{\sqrt{\tilde{T}}} \left(s_0 + \sum_i c_i J_i(m) \right), \quad (2.29)$$

$$\frac{\partial^2 \sigma}{\partial m^2} = c(m) = \frac{1}{\sqrt{\tilde{T}}} \sum_i \frac{c_i}{h_i} \kappa \left(\frac{m - \mu_i}{h_i} \right). \quad (2.30)$$

The temporal derivative $\partial\sigma/\partial T$ needed for $\mathcal{N}$ follows from (??) by the chain rule, with two sources: the time scaling of the moneyness features and the decay functions. For the first, the kernel scaling (??) makes the centres μ_i and widths h_i themselves scale with $\sqrt{\tilde{T}}$, so each moneyness feature $I_i(m)/\sqrt{\tilde{T}}$ is a *pure function* $f_i(u)$ of the standardised moneyness $u = m/\sqrt{\tilde{T}}$, with $f'_i(u) = J_i(m)$ (and the linear feature is $f_0(u) = u$). Differentiating $f_i(m/\sqrt{\tilde{T}})$ in T therefore routes entirely through u :

$$\frac{d\tilde{T}}{dT} = \frac{252}{252 + z} \cdot \frac{1}{(1 + T'/T_{\text{max}})^2}, \quad \frac{\partial}{\partial T} |\phi(m, T)\rangle = -\frac{1}{2\tilde{T}} \frac{d\tilde{T}}{dT} \begin{pmatrix} 0 \\ m/\sqrt{\tilde{T}} \\ m J_1(m)/\sqrt{\tilde{T}} \\ \vdots \\ m J_n(m)/\sqrt{\tilde{T}} \end{pmatrix},$$

each curvature row being $-\frac{1}{2\tilde{T}}(d\tilde{T}/dT) u f'_i(u) = -\frac{1}{2\tilde{T}}(d\tilde{T}/dT) m J_i(m)/\sqrt{\tilde{T}}$ —the $m J_i(m)$, *not* $I_i(m)$, being the correct chain-rule factor once μ_i, h_i scale with $\sqrt{\tilde{T}}$; the two coincide only for the linear row, where $u f'_0(u) = f_0(u)$ preserves the form $m/\sqrt{\tilde{T}}$. The second source, the decay

functions, enters through (??). In the single-expiry case (fixed T , no decay) $\partial\sigma/\partial T = 0$ and $\mathcal{N} = 1$. Inside the support of kernel i the curvature peaks at the centre,

$$c(\mu_i) = \frac{3c_i}{4w_i\sigma_{\text{ref}}\tilde{T}}, \quad (2.31)$$

the single most useful quantity for the admissibility checks below.

2.5.4 Worked Example: The Quadratic Smile

Example ?? is now worked completely: a smile quadratic in log-moneyness, i.e. constant curvature c_0 , produced by one dominant central kernel. Write

$$\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + s_0 m + \frac{1}{2}c_0 m^2, \quad \frac{\partial\sigma}{\partial m} = s_0 + c_0 m, \quad \frac{\partial^2\sigma}{\partial m^2} = c_0.$$

Substituting into the master formula (??) gives the local variance at every (m, T) ; two regimes follow.

At the money ($m = 0$). The terms in m drop and $\sigma = \sigma_{\text{ATM}}(\ell_F, T)$, so, using the completed-square form (??) below,

$$\sigma_{\text{loc}}^2(0, T) = \sigma_{\text{ATM}}^2 \frac{1 + \frac{2T}{\sigma_{\text{ATM}}} \partial_T \sigma_{\text{ATM}}}{1 + \sigma_{\text{ATM}} T c_0 - \frac{\sigma_{\text{ATM}}^2 T^2}{4} s_0^2}. \quad (2.32)$$

Positivity of the denominator is a lower bound on the ATM curvature,

$$c_0 > \frac{\sigma_{\text{ATM}} T}{4} s_0^2 - \frac{1}{\sigma_{\text{ATM}} T}, \quad (2.33)$$

which tightens with maturity: a long-dated smile with appreciable ATM skew s_0 *must* carry positive curvature to remain free of butterfly arbitrage. This is the exact quadratic-smile instance of the general sufficient condition of Proposition ??.

The purely linear (zero-curvature) sub-case. Setting $c_0 = 0$ leaves σ affine in m . The denominator at the money is then $1 - \frac{1}{4}\sigma_{\text{ATM}}^2 T^2 s_0^2$, which turns negative once $\sigma_{\text{ATM}} T |s_0| > 2$: a straight-line smile with non-zero skew inevitably violates the butterfly bound at sufficiently long maturity. Curvature is not a cosmetic refinement—it is what rescues admissibility. This is precisely why the wing tails of Section ?? are flattened to constants rather than left sloped, and why every kernel grid must retain genuine curvature in the fitted region.

2.6 The No-Arbitrage Admissibility Ladder

A positive, well-defined local variance requires both $\mathcal{N} > 0$ and $D > 0$ everywhere; together with positivity of the vol itself these are the first rungs of the admissibility ladder that continues, in the multi-name volume, into correlation and basket constraints.

Definition 2.20 (Admissible parameter set). A parameter ket $|\mathbf{b}\rangle$ is *admissible* if, for all (m, T) in the model domain:

(A1) **Positivity:** $\sigma(m, \ell_F, T) > 0$.

(A2) **Butterfly**: $D(m, T) > 0$ (strictly positive risk-neutral density).

(A3) **Calendar spread**: $\mathcal{N}(m, T) > 0$, i.e. total variance $w(m, T) = \sigma^2 T$ is strictly increasing in T .

Write $\mathcal{A}$ for the admissible set. $\mathcal{A}$ is taken *open* (strict inequalities): the strict interior is what keeps the division in the local-variance formula (??) well posed and gives the constrained optimiser of Section ?? a safe interior to sit in. The closed relaxation ($D \geq 0 \iff$ non-negative density, $\mathcal{N} \geq 0 \iff w$ non-decreasing) is arbitrage-free but sits on the boundary, where σ_{loc} may degenerate—the open-versus-closed distinction is operationally load-bearing because the optimiser will activate exactly the flat-calendar and zero-density faces.

Proposition 2.21 (Necessity for arbitrage freedom). *If $|\mathbf{b}\rangle \notin \mathcal{A}$, the surface admits static arbitrage. Precisely: if (A2) fails the risk-neutral density is negative (a butterfly arbitrage); if (A3) fails total variance decreases in T at fixed strike, a calendar-spread arbitrage; if (A1) fails the Black-Scholes map is ill-posed (σ_{loc}^2 undefined), and a negative total variance $w = \sigma^2 T$ prices options below intrinsic value, breaking monotonicity of C in σ .*

Proof. (A2): by (??), $\frac{1}{2}K^2\partial_K^2 C = F\varphi(d_1)D/(2\sigma\sqrt{T})$, so the Breeden–Litzenberger density $q(K) = \partial_K^2 C = F\varphi(d_1)D/(K^2\sigma\sqrt{T})$ is a *positive* multiple of D ; hence $q > 0 \iff D > 0$. (A3): $\partial_T w = \partial_T(\sigma^2 T) = \sigma^2 + 2\sigma T \partial_T \sigma = \sigma^2(1 + (2T/\sigma)\partial_T \sigma) = \sigma^2 \mathcal{N}$, so $\mathcal{N} > 0 \iff w$ strictly increasing in T ; a strike at which w decreases furnishes a calendar-spread arbitrage (under the standing zero rates $F = S$, fixed K is fixed m —flagged). (A1): $\sigma < 0$ does not give literally negative prices (a deep-ITM call stays above zero) but a price below intrinsic value, violating monotonicity of C in σ , and σ_{loc}^2 in (??) is undefined. The calendar-spread argument for (A3) follows Lee (2004) and Dupire (1994). $\square$

2.6.1 Wing Behaviour and Lee’s Moment Formula

Beyond all kernel supports σ is affine in m , so $w = \sigma^2 T$ grows quadratically in $|m|$ unless the wing slope vanishes. Lee’s moment formula (Lee 2004) bounds the asymptotic slope of total variance,

$$\limsup_{m \rightarrow +\infty} \frac{w(m, T)}{m} \leq 2, \quad \limsup_{m \rightarrow -\infty} \frac{w(m, T)}{|m|} \leq 2,$$

which an unconstrained affine wing violates. The remedy is to force each wing flat. With $\alpha_i = P_\kappa(-\mu_i/h_i)$ as in (??), the right- and left-wing slopes are proportional to $s_0 + \sum_i c_i(1 - \alpha_i)$ and $s_0 + \sum_i c_i J_i^-$ respectively, where

$$J_i^- = \lim_{m \rightarrow -\infty} J_i(m) = -\alpha_i = \begin{cases} 0, & \mu_i - h_i > 0, \\ -1, & \mu_i + h_i < 0, \\ -P_\kappa(-\mu_i/h_i), & |\mu_i| < h_i, \end{cases}$$

giving the two *wing constraints*

$$s_0 + \sum_{i=1}^n c_i(1 - \alpha_i) = 0 \quad (\text{right wing}), \tag{2.34}$$

$$s_0 + \sum_{i=1}^n c_i J_i^- = 0 \quad (\text{left wing}). \tag{2.35}$$

When all supports lie right of the origin ($\alpha_i = 0$), these reduce to $s_0 + \sum_i c_i = 0$ and $s_0 = 0$.

Proposition 2.22 (Compliance with Lee’s bound). *Under (??)–(??), $\partial_m \sigma = 0$ beyond all supports, so $\sigma \rightarrow \sigma_\infty^\pm$ (finite positive constants), $w = (\sigma_\infty^\pm)^2 T$ is bounded, and $w(m, T)/|m| \rightarrow 0 < 2$ as $|m| \rightarrow \infty$. Lee’s moment formula is satisfied with room to spare.*

Remark 2.23 (Flat wings under-parameterise the tails). The room to spare is not free. Flat wings are strictly stronger than Lee’s bound, which permits a total-variance wing slope up to 2. Pinning $\partial_m \sigma = 0$ beyond the supports fixes deep-OTM implied vol at the constants $\sigma_\infty^\pm$, so the model is structurally blind to legitimate wing *steepening*: it cannot represent linear wing growth up to Lee’s boundary and therefore systematically under-prices the tails and tail-hedge cost relative to a surface sitting near that boundary. This is a deliberate “precise in the centre, silent in the tails” trade; a book with material tail exposure requires a bounded non-zero wing slope within Lee’s bound rather than a forced zero.

2.6.2 Sufficient Conditions for the Butterfly Bound

Proposition 2.24 (Completing the square in D). *Writing $s = \partial_m \sigma$, $c = \partial_m^2 \sigma$, the denominator (??) admits the identity*

$$D(m, T) = \left(1 - \frac{ms}{\sigma}\right)^2 + \sigma T c - \frac{\sigma^2 T^2}{4} s^2. \quad (2.36)$$

Consequently $D(m, T) > 0$ if either holds:

$$(S1) \text{ *Curvature dominates skew (pointwise, } m\text{-free): } \sigma T c > \frac{\sigma^2 T^2}{4} s^2, \text{ i.e. } c > \frac{\sigma T}{4} s^2.*$$

$$(S2) \text{ *Joint bound (exact): } \sigma T c > -\left(1 - \frac{ms}{\sigma}\right)^2 + \frac{\sigma^2 T^2}{4} s^2.*$$

Proof. Expanding $(1 - ms/\sigma)^2 = 1 - 2ms/\sigma + m^2 s^2/\sigma^2$ and comparing with (??) establishes (??). Since the squared term is non-negative, $D \geq \sigma T c - \frac{1}{4} \sigma^2 T^2 s^2$, strict under (S1). Condition (S2) is the exact rearrangement $D > 0$, and (S1) implies (S2) for all m because dropping the non-negative squared term only weakens the requirement. $\square$

Remark 2.25 (Kernel-parameter estimate). At a kernel centre μ_i the curvature attains its peak (??), $c(\mu_i) = 3c_i/(4w_i\sigma_{\text{ref}}\tilde{T})$, so, carrying $\tilde{T}$ explicitly,

$$\sigma T c(\mu_i) = \frac{3\sigma c_i T}{4w_i\sigma_{\text{ref}}\tilde{T}} \approx \frac{3c_i\sigma}{4w_i\sigma_{\text{ref}}} \quad (T \approx \tilde{T}),$$

a genuinely dimensionless quantity, but not because c_i and σ share units. In the time-scaled feature model c_i is itself *dimensionless*: the $1/\sqrt{\tilde{T}}$ carried in the moneyness rows of the feature map absorbs the volatility dimension, so the fitted coefficients c_i are pure numbers. The product is then dimensionless term by term— c_i dimensionless, $\sigma/\sigma_{\text{ref}}$ a dimensionless volatility ratio, and $T/\tilde{T} \approx 1$ —rather than through any cancellation of a shared vol unit against σ_{ref} . (The earlier $\approx \frac{3c_i\sigma}{4w_i\sigma_{\text{ref}}}$ is the same quantity at $T \approx \tilde{T}$; it is dimensionless for the identical reason, the σ_{ref} in the denominator scaling σ , not c_i , to a ratio.) Two facts follow, and *neither is a cap* on $|c_i|$. (i) Where the curvature is negative ($c_i < 0$), the term $\sigma T c$ pulls D down, and $D > 0$ near the centre requires roughly $3|c_i|\sigma/(4w_i\sigma_{\text{ref}}) < 1 + (1 - ms/\sigma)^2$ by the completed-square form (??). (ii) Condition (S1) is a *lower* bound, $c(m) > (\sigma T/4) s(m)^2$: butterfly no-arbitrage demands *enough* positive curvature, not a bounded amount—the threat is curvature too negative, not too large. Because c is largest at the centre and vanishes between and beyond the kernels—where s^2 is nonetheless accumulated—(S1) is easiest to satisfy at the very point evaluated here; the binding check therefore lies between and beyond the supports and at the $\mu_i \pm h_i$ kinks, where the exact denominator (??) or the joint bound (S2) must be used.

2.6.3 Worked Admissibility and Local-Volatility Checks

Example 2.26 (Positivity at a kernel centre). These worked checks use the single-expiry, *un-time-scaled* model (??) (equivalently, $\tilde{T}$ folded into the bandwidth h_1), so the centre curvature is read directly as $c(\mu_1) = \kappa(0) c_1/h_1 = 3c_1/(4h_1)$, *not* from the time-scaled (??). A single put-side kernel: $c_1 = 2.0$, $\mu_1 = -0.15$, $h_1 = 0.08$, $\sigma_0 = 0.20$, $s_0 = -0.10$, $T = 0.50$. The support $[-0.23, -0.07]$ is entirely left of the origin, so $\alpha_1 = 1$. At $m = \mu_1 = -0.15$:

- $\sigma \approx 0.245$;
- $s = s_0 + c_1 J_1(\mu_1) = -0.10 + 2.0(-0.50) = -1.10$ (at the centre $J_1 = P_\kappa(0) - \alpha_1 = 0.5 - 1 = -0.5$);
- $c = 3(2.0)/(4 \cdot 0.08) = 18.75$;
- $D = 1 - 2 \frac{-0.15}{0.245}(-1.10) + \left(\frac{0.15^2}{0.245^2} - \frac{0.245^2 \cdot 0.5^2}{4} \right)(1.21) + 0.245 \cdot 0.5 \cdot 18.75 = 1 - 1.35 + 0.45 + 2.30 = 2.40 > 0$.

Flipping c_1 to -3.0 (a concave smile) and reading a butterfly violation off the curvature term alone is an invalid *ceteris paribus*: σ , s and c all depend on c_1 . Recomputing honestly at the same point $m = \mu_1 = -0.15$: $\sigma = 0.20 + 0.015 + (-3.0)(0.015) = 0.17$, $s = -0.10 + (-3.0)(-0.50) = +1.40$ (the skew flips sign), $c = 3(-3.0)/(4 \cdot 0.08) = -28.125$, giving

$$D = 1 - 2 \frac{-0.15}{0.17}(1.40) + \left(\frac{0.15^2}{0.17^2} - \frac{0.17^2 \cdot 0.5^2}{4} \right)(1.96) + 0.17 \cdot 0.5 \cdot (-28.125) = 1 + 2.47 + 1.52 - 2.39 = +2.60 > 0.$$

The centre is *rescued*: the sign flip of s makes the $-2ms/\sigma$ term strongly positive, more than offsetting the negative σTc . The lesson is precisely that negative curvature lowers D through σTc but simultaneously moves s , so a butterfly violation cannot be diagnosed from one term at one point: it must be *located* by evaluating D over the whole support and its edges. Indeed $c(m)$ is continuous but kinks at $\mu_i \pm h_i$ (extra extremum candidates), and with $c_1 = -3.0$ the vol itself turns negative in the far left tail—an (A1) violation—so both positivity and butterfly must be scanned across moneyness, never sampled at the centre alone.

Example 2.27 (Local volatility at the kernel centre). Continuing Example ?? ($\sigma = 0.245$, $s = -1.10$, $c = 18.75$, $T = 0.50$), suppose a mild term-structure decay $\partial\sigma/\partial T \approx -0.02$. Then

$$\mathcal{N} = 1 + \frac{2(0.50)}{0.245}(-0.02) = 0.918, \quad D = 2.40, \quad \sigma_{\text{loc}}^2 = \sigma^2 \frac{\mathcal{N}}{D} = 0.245^2 \cdot \frac{0.918}{2.40} = 0.0230,$$

so $\sigma_{\text{loc}} \approx 0.152$, about 15.2%—well below the implied 24.5%. The strong positive curvature inflates D ; intuitively, a hedging portfolio captures the pronounced gamma of the convex smile, lowering the local volatility the model needs to reproduce the same prices.

Remark 2.28 (The ladder continues). Definition ?? and Propositions ??–?? close the single-name rungs: positivity, butterfly, calendar spread, and Lee wing control. In the multi-name volume the same ladder acquires correlation positivity and basket-convexity rungs, with the constituent vols entering as the ATM components $\sigma_{\text{ATM},i}(\ell_{F,i}, T)$ of this chapter’s decomposition, each carrying its own forward log-level $\ell_{F,i}$ (contract bridging identity C1).

2.7 Summary

The curvature-kernel construction gives the implied surface as a single inner product, $\sigma(m, \ell_F, T) = \langle \tilde{\phi}(m, T) | \mathbf{b} \rangle$, split by the contract decomposition into an ATM level $\sigma_{\text{ATM}}(\ell_F, T)$ and a smile shape $\sigma_{\text{smile}}(m, T)$ with $\sigma_{\text{smile}}(0, T) = 0$. Four properties make the representation worth its parameters. *Linearity*: calibration is (weighted) least squares, closed-form and non-iterative. *Locality*: compactly supported Epanechnikov kernels give regional control of curvature, and non-overlapping supports yield independent contributions. *Dupire in closed form*: the local variance is a rational function of the same coefficients, its denominator encoding the butterfly bound and its numerator the calendar-spread bound. *Routed sensitivities*: vega decomposes per kernel, and—carrying the chain rule through $F(S)$ and keeping vol.slope alive rather than collapsing σ_{ATM} to a constant—the model feeds the total-gamma and shadow-Greek spine. The one structural liability, the affine wing, is disciplined by the flat-wing constraints (??)–(??); the unconstrained OLS is replaced by the vega-weighted constrained regression of Section ??, which enforces the invariants of Definition ??.

2.8 Appendix: Closed-Form Basis Functions

Standardise via $m^* = (m - \mu_i)/h_i$ and define the kernel's cumulative (single-integral) function P_κ , normalised to transition from 0 to 1:

$$P_\kappa(m^*) = \begin{cases} 0, & m^* < -1, \\ \frac{3}{4} \left(m^* - \frac{(m^*)^3}{3} + \frac{2}{3} \right), & -1 \leq m^* \leq 1, \\ 1, & m^* > 1, \end{cases} \quad P_\kappa(\mp 1) = \{0, 1\}. \quad (2.37)$$

Since (??)–(??) integrate from $m = 0$, subtract the origin value $P_\kappa(m_0^*)$ with $m_0^* = -\mu_i/h_i$, and write $\alpha = P_\kappa(m_0^*)$:

$$J_i(m) = \begin{cases} -\alpha, & m < \mu_i - h_i, \\ P_\kappa(m^*) - \alpha, & \mu_i - h_i \leq m \leq \mu_i + h_i, \\ 1 - \alpha, & m > \mu_i + h_i. \end{cases} \quad (2.38)$$

Thus J_i transitions $0 \rightarrow 1$ when the kernel is right of the origin ($\alpha = 0$), $-1 \rightarrow 0$ when left ($\alpha = 1$), and $-\frac{1}{2} \rightarrow +\frac{1}{2}$ when centred ($\alpha = \frac{1}{2}$). The double-integral function is

$$Q_\kappa(m^*) = \begin{cases} 0, & m^* < -1, \\ h_i \left(\frac{3}{16} + \frac{m^*}{2} + \frac{3(m^*)^2}{8} - \frac{(m^*)^4}{16} \right), & -1 \leq m^* \leq 1, \\ h_i m^*, & m^* > 1, \end{cases}$$

and the origin-adjusted double integral is

$$I_i(m) = Q_\kappa(m^*) - Q_\kappa(m_0^*) - \alpha m, \quad (2.39)$$

the final $-\alpha m$ absorbing the linear offset in J_i . Boundary checks confirm $Q_\kappa(-1) = 0$ and $Q_\kappa(+1) = h_i$; for a kernel left of the origin ($\alpha = 1$, $Q_\kappa(m_0^*) = -\mu_i$) one finds $I_i(m) = 0$ for $m > \mu_i + h_i$ (all mass traversed, the contribution absorbed into slope) and $I_i(m) = \mu_i - m$ as $m \rightarrow -\infty$ (affine, consistent with the linear wing).

Proposition 2.29 (Derivatives and regularity). *On the interior of each region, $dI_i/dm = J_i(m)$ and $dJ_i/dm = h_i^{-1}\kappa((m - \mu_i)/h_i)$, with the one-sided derivatives agreeing at $\mu_i \pm h_i$. Hence $I_i \in C^2$ and $J_i \in C^1$ globally, and the reconstructed smile $\sigma_{\text{smile}}(m, T)$ is globally C^2 ; the extra degree of smoothness is exactly the payoff of $\kappa(\pm 1) = 0$.*

Proof. In the support region, using $dm^*/dm = 1/h_i$,

$$\frac{dI_i}{dm} = h_i \left(\frac{1}{2} + \frac{3m^*}{4} - \frac{(m^*)^3}{4} \right) \frac{1}{h_i} - \alpha = P_\kappa(m^*) - \alpha = J_i(m), \quad \frac{dJ_i}{dm} = \frac{3}{4}(1 - (m^*)^2) \frac{1}{h_i} = \frac{\kappa(m^*)}{h_i}.$$

In the exterior both are affine, matching the vanishing kernel; the boundary checks give one-sided agreement, so $c(m) = \partial_m^2 \sigma_{\text{smile}}$ is continuous (with kinks in c' , hence jumps only in the third derivative). For Dupire's formula, which needs $\partial_m^2 \sigma$, the piecewise evaluation is well defined everywhere. $\square$

2.9 Appendix: Verification of the Black-Scholes Cross-Derivatives

Under $r = 0$ and, at the collapse points below, the standing $F = S$, the call is $C = F N(d_1) - K N(d_2)$, and $F \varphi(d_1) = K \varphi(d_2)$ (exact in the forward measure).

Strike gamma. From $\partial C / \partial K|_\sigma = -N(d_2)$ (the vega-weighted d -terms cancel by the density identity),

$$\left. \frac{\partial^2 C}{\partial K^2} \right|_\sigma = -\varphi(d_2) \frac{\partial d_2}{\partial K} = -\varphi(d_2) \left(-\frac{1}{K\sigma\sqrt{T}} \right) = \frac{\varphi(d_2)}{K\sigma\sqrt{T}}.$$

Cross-derivative. Still from $\partial C / \partial K|_\sigma = -N(d_2)$,

$$\frac{\partial^2 C}{\partial K \partial \sigma} = -\varphi(d_2) \frac{\partial d_2}{\partial \sigma} = -\varphi(d_2) \left(-\frac{d_1}{\sigma} \right) = \frac{\varphi(d_2) d_1}{\sigma}.$$

In the assembled formula (??) this multiplies $\partial \sigma / \partial K = (1/K) \partial \sigma / \partial m$, so the factor $1/K$ enters through the chain rule, not the cross-derivative.

Volga. From vega $= F\sqrt{T} \varphi(d_1)$ and $\partial d_1 / \partial \sigma = -d_2 / \sigma$,

$$\frac{\partial^2 C}{\partial \sigma^2} = F\sqrt{T} \varphi(d_1) (-d_1) \frac{\partial d_1}{\partial \sigma} = F\sqrt{T} \varphi(d_1) \frac{d_1 d_2}{\sigma}.$$

Vanna. This is the one cross-derivative entering the total gamma (??). Under $r = 0$ (and $F = S$, flagged), from $\Delta = \partial C / \partial S = N(d_1)$ and $\partial d_1 / \partial \sigma = -d_2 / \sigma$,

$$\text{Vanna} = \frac{\partial^2 C}{\partial S \partial \sigma} = \varphi(d_1) \frac{\partial d_1}{\partial \sigma} = -\frac{\varphi(d_1) d_2}{\sigma},$$

equal by the commuting of mixed partials ($C \in C^2$) to $\partial^2 C / \partial \sigma \partial S$.

The denominator cancellation. The K -to- m transition uses $K \partial_K \sigma = \partial_m \sigma$ and $K^2 \partial_K^2 \sigma = \partial_m^2 \sigma - \partial_m \sigma$; the $-\sigma T \partial_m \sigma$ from the curvature substitution cancels the $+\sigma T \partial_m \sigma$ from $2d_1 \sqrt{T} = -2m/\sigma + \sigma T$, yielding the clean $D(m, T)$ of (??).

Chapter 3

Routed Sensitivities I: Total Greeks and Profit-and-Loss

A book is long or short three things through a single vanilla position: spot, volatility, and the *coupling* between them that the surface enforces. The position's profit and loss over a day is one number; the desk's job is to say, before the day, what that number will be for a given move, and, after the day, why it was what it was. Both tasks need the same two ingredients, and this chapter builds each and then composes them. The first ingredient is the correct sensitivity: when the spot moves, the forward moves, the surface coordinates move, and the implied volatility of every fixed strike moves with them — so the derivative that predicts P&L is a *routed* total, not a frozen partial. The second ingredient is the correct bookkeeping: an exact algebraic identity that splits a price difference between two risk states into first-, second-, and third-order contributions with nothing left over. The routed calculus supplies the *content* of the coefficients; the identity supplies the *form* of the decomposition. Their composition is the attribution a risk system should print every night.

3.1 The Routed-Derivative Framework

A Greek is a derivative of a value function. What decides every derivation in this chapter — and every derivation in the multi-name theory of Volume II — is the path along which the derivative is taken. The classical Greeks are partial derivatives: they bump one argument of the value function while holding every other argument fixed. But the arguments of the value function are not independent in the market. When the spot moves, the forward moves with it; when the forward moves, the implied volatility read off the surface moves, because both the strike log-moneyness $m = \ln(K/F)$ and the forward log-level $\ell_F = \ln(F/F_{\text{ref}})$ move; and, one link deeper, when the constituent volatilities of a basket move, a quoting convention may force the implied correlation to move as well. A sensitivity that is to predict profit and loss must differentiate *along these routes*, not across them.

This section states the routing calculus once, at the level of elementary multivariable analysis: gradients, Hessians, and the chain rule, applied to a finite chain of intermediate variables. Everything is proved. The two specializations of the two volumes — the spot-volatility route $S \rightarrow F \rightarrow \sigma \rightarrow V$ of this chapter, and the deeper route $S \rightarrow F \rightarrow \sigma \rightarrow \rho \rightarrow V$ carrying the shadow Greeks of Volume II, Chapter 2 — are instances of a single operator. The first is derived in Section ??; the insertion point of the second is marked explicitly in Section ??.

3.1.1 Standing assumptions and derivative notation

Assumption 3.1 (Standing assumptions). Throughout this chapter:

1. *Rates and dividends.* Interest rates and dividend yields are zero. Consequently the forward map is the identity, $F(S) = S$. Nevertheless, every derivation below carries a general forward map $F: \mathbb{R}_+ \rightarrow \mathbb{R}_+$, assumed of class C^2 and strictly increasing, and invokes the collapse $F = S$ only at explicitly flagged points. The routed structure — not the special case — is the object of the theory.
2. *Surface decomposition.* The implied volatility of the option with strike K and expiry T is

$$\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T), \quad \sigma_{\text{smile}}(0, T) = 0, \quad (3.1)$$

with $m = \ln(K/F)$ and $\ell_F = \ln(F/F_{\text{ref}})$, where F_{ref} is the fixed reference forward anchoring the surface. The normalisation $\sigma_{\text{smile}}(0, T) = 0$ makes the decomposition unique and makes σ_{ATM} the at-the-money volatility: $\sigma(0, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T)$ exactly.

3. *Regularity.* The value function and every link map introduced below are of class C^2 on the open sets where they are used. In particular $\sigma_{\text{ATM}}(\cdot, T)$ and $\sigma_{\text{smile}}(\cdot, T)$ are twice continuously differentiable in their first arguments.
4. *Fixed expiry.* T is fixed throughout; all sensitivities are instantaneous sensitivities at fixed T . Time decay is treated separately and does not interact with the routing calculus.

The derivative notation is normative and carries the entire framework.

Definition 3.2 (Derivative notation). Let f be a function of several declared arguments.

1. $\frac{\partial f}{\partial x}$ denotes the *partial* derivative of f with respect to its argument x , all other declared arguments held fixed.
2. $\frac{df}{dx}$ denotes the *total* (routed) derivative of f with respect to x : the derivative of the composite function obtained by substituting into f every dependence of its arguments on x along the declared route.
3. When the variables held fixed are determined not by an argument list but by a constraint or a quoting convention, the held-fixed quantity is named by a subscripted bar: $\left. \frac{dp}{d\sigma} \right|_{\xi}$ is the derivative of p with respect to σ under the constraint that ξ is invariant.

For a function $V: \mathbb{R}^q \rightarrow \mathbb{R}$ of class C^2 , $\nabla V \in \mathbb{R}^q$ denotes its gradient (a column vector) and $\nabla^2 V \in \mathbb{R}^{q \times q}$ its Hessian; inner products are written $x^\top y$. By Schwarz's theorem, $\nabla^2 V$ is symmetric.

3.1.2 Routes and routed derivatives

Definition 3.3 (Route). Let $I \subset \mathbb{R}$ be an open interval and let $s \in I$ be the *driver*. A *route of depth* p over I is an ordered list of *link maps*

$$u_1: I \rightarrow \mathbb{R}, \quad u_j: I \times \mathbb{R}^{j-1} \rightarrow \mathbb{R} \quad (j = 2, \dots, p),$$

each of class C^2 , together with a *value function* $V: I \times \mathbb{R}^p \rightarrow \mathbb{R}$ of class C^2 . The j -th *intermediate* depends on the driver and on the intermediates preceding it, and on nothing else; this triangular ordering is what makes the route a route.

Definition 3.4 (Routed evaluation). The *routed intermediates* $\hat{u}_j: I \rightarrow \mathbb{R}$ are defined recursively by

$$\hat{u}_1(s) = u_1(s), \quad \hat{u}_j(s) = u_j(s, \hat{u}_1(s), \dots, \hat{u}_{j-1}(s)) \quad (j = 2, \dots, p).$$

The *augmented state* is the vector $w(s) = (s, \hat{u}_1(s), \dots, \hat{u}_p(s))^\top \in \mathbb{R}^{p+1}$, and the *routed value* is $\hat{V}(s) = V(w(s))$. The *routed derivatives* of V along the route are $\frac{d\hat{V}}{ds}$ and $\frac{d^2\hat{V}}{ds^2}$; when the route is clear from context they are written $\frac{dV}{ds}$ and $\frac{d^2V}{ds^2}$.

Every Greek in these volumes is a routed derivative for some route. The classical (partial) Greeks are the degenerate case in which every link map is constant in the driver, so that the route transmits nothing.

3.1.3 The routed differentiation operator

Theorem 3.5 (Routed chain rule). *Let a route of depth p with value function V be given, and let all maps be of class C^2 as in Definition ?? . Then:*

(i) *Each routed intermediate $\hat{u}_j$ is of class C^2 on I , and for all $s \in I$,*

$$\frac{d\hat{u}_j}{ds} = \frac{\partial u_j}{\partial s} + \sum_{i=1}^{j-1} \frac{\partial u_j}{\partial u_i} \frac{d\hat{u}_i}{ds}, \quad (3.2)$$

all partial derivatives of u_j being evaluated at $(s, \hat{u}_1(s), \dots, \hat{u}_{j-1}(s))$.

(ii) *The routed value $\hat{V}$ is of class C^2 on I , and*

$$\frac{d\hat{V}}{ds} = \nabla V(w(s))^\top w'(s) = \frac{\partial V}{\partial s} + \sum_{j=1}^p \frac{\partial V}{\partial u_j} \frac{d\hat{u}_j}{ds}. \quad (3.3)$$

(iii) *The second routed derivative is*

$$\frac{d^2\hat{V}}{ds^2} = w'(s)^\top \nabla^2 V(w(s)) w'(s) + \nabla V(w(s))^\top w''(s), \quad (3.4)$$

where the components of $w''(s)$ are 0 (the driver component is s itself) and the second routed derivatives $\frac{d^2\hat{u}_j}{ds^2}$, each of which is obtained by applying formula (??) to u_j in the role of the value function over the sub-route of depth $j - 1$.

Proof. (i). By induction on j . For $j = 1$, $\hat{u}_1 = u_1$ is C^2 by hypothesis and (??) is vacuous beyond $\frac{d\hat{u}_1}{ds} = \frac{\partial u_1}{\partial s}$. Suppose $\hat{u}_1, \dots, \hat{u}_{j-1}$ are C^2 . Then $s \mapsto (s, \hat{u}_1(s), \dots, \hat{u}_{j-1}(s))$ is a C^2 map from I into $I \times \mathbb{R}^{j-1}$, and $\hat{u}_j$ is its composition with the C^2 map u_j , hence C^2 (compositions of C^2 maps are C^2). The chain rule for a C^1 function of several variables composed with a C^1 curve gives exactly (??).

(ii). By (i), w is a C^2 curve in $\mathbb{R}^{p+1}$, so $\hat{V} = V \circ w$ is C^2 , and the same chain rule gives $\frac{d\hat{V}}{ds} = \nabla V(w(s))^\top w'(s)$. Writing out the components of ∇V and of $w'(s) = (1, \frac{d\hat{u}_1}{ds}, \dots, \frac{d\hat{u}_p}{ds})^\top$ yields the displayed sum.

(iii). Index the arguments of V by $a = 0, 1, \dots, p$, with $a = 0$ the driver slot, and write $\partial_a V$ for the corresponding partial derivative. Differentiating the component form of (??),

$$\frac{d^2\hat{V}}{ds^2} = \sum_{a=0}^p \frac{d}{ds} \left[\partial_a V(w(s)) \right] w'_a(s) + \sum_{a=0}^p \partial_a V(w(s)) w''_a(s).$$

Each $\partial_a V$ is C^1 because V is C^2 , so the chain rule applies to it along the curve w : $\frac{d}{ds} [\partial_a V(w(s))] = \sum_{b=0}^p \partial_b \partial_a V(w(s)) w'_b(s)$. Substituting,

$$\frac{d^2 \hat{V}}{ds^2} = \sum_{a,b=0}^p \partial_b \partial_a V(w(s)) w'_a(s) w'_b(s) + \sum_{a=0}^p \partial_a V(w(s)) w''_a(s),$$

which is (??) in matrix form. The component $w_0(s) = s$ has $w''_0 = 0$; the remaining components of w'' are the $\frac{d^2 \hat{u}_j}{ds^2}$. Finally, $\hat{u}_j$ is itself a routed value: it is the composition of u_j with the augmented state of the sub-route $(u_1, \dots, u_{j-1})$, so formula (??) applied to u_j over that sub-route computes $\frac{d^2 \hat{u}_j}{ds^2}$. The recursion terminates because each application strictly decreases the depth. $\square$

Remark 3.6 (One operator, applied recursively). Part (iii) is the structural heart of the framework: the second routed derivative of the value requires the second routed derivatives of the intermediates, and these are computed *by the same formula* one level down. Nothing new is needed when a route is extended by a link — one more application of Theorem ?? to the appended link map, and formulas (??)–(??) absorb it. This is precisely how the shadow-Greek route of Volume II plugs in (Section ??).

Corollary 3.7 (Depth-one routed derivatives). *Let $V = V(s, u)$ be C^2 and let $u = u(s)$ be C^2 . Then*

$$\frac{dV}{ds} = \frac{\partial V}{\partial s} + \frac{\partial V}{\partial u} \frac{du}{ds}, \quad (3.5)$$

$$\frac{d^2 V}{ds^2} = \frac{\partial^2 V}{\partial s^2} + 2 \frac{\partial^2 V}{\partial s \partial u} \frac{du}{ds} + \frac{\partial^2 V}{\partial u^2} \left(\frac{du}{ds} \right)^2 + \frac{\partial V}{\partial u} \frac{d^2 u}{ds^2}. \quad (3.6)$$

Proof. Take $p = 1$ in Theorem ?. Here $w(s) = (s, u(s))^T$, $w'(s) = (1, u'(s))^T$, $w''(s) = (0, u''(s))^T$. Formula (??) gives (??). Expanding $w^T (\nabla^2 V) w' = \frac{\partial^2 V}{\partial s^2} + (\frac{\partial^2 V}{\partial s \partial u} + \frac{\partial^2 V}{\partial u \partial s}) u' + \frac{\partial^2 V}{\partial u^2} (u')^2$ and invoking Schwarz's theorem to merge the two cross terms, formula (??) gives (??). $\square$

Remark 3.8 (What a specialization must supply). To instantiate the operator, a specialization supplies exactly three items: (a) the driver s ; (b) the ordered intermediates $u_1, \dots, u_p$ with the partial derivatives of each link map; (c) the partial derivatives of the value function in its declared arguments. Theorem ?? then *outputs* the total Greeks. In this chapter the list (b) is (F, σ) ; in Volume II, Chapter 2 it is (F_i, σ_i, ρ) — same driver role, same operator, one more link.

3.2 First Specialization: the Spot–Volatility Route

3.2.1 The route

Fix an option with strike K and expiry T . The driver is the spot S . The intermediates, in route order, are the forward $F = F(S)$ and the implied volatility σ , which depends on S only through F , via the two moneyness coordinates of the surface (??):

$$m = \ln(K/F), \quad \ell_F = \ln(F/F_{\text{ref}}), \quad \sigma = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T).$$

The value function is $V = V(S, \sigma)$, the option value as a function of spot and volatility. (Under Assumption ??(1) the direct spot argument of V and the forward coincide numerically — under zero rates, $F = S$ — but the two roles stay separate in the route so that the F -links remain visible;

they are where nonzero carry enters.) The partial derivatives of V in its declared arguments are the classical Greeks of the notation contract:

$$\Delta = \frac{\partial V}{\partial S}, \quad \Gamma = \frac{\partial^2 V}{\partial S^2}, \quad \nu = \frac{\partial V}{\partial \sigma}, \quad \text{Vanna} = \frac{\partial^2 V}{\partial S \partial \sigma}, \quad \text{Volga} = \frac{\partial^2 V}{\partial \sigma^2}.$$

Figure ?? displays the route, together with the dashed correlation link that Volume II appends.

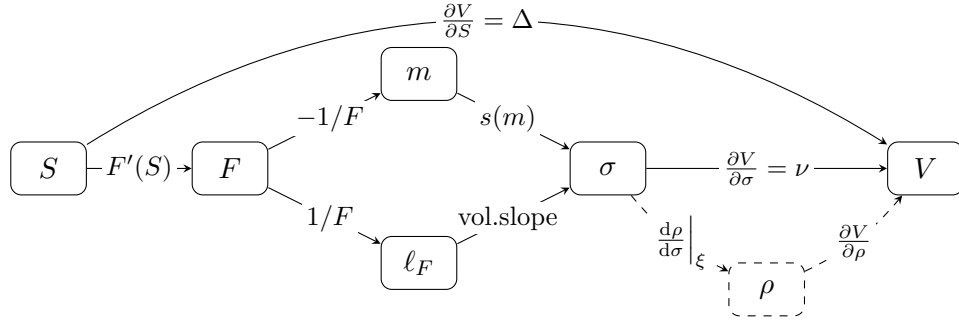


Figure 3.1: The routed dependency structure. Solid arrows: the spot-volatility route of this chapter, each edge labelled by the derivative it transmits. Dashed arrows: the correlation link appended by the sticky-CVC convention — the specialization hook of Section ??, evaluated in Volume II, Chapter 2. Under sticky- ρ the dashed link transmits nothing.

3.2.2 Moneyness derivatives through the forward

Lemma 3.9 (Moneyness links). *At fixed K , F_{ref} , and T , the moneyness coordinates satisfy, for all $F > 0$,*

$$\frac{\partial m}{\partial F} = -\frac{1}{F}, \quad \frac{\partial \ell_F}{\partial F} = \frac{1}{F}, \quad \frac{\partial^2 m}{\partial F^2} = \frac{1}{F^2}, \quad \frac{\partial^2 \ell_F}{\partial F^2} = -\frac{1}{F^2}. \quad (3.7)$$

Proof. $m = \ln K - \ln F$ gives $\frac{\partial m}{\partial F} = -1/F$ and, differentiating again, $\frac{\partial^2 m}{\partial F^2} = 1/F^2$. $\ell_F = \ln F - \ln F_{\text{ref}}$ gives $\frac{\partial \ell_F}{\partial F} = 1/F$ and $\frac{\partial^2 \ell_F}{\partial F^2} = -1/F^2$. $\square$

The two coordinates respond to the forward with *opposite signs*: when the forward rises, the option's strike moneyness m falls while the surface level coordinate ℓ_F rises. Every sign asymmetry in the formulas below descends from this single fact.

Per the contract,

$$s(m) = \frac{\partial \sigma_{\text{smile}}}{\partial m}, \quad c(m) = \frac{\partial^2 \sigma_{\text{smile}}}{\partial m^2}, \quad \text{vol.slope} = \frac{\partial \sigma_{\text{ATM}}}{\partial \ell_F},$$

the smile skew, the smile curvature, and the ATM slope, the last evaluated at the current (ℓ_F, T) ; for equity underlyings $\text{vol.slope} < 0$, and the distinction between vol.slope and $s(m)$ is that of Remark ?. The route transmits the two through different links with different signs, and conflating them destroys the delta and gamma adjustments derived below.

3.2.3 Routed volatility derivatives

Proposition 3.10 (Volatility through the forward). *Under Assumption ??, the implied volatility of the fixed option, viewed as a function of the forward through (??), satisfies*

$$\frac{d\sigma}{dF} = \frac{1}{F} [\text{vol.slope} - s(m)], \quad (3.8)$$

$$\frac{d^2\sigma}{dF^2} = \frac{1}{F^2} \left[\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} + c(m) + s(m) - \text{vol.slope} \right]. \quad (3.9)$$

Through a general C^2 forward map $F(S)$, the routed derivatives with respect to the spot are

$$\frac{d\sigma}{dS} = \frac{d\sigma}{dF} F'(S), \quad \frac{d^2\sigma}{dS^2} = \frac{d^2\sigma}{dF^2} (F'(S))^2 + \frac{d\sigma}{dF} F''(S). \quad (3.10)$$

Under zero rates, $F = S$, so $F'(S) = 1$ and $F''(S) = 0$, and

$$\frac{d\sigma}{dS} = \frac{\text{vol.slope} - s(m)}{S}, \quad \frac{d^2\sigma}{dS^2} = \frac{1}{S^2} \left[\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} + c(m) + s(m) - \text{vol.slope} \right]. \quad (3.11)$$

Proof. The map $F \mapsto \sigma$ is the sum of the two compositions $F \mapsto \ell_F \mapsto \sigma_{\text{ATM}}(\ell_F, T)$ and $F \mapsto m \mapsto \sigma_{\text{smile}}(m, T)$. By the chain rule and Lemma ??,

$$\frac{d\sigma}{dF} = \text{vol.slope} \cdot \frac{1}{F} + s(m) \cdot \left(-\frac{1}{F} \right) = \frac{\text{vol.slope} - s(m)}{F},$$

which is (??). Differentiate once more in F by the product rule applied to $F^{-1}[\text{vol.slope} - s(m)]$:

$$\frac{d^2\sigma}{dF^2} = -\frac{1}{F^2} [\text{vol.slope} - s(m)] + \frac{1}{F} \left[\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} \cdot \frac{1}{F} - c(m) \cdot \left(-\frac{1}{F} \right) \right],$$

where the bracketed derivatives again use Lemma ?? through the chain rule. Collecting over F^{-2} gives (??). Since σ has no direct dependence on S outside of F , the composite $S \mapsto \sigma(F(S))$ obeys the univariate chain and product rules, yielding (??); the zero-rate collapse $F = S$ (flagged here per Assumption ??(1)) reduces it to (??). $\square$

Remark 3.11 (Sign asymmetry). The skew $s(m)$ enters the first derivative (??) with a minus sign and the second derivative (??) with a plus sign, while vol.slope does the opposite. Both asymmetries are forced by Lemma ??: the coordinate m carries $\frac{\partial m}{\partial F} = -1/F$ but $\frac{\partial^2 m}{\partial F^2} = +1/F^2$, and ℓ_F carries the mirror-image pair. For an equity surface with $\text{vol.slope} < 0$ and $s(m) < 0$ near the money, the two routes *partially offset* in $\frac{d\sigma}{dS}$ and *reinforce* in the curvature bracket of $\frac{d^2\sigma}{dS^2}$ — a structural fact with direct hedging consequences, quantified in Example ??.

3.2.4 Total delta and total gamma

Theorem 3.12 (Total Greeks along the spot–volatility route). *Under Assumption ??, the routed derivatives of the value $V = V(S, \sigma)$ along the route of Section ?? are*

$$\frac{dV}{dS} = \Delta + \nu \frac{d\sigma}{dS}, \quad (3.12)$$

$$\Gamma_{\text{adj}} := \frac{d^2V}{dS^2} = \Gamma + 2 \text{Vanna} \frac{d\sigma}{dS} + \text{Volga} \left(\frac{d\sigma}{dS} \right)^2 + \nu \frac{d^2\sigma}{dS^2}, \quad (3.13)$$

with $\frac{d\sigma}{dS}$ and $\frac{d^2\sigma}{dS^2}$ given by Proposition ?? . Fully expanded — under zero rates, $F = S$ — these read

$$\frac{dV}{dS} = \Delta + \frac{\nu}{S} [\text{vol.slope} - s(m)], \quad (3.14)$$

$$\begin{aligned} \Gamma_{\text{adj}} = \Gamma &+ \frac{2 \text{Vanna}}{S} [\text{vol.slope} - s(m)] + \frac{\text{Volga}}{S^2} [\text{vol.slope} - s(m)]^2 \\ &+ \frac{\nu}{S^2} \left[\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} + c(m) + s(m) - \text{vol.slope} \right]. \end{aligned} \quad (3.15)$$

Proof. Apply Corollary ?? with driver $s = S$ and single effective intermediate $u = \sigma$ (the forward link is already absorbed into $\frac{d\sigma}{dS}$ and $\frac{d^2\sigma}{dS^2}$ by Proposition ??, and V has no direct F argument). Identifying the partial derivatives of V with the classical Greeks — $\frac{\partial V}{\partial S} = \Delta$, $\frac{\partial V}{\partial \sigma} = \nu$, $\frac{\partial^2 V}{\partial S^2} = \Gamma$, $\frac{\partial^2 V}{\partial S \partial \sigma} = \text{Vanna}$, $\frac{\partial^2 V}{\partial \sigma^2} = \text{Volga}$ — formula (??) gives (??) and formula (??) gives (??); the factor 2 on Vanna is the Schwarz merge of the two cross-pathways (S -then- σ and σ -then- S). Substituting (??) — the zero-rate collapse $F = S$, flagged — yields the expanded forms. $\square$

Each term of (??) has one route reading:

Term	Route pathway	Content
Γ	$S \rightarrow V$ twice	direct spot convexity
$2 \text{Vanna} \frac{d\sigma}{dS}$	$S \rightarrow V, S \rightarrow F \rightarrow \sigma \rightarrow V$ (both orders)	spot-vol cross-coupling
$\text{Volga} \left(\frac{d\sigma}{dS}\right)^2$	$S \rightarrow F \rightarrow \sigma \rightarrow V$ twice	vol convexity through the surface slope
$\nu \frac{d^2\sigma}{dS^2}$	curvature of the link itself	vega weighted by routed surface curvature

Remark 3.13 (Consistency of the two viewpoints). The routed derivatives can equivalently be computed by keeping (S, σ) as a two-dimensional state and contracting the full gradient and Hessian of V against the augmented velocity $w'(S) = (1, \frac{d\sigma}{dS})^\top$ and acceleration $w''(S) = (0, \frac{d^2\sigma}{dS^2})^\top$, as in Theorem ??(iii). The reduced formulas (??)–(??) and the matrix contraction agree identically — this is a theorem, not a check — but computing both in an implementation and verifying agreement is a sharp diagnostic for surface-model consistency.

3.2.5 Worked example

Example 3.14 (At-the-money call on an equity surface). Fix $T = 1$, $K = 100$, $F_{\text{ref}} = 100$, and a local surface model

$$\sigma_{\text{ATM}}(\ell_F, T) = 0.20 - 0.04 \ell_F, \quad \sigma_{\text{smile}}(m, T) = -0.10 m + 0.05 m^2, \quad (3.16)$$

so that $\text{vol.slope} = -0.04$, $\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} = 0$, $s(0) = -0.10$, $c(0) = 0.10$, and the normalisation $\sigma_{\text{smile}}(0, T) = 0$ holds. Both slopes are negative, the equity-index configuration of Remark ?? . Evaluate at spot $S = 100$; under zero rates, $F = S = 100$, hence $\ell_F = 0$, $m = 0$, and $\sigma = \sigma_{\text{ATM}}(0, 1) = 0.20$.

Classical Greeks. With $d_1 = -m/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2 = 0.1$ and $d_2 = d_1 - \sigma\sqrt{T} = -0.1$, the Black–Scholes call values are (six significant figures, derived analytically from the closed forms

$\Delta = N(d_1)$, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{T})$, $\nu = S\varphi(d_1)\sqrt{T}$, $\text{Vanna} = -\varphi(d_1)d_2/\sigma$, $\text{Volga} = \nu d_1 d_2/\sigma$, with $\varphi(0.1) = 0.396953$:

$$\Delta = 0.539828, \quad \Gamma = 0.0198476, \quad \nu = 39.6953, \quad \text{Vanna} = 0.198476, \quad \text{Volga} = -1.98476.$$

Routed volatility derivatives. By Proposition ??, under zero rates ($F = S$):

$$\frac{d\sigma}{dS} = \frac{-0.04 - (-0.10)}{100} = 6.0 \times 10^{-4}, \quad \frac{d^2\sigma}{dS^2} = \frac{0 + 0.10 + (-0.10) - (-0.04)}{100^2} = 4.0 \times 10^{-6}.$$

The offset of Remark ?? is visible: the smile route contributes $-s(0)/S = +10^{-3}$ to $\frac{d\sigma}{dS}$ while the ATM route contributes $\text{vol.slope}/S = -4 \times 10^{-4}$, cancelling forty per cent of it; in the curvature bracket, $s(0)$ and $c(0)$ cancel exactly and the surviving term is $-\text{vol.slope} > 0$.

Total Greeks. By Theorem ??:

$$\frac{dV}{dS} = 0.539828 + 39.6953 \times 6.0 \times 10^{-4} = 0.539828 + 0.023817 = 0.563645,$$

$$\begin{aligned} \Gamma_{\text{adj}} &= 0.0198476 + 2(0.198476)(6.0 \times 10^{-4}) + (-1.98476)(6.0 \times 10^{-4})^2 + 39.6953 \times 4.0 \times 10^{-6} \\ &= 0.0198476 + 0.000238172 - 0.000000715 + 0.000158781 = 0.0202439. \end{aligned}$$

The delta adjustment, $+0.023817$, is 4.4% of the classical delta — a material hedging correction. The gamma adjustment, $+0.000396$, is +2.0% of the classical gamma, and its dominant contributors are the vanna cross-term and the vega-curvature term, in the ratio 3 : 2; the volga term is three orders of magnitude smaller at this slope.

Flagged comparison. Computed under the $\sigma_{\text{ATM}} = \text{const}$ convention (C4) — *not* the surface model of this chapter — $\text{vol.slope} = 0$ and $\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} = 0$, giving $\frac{d\sigma}{dS} = 10^{-3}$, $\frac{d^2\sigma}{dS^2} = 0$, total delta 0.579523, and $\Gamma_{\text{adj}} = 0.0202426$. The total deltas differ by $0.015878 = \nu \cdot \text{vol.slope}/S$ — exactly the ATM-route contribution that the constant-ATM convention deletes. That the two Γ_{adj} values nearly coincide here is a numerical accident of the parameter choice ($c(0) = -s(0)$, which zeroes the smile part of the curvature bracket), not a structural fact; the delta discrepancy is the honest measure of what the convention discards.

3.3 The Specialization Hook: Convention-Completed Routes

The route of Section ?? is closed: every intermediate is a function of the driver by construction. A multi-name book is not closed in this way. Its value depends on a parameter — the implied correlation ρ — that no market map determines from the spot. What determines it is a *quoting convention*: a declaration of which quantity the market holds invariant as volatilities move. A convention closes the system, and a closed system is a route; the operator of Theorem ?? then applies without modification.

Definition 3.15 (Convention-completed route). Let a route of depth p with intermediates $u_1, \dots, u_p$ be given, and let θ be a parameter of the value function $V = V(s, u_1, \dots, u_p, \theta)$ that is not a declared function of the driver. A *quoting convention* for θ is a choice of a C^2 function g and a declaration that the quantity $g(u_1, \dots, u_p, \theta)$ is invariant along the route; when $\frac{\partial g}{\partial \theta} \neq 0$, the implicit function theorem determines $\theta = \theta(u_1, \dots, u_p)$ locally as a C^2 function of the intermediates. The *convention-completed route* is the route of depth $p+1$ obtained by appending $u_{p+1} = \theta$ with this link map. All routed derivatives of V are then computed by Theorem ?? on the completed route.

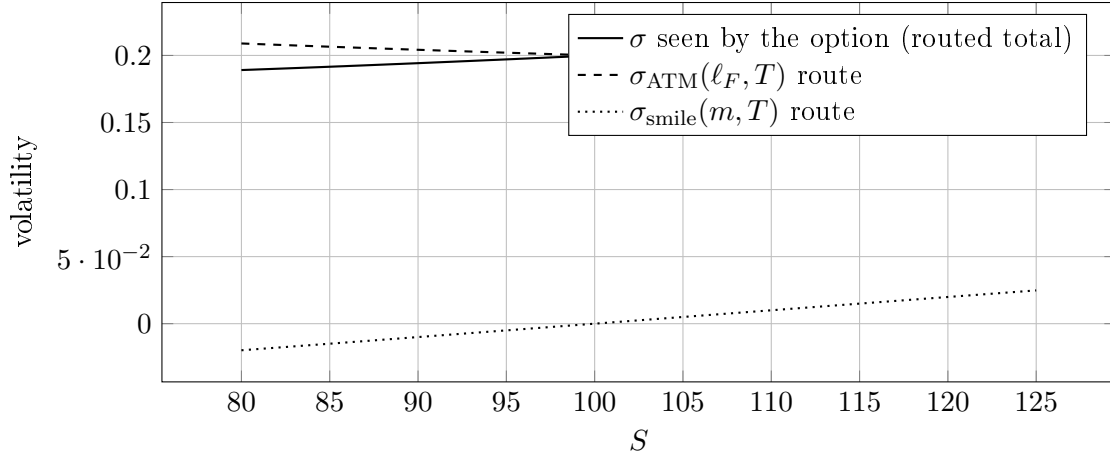


Figure 3.2: The two routes of Example ?? for the fixed-strike option $K = 100$, as functions of spot; under zero rates, $F = S$, so $m = \ln(100/S)$ and $\ell_F = \ln(S/100)$ here. The routed volatility (solid) is the sum of the ATM route (dashed) and the smile route (dotted). The slopes of the two components at $S = 100$ have the same sign in moneyness terms ($\text{vol.slope} < 0$, $s(0) < 0$) yet contribute with *opposite* signs to $\frac{d\sigma}{dS}$, by Lemma ??.

Remark 3.16 (Specialization hook: the correlation link). This is the insertion point of the shadow-Greek theory. In the two-name setting of Volume II, the constituent volatilities are the single-name at-the-money volatilities $\sigma_i \equiv \sigma_{\text{ATM},i}(\ell_{F,i}, T)$, $i = 1, 2$, where $\ell_{F,i}$ and $F_{\text{ref},i}$ extend the ratified ℓ_F and F_{ref} by a name index; each σ_i is routed from its spot through the σ_{ATM} route of Proposition ?? *alone*, with link derivative $\frac{d\sigma_i}{dS_i} = (\text{vol.slope}_i/F_i) F'_i(S_i)$ and per-name slope $\text{vol.slope}_i = \frac{\partial \sigma_{\text{ATM},i}}{\partial \ell_{F,i}}$ — the smile coordinate transmits nothing here, because σ_i is the floating at-the-money point, not a fixed-strike volatility — and forward collapse $F_i = S_i$ under zero rates, flagged as always. Two conventions close the correlation:

- sticky- ρ : the invariant is ρ itself ($d\rho = 0$). The appended link is inert, $\frac{d\rho}{dS} = 0$, and the completed route collapses to the depth-two route of this chapter, name by name.
- sticky-CVC: the invariant is the basket vol spread $\xi = \bar{\sigma} - \sigma_B$ ($d\xi = 0$). By Definition ??, this determines $\rho = \rho(\sigma_1, \sigma_2, \xi)$ and appends a live link whose routed derivative is

$$\frac{d\rho}{dS} = \sum_{i=1}^2 \frac{\partial \rho}{\partial \sigma_i} \Big|_{\xi} \frac{d\sigma_i}{dS}. \quad (3.17)$$

The difference between the two completed routes, applied to the first-order formula (??), is a single term:

$$\frac{dV}{dS} \Big|_{\text{sticky-CVC}} - \frac{dV}{dS} \Big|_{\text{sticky-}\rho} = \frac{\partial V}{\partial \rho} \frac{d\rho}{dS}, \quad (3.18)$$

and the *shadow Greeks are by definition this difference*, expressed on the contract's normalised bases (the ShadowDelta form below is the symmetric parallel move under zero rates, $F_i = S_i$, flagged; Volume II, Chapter 2 states the general per-name form):

$$\text{ShadowVega} = \frac{1}{100} \frac{\partial V}{\partial \rho} \frac{d\rho}{d\sigma} \Big|_{\xi}, \quad \text{ShadowDelta} = 100 \text{ vol.slope} \text{ ShadowVega}. \quad (3.19)$$

They vanish identically under sticky- ρ . Volume II, Chapter 2 owns the evaluation of the link derivatives $\left. \frac{\partial \rho}{\partial \sigma_i} \right|_{\xi}$ from the basket variance identity, the second-order (shadow-gamma) terms generated by Theorem ??(iii) on the completed route, and the Cega normalisation; the present section fixes, once and for all, the operator those computations instantiate and the interface (Remark ??) they must supply. Nothing else about the framework changes when the link is appended: this is Remark ?? in action.

Remark 3.17 (Why the hook carries the forward). The per-name routed derivatives $\frac{d\sigma_i}{dS}$ entering (??) contain the factors $F'_i(S_i)/F_i$ of Proposition ??. Under the standing zero-rate assumption these collapse to $1/S_i$ — under zero rates, $F_i = S_i$, flagged — but the shadow-Greek chain $S \rightarrow \sigma \rightarrow \rho \rightarrow V$ is stated, and must be implemented, through $F(S)$: the correlation link composes with the forward links, it does not bypass them.

3.4 The Cubic Profit-and-Loss Identity

The preceding sections settled which derivative is correct. This section settles how the price difference between two successive risk states decomposes — exactly, with no error term — into contributions attributable to each order. The answer is an algebraic identity about cubic polynomials, deliberately model-free: nothing in it knows about Black–Scholes, surfaces, or routes. The surface content enters only through the coefficients, in Section ??.

Throughout this section and the next, x denotes the vector of declared risk factors of the book, x_1 and x_2 two successive states (yesterday’s close and today’s, or two intraday marks), and $h = x_2 - x_1$ the move. For a state x ,

$$\Delta_x := \nabla V(x), \quad \Gamma_x := \nabla^2 V(x),$$

the gradient and Hessian of the book value V in the declared risk factors. The subscripted-state symbols Δ_x, Γ_x are scoped to the P&L identity and are distinct from the unsubscripted Black–Scholes partials Δ, Γ of Section ??; which concrete derivatives must populate them — partials or routed totals — is not a matter of taste, and is settled in Section ??. (Notation-contract exemption, recorded here in full: Δ_x and Γ_x are chapter-local aliases for the $\nabla V(x)$ and $\nabla^2 V(x)$ of Definition ??; their components include vegas and, in Volume II, Cega as much as deltas, and a *subscripted-state* Δ or Γ in this chapter never denotes a Black–Scholes partial.)

3.4.1 The univariate identity

Model the book value as a polynomial of degree three in a single risk factor $x \in \mathbb{R}$. Degree three is the lowest order that carries a *varying* gamma: V''' is constant (the desk’s “speed”), $\Gamma_x = V''(x)$ is affine in x , and $\Delta_x = V'(x)$ is quadratic. Within this class every identity below is exact — not a truncation.

Taylor’s theorem at x_1 terminates:

$$V(x_2) - V(x_1) = \Delta_{x_1} h + \frac{1}{2} \Gamma_{x_1} h^2 + \frac{1}{6} V''' h^3. \quad (3.20)$$

The constant third derivative is observable from the endpoint gammas: by the fundamental theorem of calculus,

$$\Gamma_{x_2} - \Gamma_{x_1} = \int_{x_1}^{x_2} V''' dx = V''' h. \quad (3.21)$$

Substituting (??) into (??) eliminates V''' :

$$V(x_2) - V(x_1) = \Delta_{x_1} h + \frac{1}{2} \Gamma_{x_1} h^2 + \frac{1}{6} (\Gamma_{x_2} - \Gamma_{x_1}) h^2. \quad (3.22)$$

Split the frozen gamma symmetrically about the two states, $\Gamma_{x_1} = \frac{1}{2}(\Gamma_{x_1} + \Gamma_{x_2}) - \frac{1}{2}(\Gamma_{x_2} - \Gamma_{x_1})$, and collect the two coefficients of $(\Gamma_{x_2} - \Gamma_{x_1})h^2$ ($-\frac{1}{4} + \frac{1}{6} = -\frac{1}{12}$):

$$V(x_2) - V(x_1) = \Delta_{x_1}h + \frac{1}{4}(\Gamma_{x_1} + \Gamma_{x_2})h^2 - \frac{1}{12}(\Gamma_{x_2} - \Gamma_{x_1})h^2. \quad (3.23)$$

Finally, because Γ_x is affine, the trapezoidal rule integrates it exactly:

$$\Delta_{x_2} - \Delta_{x_1} = \int_{x_1}^{x_2} \Gamma_x dx = \frac{1}{2}(\Gamma_{x_1} + \Gamma_{x_2})h. \quad (3.24)$$

Substituting (??) into (??) absorbs the average gamma into the average delta and yields the identity that carries the rest of the chapter.

Theorem 3.18 (Fundamental P&L identity). *For any polynomial V of degree at most three and any $x_1, x_2 \in \mathbb{R}$, with $h = x_2 - x_1$,*

$$V(x_2) - V(x_1) = \frac{1}{2}(\Delta_{x_1} + \Delta_{x_2})h - \frac{1}{12}(\Gamma_{x_2} - \Gamma_{x_1})h^2. \quad (3.25)$$

The identity is exact; no remainder term exists.

Proof. The chain (??)–(??) above, each step an identity for cubics. $\square$

Read as a desk statement: the P&L is the move times the *average* of the entry and exit deltas, minus one-twelfth of the gamma change times the move squared. Regrouping exposes the three attributable orders:

$$V(x_2) - V(x_1) = \underbrace{\Delta_{x_1}h}_{\text{1st order}} + \underbrace{\frac{\Delta_{x_2} - \Delta_{x_1}}{2}h}_{\text{2nd order}} - \underbrace{\frac{1}{12}(\Gamma_{x_2} - \Gamma_{x_1})h^2}_{\text{3rd order}}. \quad (3.26)$$

The first order is the delta P&L on the entry hedge; the second is the half-of-delta-change term — gamma P&L in its endpoint form; the third is the correction for the fact that gamma itself moved.

Remark 3.19 (Constant-gamma reduction, and where the classical formula breaks). If $\Gamma_{x_1} = \Gamma_{x_2} = \Gamma_x$ (degree at most two), the third-order term vanishes and two familiar forms coincide:

$$V(x_2) - V(x_1) = \frac{\Delta_{x_2} + \Delta_{x_1}}{2}h = \Delta_{x_1}h + \frac{1}{2}\Gamma_x h^2,$$

the trapezoid on delta and the classical delta–gamma expansion. Their equivalence holds *only* at constant gamma. When gamma is affine, the trapezoid rule (??) shows the average-delta form silently replaces Γ_x by the average $\frac{1}{2}(\Gamma_{x_1} + \Gamma_{x_2})$ — and the $-\frac{1}{12}$ correction of (??) must then be carried. Booking the classical formula against endpoint deltas while gamma is drifting is a standing source of spurious unexplained P&L; the identity prices that error exactly.

3.4.2 The multivariate identity

A real book has many risk factors — spots, volatilities by name and bucket, and, from Volume II onward, correlations. Let $x_1, x_2 \in \mathbb{R}^n$ and $h = x_2 - x_1$, and let V be a polynomial of total degree at most three on $\mathbb{R}^n$. Restrict V to the segment between the states:

$$V_h(t) = V(x_1 + th), \quad t \in [0, 1], \quad V_h(0) = V(x_1), \quad V_h(1) = V(x_2)$$

(the letter g stays reserved for the quoting-convention function of Definition ??, so the segment restriction gets its own symbol), with derivatives, by the chain rule,

$$V'_h(t) = \Delta_{x_1+th}^\top h, \quad V''_h(t) = h^\top \Gamma_{x_1+th} h.$$

Since V has degree three, V_h is a univariate cubic in t , and Theorem ?? applied to V_h on $[0, 1]$ (driver t , move 1) translates term by term:

Corollary 3.20 (Multivariate P&L identity). *For any polynomial V of total degree at most three on $\mathbb{R}^n$ and any states x_1, x_2 ,*

$$\boxed{V(x_2) - V(x_1) = \left(\frac{\Delta_{x_1} + \Delta_{x_2}}{2} \right)^\top h - \frac{1}{12} h^\top (\Gamma_{x_2} - \Gamma_{x_1}) h.} \quad (3.27)$$

Here Δ_x is the full vector of first-order sensitivities of the book — deltas, vegas, and in Volume II the correlation sensitivity Cega — and Γ_x the matrix of second-order ones: gammas on the diagonal blocks, vanna and volga off and on the volatility block, cross-gammas between names. By Schwarz's theorem Γ_x is symmetric; an asymmetric Hessian in an implementation is a bug, not a convention (Section ??).

Remark 3.21 (Hessian coverage). The Hessian has $n(n+1)/2$ independent entries. At $n = 50$ risk factors that is 1,275 second-order sensitivities, most of them cross-gammas. Any implementation populates only the material entries; the neglected remainder does not disappear — it lands, move by move, in the unexplained residual, which is why the residual must be monitored rather than assumed small (Section ??).

3.4.3 Degree three as a working boundary

Degree three is the lowest order at which the expansion carries delta, gamma, *and* the leading variation of gamma — the three quantities a desk hedges, rebalances, and disputes. For a real book the value function is not a polynomial, and the identity (??), evaluated with measured endpoint sensitivities, explains the reprice only up to a residual

$$R(x_1, x_2) := [V(x_2) - V(x_1)] - \left(\frac{\Delta_{x_1} + \Delta_{x_2}}{2} \right)^\top h + \frac{1}{12} h^\top (\Gamma_{x_2} - \Gamma_{x_1}) h, \quad (3.28)$$

the quartic-and-above content of the price function over the move. For well-behaved products over realistic moves, R is small — the worked example of Section ?? puts it near one basis point of the P&L for a ten-percent spot move. The doctrine for products whose R is structurally large — what they are, why they cannot be risk-managed inside this bookkeeping, and what a desk should do about them — is developed in Section ??.

3.5 Ex Ante and Ex Post Attribution

The identity (??) needs sensitivities at *both* states. Before the move only x_1 has been risk-run; after it, both have. This asymmetry splits attribution into an ex ante estimate and an ex post decomposition, and their difference is itself an exactly computable diagnostic.

3.5.1 Ex ante: the projected gradient

At x_1 the desk holds Δ_{x_1} and (the computed part of) Γ_{x_1} . To use the average-delta form, project the exit gradient with the available Hessian:

$$\tilde{\Delta}_{x_2} := \Delta_{x_1} + \Gamma_{x_1} h. \quad (3.29)$$

This is the gradient transported to x_2 under a locally-constant-gamma assumption — exactly the delta a trader pre-computes when laddering hedges against a gap scenario. The resulting estimate collapses to the classical delta–gamma form:

$$V(x_2) - V(x_1) \approx \left(\frac{\tilde{\Delta}_{x_2} + \Delta_{x_1}}{2} \right)^\top h = \underbrace{\Delta_{x_1}^\top h}_{\text{1st order}} + \underbrace{\frac{1}{2} h^\top \Gamma_{x_1} h}_{\text{2nd order}}. \quad (3.30)$$

Two qualifications. First, Γ_{x_1} is never complete (Remark ??), so $\tilde{\Delta}_{x_2}$ inherits the coverage gaps of the Hessian. Second, the construction is a control variate, not a dressed-up forecast: the cubic polynomial is the control whose increment is known in closed form, the repriced P&L is the target, and the residual against reprice is the quantity whose smallness must be demonstrated, not assumed. Using ex ante Greeks does not inflate the residual, but the estimate itself misses the reprice by a *deterministic, signed* error — the reconciliation gap $\mathcal{E}$ of Section ??, the truncated third-order term plus the delta projection error — which dominates the cubic residual by two orders of magnitude in the worked example (a factor of 139). The only requirement on the *control* is that it track the true price function over plausible moves, which is exactly the boundary question of Section ??.

3.5.2 Ex post: the exact decomposition

Once x_2 has been risk-run, Δ_{x_2} and Γ_{x_2} are known and the identity applies with measured endpoints:

$$V(x_2) - V(x_1) = \underbrace{\Delta_{x_1}^\top h}_{\text{1st order}} + \underbrace{\left(\frac{\Delta_{x_2} - \Delta_{x_1}}{2} \right)^\top h}_{\text{2nd order}} - \underbrace{\frac{1}{12} h^\top (\Gamma_{x_2} - \Gamma_{x_1}) h}_{\text{3rd order}} + R(x_1, x_2), \quad (3.31)$$

with R the cubic residual (??) — identically zero for a cubic book, small and monitored for a real one. The ex post form buys one thing: the second-order term uses the *measured* delta change, not a Hessian model of it, so Hessian coverage gaps that contaminate (??) do not contaminate the first two orders here; they are squeezed into the third-order term and R .

3.5.3 Reconciliation

Subtracting (??) from the exact part of (??) gives the reconciliation gap

$$\mathcal{E} := \left(\frac{\Delta_{x_2} - \tilde{\Delta}_{x_2}}{2} \right)^\top h - \frac{1}{12} h^\top (\Gamma_{x_2} - \Gamma_{x_1}) h, \quad (3.32)$$

a sum of two separately reportable pieces: the *delta projection error* — how far the Hessian-transported gradient landed from the measured one — and the *third-order correction* absent from any delta–gamma estimate.

Proposition 3.22 (The gap is pure gamma variation). *For a book that is exactly cubic in its risk factors,*

$$\Delta_{x_2} - \tilde{\Delta}_{x_2} = \frac{1}{2} (\Gamma_{x_2} - \Gamma_{x_1}) h, \quad \text{hence} \quad \mathcal{E} = \frac{1}{6} h^\top (\Gamma_{x_2} - \Gamma_{x_1}) h. \quad (3.33)$$

Proof. Along the segment, $t \mapsto \Gamma_{x_1+th}$ is affine, so $\Delta_{x_2} - \Delta_{x_1} = \int_0^1 \Gamma_{x_1+th} h dt = \frac{1}{2}(\Gamma_{x_1} + \Gamma_{x_2})h$; subtracting $\tilde{\Delta}_{x_2} - \Delta_{x_1} = \Gamma_{x_1}h$ gives the first claim, and substituting into (??) collapses $\frac{1}{4} - \frac{1}{12} = \frac{1}{6}$. Sanity check in one dimension: $\mathcal{E} = \frac{1}{6}V'''h^3$ by (??) — exactly the cubic Taylor term that the ex ante estimate (??) truncates. The signs and the limiting case ($\mathcal{E} = 0$ at constant gamma) both close. $\square$

The reconciliation gap therefore depends *entirely* on the variation of gamma over the move — there is no other source. A small $\mathcal{E}$, day after day, certifies that the constant-gamma projection underlying every intraday hedge ladder is adequate at the book's actual gamma turnover. A persistently large $\mathcal{E}$ has exactly two explanations: material Hessian entries are missing from Γ_{x_1} , or the book's gamma is turning over so fast that the position is drifting toward the boundary of Section ?? . Both demand action, and the decomposition (??) says which.

3.6 The Composition: Total Greeks inside the Identity

The identity of Section ?? is agnostic about what Δ_x and Γ_x are (recall the scoping declared there: $\Delta_x = \nabla V(x)$ is the full book gradient, not the Black–Scholes delta, which appears unsubscripted below); the routing calculus of Sections ??–?? is agnostic about how sensitivities are aggregated into P&L. This section joins them, and the joint carries one non-negotiable rule.

3.6.1 Which Greeks enter

Fix a single name and let the declared risk factor be the spot, $x = S$. The implied volatility of every position on the book rides the surface: through the forward $F(S)$, both moneyness coordinates move, and $\sigma = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ moves with them (Proposition ??). The book value between marks is therefore the *routed* value $S \mapsto V(S, \sigma(S))$, and the gradient and Hessian entries that the identity (??) consumes are the routed totals of Theorem ??:

$$\Delta_S^{\text{total}} = \frac{dV}{dS} = \Delta + \nu \frac{d\sigma}{dS}, \quad \Gamma_S^{\text{total}} = \Gamma_{\text{adj}} = \Gamma + 2 \text{Vanna} \frac{d\sigma}{dS} + \text{Volga} \left(\frac{d\sigma}{dS} \right)^2 + \nu \frac{d^2\sigma}{dS^2},$$

with $\frac{d\sigma}{dS} = \frac{d\sigma}{dF} F'(S)$ and its second-order companion carried through the forward map exactly as in (??); under zero rates, $F = S$, and the expanded forms (??) and (??) apply. Substituting into Theorem ??:

Theorem 3.23 (Unified attribution identity). *For a book cubic in the routed variable S , with $h = S_2 - S_1$,*

$$\boxed{V(S_2) - V(S_1) = \frac{1}{2} \left(\left(\frac{dV}{dS} \right)(S_1) + \left(\frac{dV}{dS} \right)(S_2) \right) h - \frac{1}{12} \left((\Gamma_{\text{adj}})(S_2) - (\Gamma_{\text{adj}})(S_1) \right) h^2,} \quad (3.34)$$

where each total is evaluated on the surface at its own state — its own forward, its own (m, ℓ_F) , its own classical Greeks. The parenthesised argument denotes point evaluation at the state; the held-fixed bar $\cdot|_{\xi}$ remains reserved for genuine constraints per the contract and is never used for evaluation in this chapter.

Using partial Greeks in (??) — classical Δ and Γ with the smile adjustments dropped — is not a smaller model; it is an inconsistent one. The reprice $V(S_2)$ is computed on the moved surface, so an explanation built from frozen- σ sensitivities attributes the vol-through-spot P&L to nothing,

and it surfaces as a spurious residual that no amount of Hessian coverage will remove. The worked example below prices this error: on a ten-percent move it is first order in the move, roughly 260 times the honest cubic residual, and nearly twice the total-Greek ex ante error, of the opposite sign (Figure ??).

3.6.2 Two equivalent viewpoints

The same P&L can be booked two ways, and a sound system computes both.

Option A — full multivariate. Keep $x = (S, \sigma)^\top \in \mathbb{R}^2$ as separate risk factors. Then $\Delta_x = (\Delta, \nu)^\top$, the Hessian holds $(\Gamma, \text{Vanna}, \text{Volga})$, and Corollary ?? applies with the realized move $h = (S_2 - S_1, \sigma_2 - \sigma_1)^\top$, where $\sigma_2 - \sigma_1$ is the volatility move the surface actually delivered. This viewpoint attributes P&L to spot and vol *separately* — the granular report.

Option B — reduced univariate. Substitute the route into the value first, reduce to the single driver S , and apply Theorem ?? with total Greeks — the compact report.

When the realized vol move equals the routed one, $\sigma_2 - \sigma_1 = \sigma(S_2) - \sigma(S_1)$ along the surface, the two agree — but only up to the difference of their cubic residuals, not to machine precision. Write R_A and R_B for the residual (??) of each representation; both explanations target the same reprice, so

$$\text{explanation}_A - \text{explanation}_B = R_B - R_A = O(\|h\|^4),$$

the bound following from Proposition ?? applied to each representation separately. The agreement is *exact* only when the route is affine and the book cubic in (S, σ) : in general the two truncate *different* quartic content, because a book cubic in (S, σ) is not cubic in the routed S when $\sigma(S)$ is nonlinear. Remark ?? is exact for *derivatives* — the chain rule — and lifts to P&L only to fourth order. The worked example of Section ?? prices the gap: with the vol move set exactly on-route at $h = 10$, $R_A = 0.000589$, $R_B = 0.000678$, and the A-minus-B gap is 8.9×10^{-5} — thirteen percent of the residual itself, invisible on the report, and fatal only to a test that demands machine-precision equality (Section ?? files the comparison accordingly).

Their difference in production, where the realized vol move contains a component the route did not predict, is itself informative: it is the book's P&L on *surface repricing beyond the sticky dynamics* — vega P&L on the residual vol move — and Option A isolates it while Option B buries it in R . But the sensor has a floor: even with the vol move exactly on-route the gap is $R_B - R_A$, so the A-minus-B difference isolates unrouted vol P&L only *above* a baseline of order $\max(|R_A|, |R_B|)$ — fourth order in the move, sixteen-folding when the move doubles, and capable of printing as spurious unrouted vol on a large day if it is not netted out first. Both are run: A for the attribution report, B for the compact one, with the A-minus-B gap, read against its fourth-order floor, as the measure of how much of the day's vol move the surface dynamics failed to route.

3.6.3 Ex ante with total Greeks

The ex ante machinery of Section ?? composes the same way: the projected gradient must be transported with the *total* gamma,

$$\tilde{\Delta}_{S_2}^{\text{total}} = \left(\frac{dV}{dS}\right)(S_1) + (\Gamma_{\text{adj}})(S_1) h, \quad V(S_2) - V(S_1) \approx \left(\frac{dV}{dS}\right)(S_1) h + \frac{1}{2} (\Gamma_{\text{adj}})(S_1) h^2. \quad (3.35)$$

This is where the smile adjustment stops being an accounting nicety and becomes hedge slippage: the delta a trader expects to hold *after* the move is $\tilde{\Delta}^{\text{total}}$, and every term of Γ_{adj} — vanna coupling, volga through the slope squared, vega times routed curvature — feeds it. Omit them and the projected delta is systematically wrong by an amount growing linearly in the move and in the steepness of the surface; on a desk running real vega this is rehedging cost, paid daily, not a rounding error.

Remark 3.24 (Fixed expiry, and where theta sits). Per Assumption ??(4), every sensitivity in this chapter is instantaneous at fixed T ; the identity decomposes the P&L of the *move*, not of the calendar. In production the state vector x of Corollary ?? carries calendar time as one more coordinate: theta enters the gradient as a first-order sensitivity, and the gamma/theta balance the desk lives on is then read directly off the attribution — the second-order spot row against the first-order time row. Two cautions before this is wired in. First, the licence of Proposition ?? is an asymptotic in the move: a one-day step in the time coordinate is a *fixed finite* component of h , not a shrinking one, so the $O(\|h\|^4)$ residual control does not cover the time direction — near expiry, where theta accelerates, V is nothing like a low-degree polynomial in calendar time, and the time-direction content lands in R at full size. Second, the gamma-row-versus-theta-row read-off closes only if the spot–time cross entries of the Hessian — charm and colour — are populated; both blow up near expiry, exactly where the first caution already bites, so the read-off is weakest precisely where the desk leans on it hardest. The routing calculus is untouched by all of this, because time is a driver of its own, not a link on the spot route.

Remark 3.25 (Nonzero carry). Nothing in (??) used $F = S$ except through the expanded forms of the totals. With nonzero rates or dividends the same identity holds verbatim; only the link derivatives change, via $F'(S) \neq 1$ and $F''(S) \neq 0$ in (??). The attribution framework and the surface dynamics are orthogonal by construction — that is the point of routing through F rather than collapsing it.

3.7 Worked Example: a Quadratic Surface End to End

The linear surface of Example ?? exercises the first-order routing. A *quadratic* surface exercises everything this chapter has built: both curvature sources of Proposition ?? are live, the total gamma varies with spot, the third-order term of the identity is nonzero, and the ex ante / ex post machinery has something real to reconcile. All numbers below are derived analytically under this surface; the $\sigma_{\text{ATM}} = \text{const}$ convention (C4), which would zero the vol.slope and ATM-curvature content on display here, is not used.

Example 3.26 (Quadratic surface: states, totals, attribution). Fix $T = 1$, $K = 100$, $F_{\text{ref}} = 100$, and the surface

$$\sigma_{\text{ATM}}(\ell_F, T) = 0.20 - 0.04\ell_F + 0.03\ell_F^2, \quad \sigma_{\text{smile}}(m, T) = -0.10m + 0.05m^2, \quad (3.36)$$

so that, at general (ℓ_F, m) ,

$$\text{vol.slope} = -0.04 + 0.06\ell_F, \quad \frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} = 0.06, \quad s(m) = -0.10 + 0.10m, \quad c(m) = 0.10,$$

and the normalisation $\sigma_{\text{smile}}(0, T) = 0$ holds. This quadratic surface is a *local test function* for the attribution mechanics, valid on the moneyness band exercised here (total vol stays positive, minimum ≈ 0.137); it is not globally admissible — quadratic growth of the smile violates the Lee moment bounds far from the money and carries static arbitrage there — and it must not be reused as a calibration form (Chapter 4 develops the admissibility machinery). The smile block reproduces Example ??; the ATM block adds convexity ($\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} = 0.06 > 0$, the ATM vol reaccelerating away from the anchor in both directions) and lets vol.slope itself move with the forward. The book is one European call, long.

State 1: the anchor. At $S_1 = 100$ — under zero rates, $F = S = 100$ — $\ell_F = 0$, $m = 0$, $\sigma = 0.20$, and the classical Greeks of Example ?? apply unchanged: $\Delta = 0.539828$, $\Gamma = 0.0198476$,

$\nu = 39.6953$, Vanna = 0.198476, Volga = -1.98476 , and value $V(S_1) = 7.96557$. The routed derivatives, by Proposition ?? with $F = S$ (flagged),

$$\frac{d\sigma}{dS} = \frac{-0.04 - (-0.10)}{100} = 6.0 \times 10^{-4}, \quad \frac{d^2\sigma}{dS^2} = \frac{0.06 + 0.10 + (-0.10) - (-0.04)}{100^2} = 1.0 \times 10^{-5},$$

give, by Theorem ??,

$$\left(\frac{dV}{dS}\right)(S_1) = 0.563645, \quad (\Gamma_{\text{adj}})(S_1) = 0.0204820.$$

Only the curvature differs from the linear surface: the ATM convexity adds $\nu \cdot 0.06/S^2 = 2.38 \times 10^{-4}$ to the $\Gamma_{\text{adj}} = 0.0202439$ of Example ?? — the entire effect of the new parameter sits in one term, exactly where the routing says it must.

State 2: a ten-percent rally. At $S_2 = 110$ — under zero rates, $F = S = 110$, flagged — the coordinates are $\ell_F = \ln 1.1 = 0.0953102$ and $m = \ln(100/110) = -0.0953102$, so the option has gone in the money while the surface level has fallen:

$$\sigma = 0.206445, \quad \text{vol.slope} = -0.0342814, \quad s(m) = -0.109531.$$

Both slopes have decayed in magnitude relative to the anchor — the quadratic terms at work. With $d_1 = -m/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2 = 0.564895$ and $d_2 = 0.358450$, the classical Greeks and routed derivatives are (six significant figures)

$$\Delta = 0.713928, \quad \Gamma = 0.0149768, \quad \nu = 37.4119, \quad \text{Vanna} = -0.590528, \quad \text{Volga} = 36.6945,$$

$$\begin{aligned} \frac{d\sigma}{dS} &= \frac{-0.0342814 + 0.109531}{110} = 6.84088 \times 10^{-4}, \\ \frac{d^2\sigma}{dS^2} &= \frac{0.06 + 0.10 - 0.109531 + 0.0342814}{110^2} = 7.00416 \times 10^{-6}, \end{aligned}$$

with value $V(S_2) = 14.5324$, and totals

$$\left(\frac{dV}{dS}\right)(S_2) = 0.713928 + 37.4119 \times 6.84088 \times 10^{-4} = 0.739521,$$

$$\begin{aligned} (\Gamma_{\text{adj}})(S_2) &= 0.0149768 + 2(-0.590528)(6.84088 \times 10^{-4}) \\ &\quad + 36.6945(6.84088 \times 10^{-4})^2 + 37.4119 \times 7.00416 \times 10^{-6} \\ &= 0.0149768 - 0.000807940 + 0.0000171731 + 0.000262035 = 0.0144481. \end{aligned}$$

At the anchor the smile adjustment *adds* 3.2% to gamma; ten percent higher it *subtracts* 3.5%. The driver is vanna, which changed sign as d_2 crossed zero — in the money, rising vol now hurts the delta — and at this moneyness the vanna pathway overwhelms the (still positive) vega-curvature term. A risk system carrying the anchor's gamma adjustment as a static markup would have the sign of the correction wrong after the rally; the adjustment is a routed quantity and must be re-evaluated with the state, which is the entire operational content of Theorem ??.

Ex post attribution. With $h = 10$, the identity (??) decomposes the reprice as

Order	Contribution	Formula
1st	+5.63645	$(\frac{dV}{dS})(S_1)h$
2nd	+0.879378	$\frac{1}{2}((\frac{dV}{dS})(S_2) - (\frac{dV}{dS})(S_1))h$
3rd	+0.050283	$-\frac{1}{12}((\Gamma_{\text{adj}})(S_2) - (\Gamma_{\text{adj}})(S_1))h^2$
Identity total	+6.56611	
Reprice $V(S_2) - V(S_1)$	+6.56679	
Cubic residual R	+0.000678	about 1 bp of the P&L

The third-order term is *positive* 5 cents because the total gamma fell over the rally: the long-gamma profile decayed into the move (the vanna sign flip again), and the identity books that decay explicitly instead of leaving it in the residual. The residual itself — the quartic-and-above content of Black–Scholes plus surface over a ten-percent move — is under seven-hundredths of a cent on a 6.57 P&L.

Ex ante, and the reconciliation. Before the move, the desk’s estimate (??) reads

$$0.563645 \times 10 + \frac{1}{2} (0.0204820)(100) = 6.66056,$$

an overshoot of -0.093763 against the reprice. The reconciliation gap (??) accounts for it: the projected total delta $\tilde{\Delta}_{S_2}^{\text{total}} = 0.563645 + 0.0204820 \times 10 = 0.768465$ landed -0.028945 above the measured 0.739521 (delta projection error $\frac{1}{2}(-0.028945)(10) = -0.144723$), partially offset by the third-order correction $+0.050283$, for $\mathcal{E} = -0.094441$; the remaining $+0.000678$ is R . Under an exactly cubic book, Proposition ?? would put the gap at $\frac{1}{6}((\Gamma_{\text{adj}})(S_2) - (\Gamma_{\text{adj}})(S_1))h^2 = -0.100566$; the measured -0.094441 differs from it by 0.0061 — a tenth of a percent of the P&L, quartic-order like R but not equal to it (it is the trapezoid error on the non-affine gamma path, a different fourth-order functional) — which is the proposition holding to exactly the accuracy the cubic model claims and no more.

The cost of partial Greeks. Rerun the ex ante estimate with classical Greeks — $\Delta = 0.539828$, $\Gamma = 0.0198476$, smile adjustments dropped: the estimate is 6.39066 , an error of $+0.176129$, nearly twice the total-Greek error and of the opposite sign. Worse is the small-move behaviour: the partial-Greek error is *first order* in h , with slope $\nu \frac{d\sigma}{dS} = 0.0238$ per unit of spot (± 4.8 cents at $h = \pm 2$), because the frozen- σ delta is the wrong hedge ratio; the total-Greek ex ante error is third order in h by construction (Proposition ??) and is under a tenth of a cent there. A desk that hedges the partial delta pays that first-order bias on every rebalance — this is the delta adjustment of Example ?? recurring as realized slippage rather than as a Greek.

3.8 Attribution Protocol and Structural Checks

The results above are identities, so an implementation can be tested against them with no tolerance for style: a failure is a defect, not a calibration issue. The protocol has two layers — structural checks that must hold to numerical precision, and a daily reconciliation loop whose outputs are diagnostics, not pass/fail.

3.8.1 Structural identities

1. **Constant-gamma limit.** When $\Gamma_{x_1} = \Gamma_{x_2}$, the third-order term of (??) vanishes and the average-delta form must equal the classical delta–gamma form (Remark ??), to machine

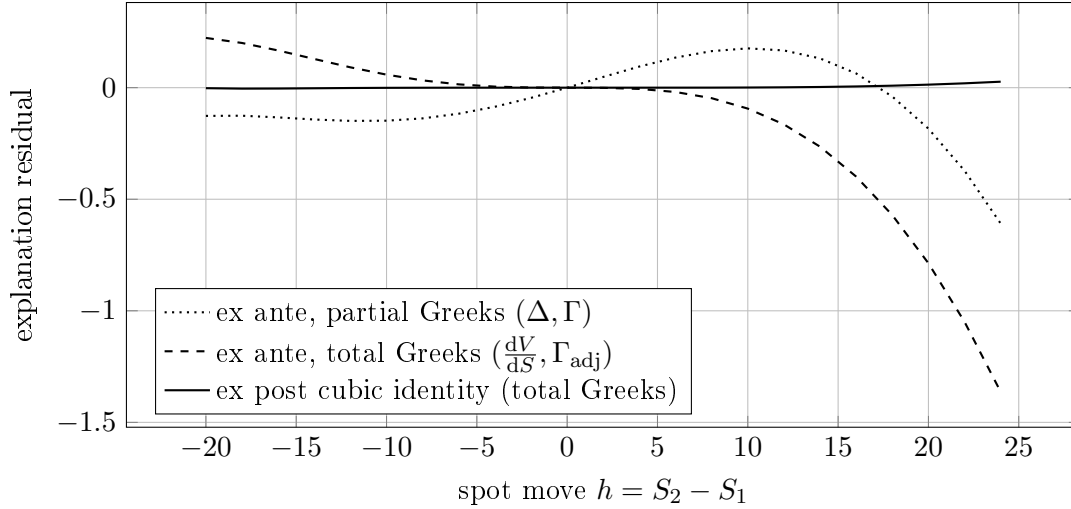


Figure 3.3: Explanation residuals (reprice minus explanation) for the quadratic surface of Example ??, as functions of the spot move from the anchor $S_1 = 100$; under zero rates, $F = S$ throughout. The partial-Greek estimate (dotted) is biased at *first* order in h — slope $\nu \frac{d\sigma}{dS} = 0.0238$ through the origin — and its seemingly benign behaviour at large positive h is an accidental cancellation, not accuracy. The total-Greek ex ante residual (dashed) is third order in h near the origin, per Proposition ?. The ex post identity residual (solid) is the cubic residual R : quartic near the origin, at most 0.0136 over a $\pm 20\%$ move, and 0.027 at the plotted extreme $h = +24$ — indistinguishable from zero at the scale of the other two.

precision.

2. **Flat-surface limit.** When $\text{vol.slope} = 0$, $s(m) = 0$, $c(m) = 0$, and $\frac{\partial^2 \sigma_{\text{ATM}}}{\partial \ell_F^2} = 0$, all routed adjustments vanish: $\frac{dV}{dS} = \Delta$ and $\Gamma_{\text{adj}} = \Gamma$ exactly (Theorem ??).
3. **Zero-move limit.** $x_2 = x_1$ must return zero in every formula — including (??), where a nonzero value at zero move indicates state leakage between the two risk runs.
4. **Hessian symmetry.** Γ_x must be symmetric (Schwarz). In routed terms, the two cross-pathways that merge into the factor 2 Vanna in (??) must be computed identically; asymmetry is a bug in one of them.

One comparison deliberately does *not* appear on this list: Option A versus Option B. The two attributions agree only up to $R_B - R_A = O(\|h\|^4)$ (Section ??), so demanding machine-precision equality would fail a *correct* implementation; the comparison is a diagnostic with a tolerance, and it lives in the daily loop below.

3.8.2 The daily loop

Each day, book three numbers per portfolio: the ex ante estimate (??) at the prior close; the ex post decomposition (??) at the current close; and the reprice. From these, report the reconciliation gap $\mathcal{E}$ split per (??) into delta projection error and third-order correction, and the cubic residual R from (??). The reading discipline is:

- *Delta projection error large, third-order small:* the Hessian transported the gradient badly — missing cross terms or stale surface derivatives in Γ_{adj} . Fix coverage.

- *Third-order persistently material*: gamma is turning over fast relative to the rebalancing grid; the constant-gamma intraday ladder is mispricing hedges even when end-of-day attribution closes. Tighten the ladder or cut convexity.
- *R trending, not mean-zero*: the book's quartic-and-above content is systematic. The cubic control no longer tracks the position, and the product-level question of Section ?? is now open.
- *Option A versus Option B* (Section ??), with the vol move computed consistently along the surface: the gap must sit within a tolerance of order $\max(|R_A|, |R_B|)$ — fourth order in the move, *not* machine precision. A gap at that floor jointly certifies the routing derivatives and the identity plumbing; a gap above it, in production, is unrouted vol P&L — but only the excess over the floor, which scales like the fourth power of the move and must be netted out before anything is attributed to the surface reshaping.

The magnitudes matter as much as the trends, and the worked example calibrates intuition for a vanilla book: R near a basis point of P&L on a ten-percent move, $\mathcal{E}$ two orders of magnitude above R and entirely explained by gamma variation. A vanilla book printing R comparable to $\mathcal{E}$ is not a vanilla book, whatever the term sheet says.

3.9 Toxic products and the cubic boundary

The identity of Section ?? is exact for a cubic book and, as the next proposition records, accurate to fourth order in the move for a smooth one. That accuracy is a *licence*, and the licence has conditions. This section states the conditions, names the products that violate them, and says what a desk should do when they do. The organising object is the residual R of (??). Through the chapter it has been a small number to monitor; here it becomes the instrument that says whether the apparatus applies at all. The failure mode this section guards against is not a wrong number — the identity cannot produce one — but a right number that is trusted past the point where it means anything.

3.9.1 The residual is a sensor, not a rounding error

Everything downstream rests on how R behaves for a well-posed book.

Proposition 3.27 (Quartic residual for smooth books). *Let V be four times continuously differentiable in a neighbourhood of x_1 . Then the cubic residual of (??) satisfies*

$$R(x_1, x_2) = O(\|h\|^4) \quad \text{as } h = x_2 - x_1 \rightarrow 0.$$

Proof. The explanation $\frac{1}{2}(\Delta_{x_1} + \Delta_{x_2})^\top h - \frac{1}{12}h^\top(\Gamma_{x_2} - \Gamma_{x_1})h$ is built from the *true* endpoint gradients and Hessians. Expand each about x_1 : since $\nabla V \in C^3$ and $\nabla^2 V \in C^2$ near x_1 ,

$$\begin{aligned} \Delta_{x_2} &= \Delta_{x_1} + \Gamma_{x_1}h + \frac{1}{2}\nabla^3 V(x_1)[h, h] + O(\|h\|^3), \\ \Gamma_{x_2} - \Gamma_{x_1} &= \nabla^3 V(x_1)[h] + O(\|h\|^2). \end{aligned}$$

Contracting with h , the averaged-gradient term contributes $\Delta_{x_1}^\top h + \frac{1}{2}h^\top \Gamma_{x_1}h + \frac{1}{4}\nabla^3 V(x_1)[h, h, h] + O(\|h\|^4)$ and the Hessian-difference term contributes $-\frac{1}{12}\nabla^3 V(x_1)[h, h, h] + O(\|h\|^4)$; the third-order coefficients close, $\frac{1}{4} - \frac{1}{12} = \frac{1}{6}$, so the explanation reproduces the degree-three Taylor polynomial of $V(x_2) - V(x_1)$ term by term. Subtracting it from the reprice, the first surviving term is the quartic Taylor remainder, so $R = O(\|h\|^4)$. $\square$

The exponent, not the constant, is the message. The residual is fourth order in the move: *doubling the move multiplies its leading part by sixteen*. The worked example of Section ?? shows the mechanism live — R runs 0.000678 at a ten-percent move and 0.0136 at twenty (Figure ??), a factor near twenty, the excess over sixteen being the fifth-and-higher orders the estimate also drops. This is the property a risk manager distrusts on sight: the control is most accurate exactly where nothing happens to the book, and degrades fastest exactly in the tail where the book loses money. Precise in the centre, silent where it counts: a residual invisible on everyday moves says almost nothing about the extreme move.

So the smallness of R on a quiet day is not, by itself, evidence of anything. Three conditions together define the domain in which the bookkeeping is a control variate rather than a story:

1. **Small** relative to the P&L it explains — the vanilla baseline is R near a basis point on a ten-percent move.
2. **Mean-zero** through time — a smoothness remainder has no preferred sign; a biased R is not a remainder, it is an unmodelled exposure.
3. **Regime-independent** — R that is small in calm markets and large in stress is measuring an exposure the calm regime never exhibited.

Definition 3.28 (Cubic-toxic product). A product is *toxic to the cubic bookkeeping* if its residual R violates any of the three conditions above: structurally large relative to its own P&L, systematically biased in sign, or regime-dependent (small in calm, large in stress). For such a product the identity is no longer a control variate whose validity is the demonstrated smallness of R ; it has become, silently, a forecast of the one part of the P&L it cannot see.

The word “toxic” is deliberate and narrow. It is not a claim that the product is a bad trade or badly priced — the pricer may be flawless. It is a claim about the *attribution*: that the nightly decomposition the desk relies on to know why it made or lost money has stopped being trustworthy for this position, and will be most wrong on the day it matters most.

3.9.2 Three ways the boundary is breached

Proposition ?? needed two things: a value function smooth to fourth order, and a residual that is genuinely the quartic remainder rather than something else hiding in it. Each assumption fails in a characteristic way, and each failure names a class of products.

Breach A — the value is not smooth. Digitals, barriers, autocallables, range accruals, any coupon that switches on a level: their value functions are not C^4 ; often not C^1 . No polynomial of any finite degree approximates a step, so R is not $O(\|h\|^4)$ — it is $O(1)$, the size of the discontinuity, the instant a move straddles the feature.

Example 3.29 (The digital near expiry). Take a cash-or-nothing digital call paying one unit if $S_T > K$, on notional $\mathcal{N}$. Under the standing zero-rate assumption, $F = S$ (flagged), and its value per unit is $N(d_2)$ with $d_2 = -m/(\sigma\sqrt{T}) - \sigma\sqrt{T}/2$ and $m = \ln(K/F)$. Differentiating through spot, the leading term as $T \rightarrow 0$ comes from $\partial m/\partial S = -1/S$,

$$\frac{d}{dS}N(d_2) = \varphi(d_2) \frac{dd_2}{dS} = \frac{\varphi(d_2)}{S\sigma\sqrt{T}} + \varphi(d_2) \frac{\partial d_2}{\partial \sigma} \frac{d\sigma}{dS}, \quad \frac{\partial d_2}{\partial \sigma} = \frac{m}{\sigma^2\sqrt{T}} - \frac{\sqrt{T}}{2}.$$

The routed- σ correction requires scoping, not neglect: *at the money* ($m = 0$) it vanishes like $\sqrt{T}$ against the $1/\sqrt{T}$ leading term, since only $-\sqrt{T}/2$ survives in $\frac{\partial d_2}{\partial \sigma}$; *off the money* the $m/(\sigma^2\sqrt{T})$ piece makes it the same order $1/\sqrt{T}$ as the leading term, with $\frac{d\sigma}{dS} = (\text{vol.slope} - s(m))/S$ under zero rates, smaller only by the T -independent factor $m(\text{vol.slope} - s(m))/\sigma$. The divergence conclusion below therefore holds everywhere on the strike axis; neglecting the correction is licensed only at the strike, which is where this example sits. The dollar delta per unit notional, $S \frac{d}{dS} N(d_2) = \varphi(d_2)/(\sigma\sqrt{T})$ to leading order, *diverges like* $1/\sqrt{T}$. At the money ($m = 0$, so $\sigma = \sigma_{\text{ATM}}(\ell_F, T)$; take $\sigma = 0.20$) a week from expiry, on a calendar-year basis $T = 1/52$, $\sigma\sqrt{T} = 0.20\sqrt{1/52} = 0.02774$ and $\varphi(d_2) \approx \varphi(0) = 0.3989$, giving a dollar delta of

$$\frac{0.3989}{0.02774} \approx 14.4 \text{ per unit} \implies \$144\text{mm on a } \$10\text{mm digital},$$

fourteen times the notional. One day out, on the same calendar basis ($T = 1/364$), it is \$381mm; at the closing bell it is unbounded. Meanwhile the value $N(d_2)$ is convex in S below the strike and concave above, so gamma *reverses sign* across K — the book flips from long convexity to short convexity over a single point of spot. The identity has no good reading of this position. If the two marks sit near K , the endpoint Greeks are enormous and unstable. If they straddle a near-step, the reprice is the whole coupon while the endpoint deltas, evaluated where the value has already flattened, are near zero: the cubic explanation draws a smooth ramp, reality delivers a cliff, and the entire notional lands in R . In the limit $T \rightarrow 0$ the value tends to $\mathcal{N} \cdot \mathbf{1}\{S > K\}$ and R tends to the full step. No gamma budget contains a delta that grows without bound; this product cannot be risk-managed as a convexity at all.

Breach B — the convexity lives between the marks. The identity is a two-point stencil: it sees Γ_{x_1} and Γ_{x_2} and nothing in between, and the trapezoid step (??) is exact only because a cubic's gamma is affine along the segment. Let the true gamma have a spike or a zero-crossing *strictly inside* (x_1, x_2) and the endpoints go blind: they can be equal, or both positive, while the interior dips sharply the other way. Pin risk near expiry is the clean case — an at-the-money option's gamma is a tall narrow spike at the strike; two marks bracketing it both read small gamma, the third-order term reports near zero, and yet the P&L over the move rode the entire spike. This is Breach A's smooth cousin: not a discontinuity in the value, but curvature concentrated on a scale finer than the move, invisible to a stencil whose feet straddle it. Shrinking the move — rebalancing finer — resolves it, right up until the move is a gap that admitted no subdivision.

Breach C — the smile moves off the route. Proposition ?? silently assumed one more thing: that the realised volatility move equals the routed one, $\sigma_2 - \sigma_1 = \sigma(S_2) - \sigma(S_1)$ along the surface. The total gamma Γ_{adj} inside the identity is computed under an *assumed* surface dynamic — the decomposition $\sigma = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$, its slope vol.slope , and the sticky rule that maps a spot move to a vol move. When the smile *reshapes* instead of merely translating — the slope steepens in a sell-off, the ATM convexity jumps, the term structure twists — the realised vol move parts company with the routed one, and the difference is vega P&L that is *first order* in the vol surprise, not fourth order in h . It lands in R disguised as a smoothness remainder. This is the most dangerous breach because it respects the first two health conditions until it fails the third: R stays small and mean-zero for as long as the surface obeys the assumed dynamics, and turns biased and large on the day it does not. The sensor is already in the chapter: the Option A versus Option B gap of Section ?? isolates the unrouted vol P&L — above the fourth-order routed baseline of order $\max(|R_A|, |R_B|)$ that the gap carries even on-route — before it hides in R . In Volume II the same

breach wears correlation — a basket routed under sticky- ρ ($d\rho = 0$, Remark ??) is short the crash in which $\rho \rightarrow 1$, and the realised $d\rho$ the route held fixed lands, once again, in the residual.

3.9.3 The convexity sign flip: benign twin and toxic twin

The worked example of Section ?? already showed a total gamma change sign: Γ_{adj} ran from a +3.2% smile adjustment at the anchor to -3.5% ten percent higher, driven by vanna crossing zero as d_2 did. That flip was *harmless*. It happened *at* the two marks, the identity read it off measured endpoint gammas, and it was booked in the third-order term to the cent (the +5-cent line of the attribution table). The same convexity flip turns toxic only when it is moved to where the stencil cannot see it: between the marks (Breach B), or onto a route that reality does not follow (Breach C). Same picture, opposite consequence, and the only difference is whether the two evaluation points *bracket* the structure or the structure hides between them — or off-axis, in a vol move the route never predicted.

The doctrine compresses to one sentence: a book can be long gamma at yesterday's close and long gamma at today's, and short gamma every hour in between. Its attribution closes cleanly every night and its hedges bleed every day. That is short optionality the endpoint Greeks never name — and naming it is not a matter of a finer pricer or a fuller Hessian, but of recognising that the two-point identity was never built to see it.

3.9.4 What a desk does at the boundary

The response is specific to the breach, and in no case is it “improve Hessian coverage” — that answer belongs to Section ??, on the right side of the boundary.

- *Non-smooth value (Breach A)*. Do not carry a digital or a barrier as a gamma. Replicate the discontinuity with a spread and manage the residual step as a *scenario* — a gap P&L with a probability attached, not a Greek. The replication width is itself the risk decision: a tighter spread cuts the un-hedged step but raises the local delta to be carried (the $1/\sqrt{T}$ of Example ??); the trade is hedgeability against slippage, and R prices it either way.
- *Interior convexity (Breach B)*. Shrink the rebalance grid until it resolves the interior structure — but a gap admits no subdivision, so the position is sized to survive the un-subdivided move rather than hedged through it.
- *Off-route smile (Breach C)*. Stress the surface *off* the route: bump vol.slope, steepen and reshape $\sigma_{\text{smile}}(m, T)$, twist the term structure, and reprice. If that reprice dwarfs the R observed in calm markets, the book is short smile-reshaping optionality — or, in Volume II, correlation optionality — that the attribution is hiding. Option A run alongside Option B, with the gap read against its fourth-order floor, is the early sensor before the loss.

This closes the reading discipline of Section ?. An R that trends, carries a sign, or wakes up in a crisis is not a call for a better Hessian; it is the book reporting that this product has crossed the boundary, and on the far side the honest attribution is a scenario, not a Greek. The cubic identity earns its trust by making its own failure measurable: R small, mean-zero and regime-independent certifies that the control still holds; R large, biased, or regime-switching reports that it no longer does — and a bookkeeping that announces when to stop believing it is worth more than one that is merely elegant while it lasts.

Chapter 4

Calibration and Market Data

The only source of truth is the observed price. Everything else is a model.

Every surface in this book begins its life as a fit to attested prices. This chapter is where those prices enter: how a stream of noisy, asynchronous market observations is turned into a calibrated parameter vector, how that vector is kept honest against no-arbitrage, and how bad data is caught before it contaminates the state. Calibration is *inference*: the object carried forward is not a point estimate but a posterior distribution over parameters, updated one accepted observation at a time.

The register is Bayesian statistics and linear algebra, with the linear-Gaussian (Kalman) case worked out in full because it is the one setting where the whole machinery – prediction, update, gating, structural-break detection – can be written down in closed form and checked by hand. The surface parameterisation of the preceding chapters supplies the state vector: the parameter ket $|\beta\rangle = (\sigma_0, s_0, c_1, \dots, c_n)^\top$ and the feature ket $|\phi(m)\rangle$ are exactly what the filter estimates, and the design matrix Φ is exactly its observation operator.

Remark 4.1 (Local notation, declared once). This chapter introduces several symbols scoped to calibration. Three of them would collide with ratified glyphs if left bare, and are disambiguated here explicitly (Contract §5.3):

- The *calibrated parameter vector* is written ϑ (ϑ), *not* θ : by the Notation Contract θ is Black–Scholes theta and nothing else. The identification with the surface ket is $\vartheta = |\beta\rangle$.
- The *Gaussian law* “distributed as Normal with mean μ and covariance Σ ” is written $\mathcal{N}(\mu, \Sigma)$ ($\mathcal{N}$). The macro N (glyph N) is reserved by ruling R5 for the scalar standard-normal *CDF* $N(x)$ and is never used as a law in this chapter.
- The number of instruments is written N_{ins} (N_{ins}), *not* bare N , to keep clear of the CDF $N(x)$.

The remaining local symbols collide with nothing in the ratified table: the auxiliary attestation metadata g (the Durrleman butterfly function of Section ?? is a *different* symbol, the fraktur $\mathfrak{g}(m)$); the total implied variance $w(m, T)$, whose subscripts denote partials in m (so the process noise is written ζ_t , never w_t , to break the subscript clash); the two weight families of Axiom ??, ω_i^{price} (statistical, price space) and ω_i^{vol} (vol space); the Kalman gain matrix G_t ; the innovation covariance $\Sigma_{e,t}$ (written with Σ , never S , which is spot); the predicted / updated state covariances $\Sigma_{t|t-1}, \Sigma_{t|t}$; and the squared Mahalanobis statistic $\mathcal{M}_t^2$ (calligraphic, never D , which is the discount factor). The innovation vector is $e_t := y_t - H_t \mu_{t|t-1}$ throughout (it is *not* ν ; ν is Black–Scholes vega). The number of market observations processed at knowledge time t is J_t , so $y_t \in \mathbb{R}^{J_t}$; the total number of smile observations in a batch fit is J , indexed $j = 1, \dots, J$.

Remark 4.2 (Rates and dividends are carried explicitly here – $F = S$ flag). The standing assumption of this volume is zero rates and zero dividends, under which the forward collapses to the spot,

$F = S$; every derivation elsewhere in Volume I that uses this collapse flags it at its own point of use. Calibration is the one place where the rate, dividend, and repo curves are *themselves the objects being inferred*. The collapse $F = S$ is therefore unavailable: throughout this chapter the discount factor $D(T)$, the dividend yield $q(t)$, and the forward $F(T)$ are carried generally, and every pricing map π_i that consumes those curves is written with a general forward. The single exception is flagged inline where it occurs.

4.1 The Calibration Problem

4.1.1 Instrument universe and Oracle

Fix a finite set of instruments $\mathcal{I} = \{1, \dots, N_{\text{ins}}\}$: swap tenors, option strikes and maturities, equity underlyings, indices, credit names. An *Oracle* is any external, independently observable source of market truth – a listed exchange, a broker run, a consensus service, a fixing provider, or an electronically attested feed.

Definition 4.3 (Attestation). An attestation is a signed message from an Oracle,

$$o = (i, v, t_{\text{event}}, t_{\text{seen}}, g),$$

where $i \in \mathcal{I}$ is the instrument, v the observed value (price, rate, spread), t_{event} the time the market event occurred, t_{seen} the time the Oracle delivered it, and g any auxiliary metadata (source ID, quote side, size, venue).

The metadata slot is written g , not m : in this book $m = \ln(K/F)$ is strike log-moneyness and carries no other meaning. Attestations arrive as asynchronous streams: the same instrument is quoted on several venues with different latencies, and different instruments update at different times and frequencies. At any wall-clock time the system's view of the market is a collection of latest-known attestations with heterogeneous event times and sources. This asynchrony is a basic feature, not an exception, and it is the source of the broken states of Section ??.

4.1.2 Parameter space and pricing map

All calibrated quantities are collected into a finite-dimensional parameter vector $\vartheta \in \Theta \subseteq \mathbb{R}^d$: zero rates at selected tenors, volatility-surface coefficients, hazard rates, correlations. The *pricing map*

$$\pi : \Theta \rightarrow \mathbb{R}^{\mathcal{I}}, \quad \vartheta \mapsto (\pi_i(\vartheta))_{i \in \mathcal{I}},$$

sends parameters to model-implied prices. Calibration is inference about ϑ : the system carries a posterior distribution P on Θ , represented by its mean vector, its covariance Σ_{ϑ} , and selected quantiles.

For a volatility surface the parameter vector *is* the surface ket of the kernel model: $\vartheta = |\beta\rangle = (\sigma_0, s_0, c_1, \dots, c_n)^\top$, and the model-implied vol at strike log-moneyness m is the linear form $\langle \phi(m) | \beta \rangle$. This identification is what lets the abstract filter below run on a concrete surface.

4.2 Axioms of Calibration

The following seven assumptions are the load-bearing structure; every constraint and invariant later in the chapter is derived from them.

Assumption 4.4 (Observations are noisy samples). For each instrument i and event time t_{event} there is a latent true price $p_i(t_{\text{event}})$, and an attested value relates to it by

$$v_i = p_i + \varepsilon_i.$$

The noise ε_i is allowed to be heteroscedastic ($\text{Var}[\varepsilon_i]$ depends on i , time, liquidity, source), non-Gaussian (fat tails, skew), and cross-correlated across instruments and sources. Noise models may read the metadata g , but no discrete “class” structure is assumed.

Assumption 4.5 (Finite-dimensional parameter space). There is a finite-dimensional $\Theta \subseteq \mathbb{R}^d$ and a pricing map π with $p_i(t) = \pi_i(\vartheta(t))$ for a latent parameter process $t \mapsto \vartheta(t)$.

Assumption 4.6 (Arbitrage-free and smooth). There is an admissible region $\Theta_{\text{AF}} \subseteq \Theta$ encoding all structural no-arbitrage constraints (Section ??). Every acceptable ϑ lies in Θ_{AF} ; on Θ_{AF} the map π is continuously differentiable, and within a regime the path $t \mapsto \vartheta(t)$ has bounded variation. At declared regime changes bounded variation may be violated, but the regime boundary and the change in prior must be recorded explicitly.

Assumption 4.7 (Price-space primacy). Calibration quality is judged in price space. For observed instrument i with attested value v_i and candidate ϑ , the residual is $r_i(\vartheta) = \pi_i(\vartheta) - v_i$. With per-instrument *price-space* weights $\omega_i^{\text{price}} > 0$ (the statistical weight, for example $\omega_i^{\text{price}} = 1/\text{Var}[\varepsilon_i]$, heteroscedastic by Axiom ??), the weighted RMS error is

$$\text{WRMSE}(\vartheta) = \sqrt{\frac{\sum_i \omega_i^{\text{price}} r_i(\vartheta)^2}{\sum_i \omega_i^{\text{price}}}}.$$

A calibration to ϑ^* is *certified* only if (i) $\vartheta^* \in \Theta_{\text{AF}}$, (ii) $|r_i(\vartheta^*)| \leq \tau_i$ for every observed i , and (iii) $\text{WRMSE}(\vartheta^*) \leq \tau_{\text{agg}}$. The tolerances $\tau_i, \tau_{\text{agg}}$ are specified in the Model Configuration Attestation (MCA).

Assumption 4.8 (Posterior mean is a martingale). Let P_n be the posterior after all accepted attestations up to epoch n , with mean $\vartheta_n := \mathbb{E}_{P_n}[\vartheta]$, and let I_n be the information set of those attestations. Then

$$\mathbb{E}[\vartheta_{n+1} \mid I_n] = \vartheta_n.$$

Parameters move, on average, only because new information arrives; the posterior at epoch n is the prior at epoch $n+1$. At structural breaks the prior is explicitly widened, after which the martingale property resumes in the new regime.

Assumption 4.9 (Reproducible, auditable, traceable). Given the ordered input attestations, the MCA (model, hyperparameters, constraints, algorithm), and the numerical environment, any conforming implementation produces the same posterior up to a declared numerical tolerance. Every step emits an audit record (inputs, innovations, residuals, gate decisions, outputs) and a provenance chain back to all inputs and the MCA.

Assumption 4.10 (Attestations can be wrong). Some attestations are not merely noisy but incorrect (stale quotes mislabelled as current, fat-finger trades, feed corruption). The engine classifies each attestation into *accepted*, *down-weighted* (used with inflated observation noise), or *rejected* (not used to update the posterior). The classification is decided by the innovation-based tests of Section ??.

4.2.1 Price-space tolerance in vol space, via vega

Axiom ?? certifies in price space, but a volatility surface is fit most naturally in vol space. The two are reconciled by the vega $\nu = \frac{\partial C}{\partial \sigma}$. For a single option a small vol error $\delta\sigma$ produces a price error $\delta C \approx \nu \delta\sigma$, so a price residual maps to a vol residual by

$$r_i^{\text{price}} \approx \nu_i (\sigma_i^{\text{model}} - \hat{\sigma}_i), \quad \tau_i^{\text{vol}} = \frac{\tau_i}{\nu_i}.$$

The correct vol-space weight that *reproduces* the price-space objective is therefore

$$\omega_i^{\text{vol}} = \nu_i^2 \omega_i^{\text{price}},$$

because squaring the residual map carries the ν_i^2 Jacobian on top of the statistical price weight. The shortcut $\omega_i^{\text{vol}} \propto \nu_i^2$ (unit price weights) is the special case $\omega_i^{\text{price}} = \text{const}$, i.e. *homoscedastic price noise* – which Axiom ?? refuses in general, so this collapse is a flagged approximation, not an identity. Under it, a vol-space least-squares fit weighted by ν_i^2 is a price-space fit to first order and the certification gate reads off in either space; away from it, one must carry $\omega_i^{\text{vol}} = \nu_i^2 \omega_i^{\text{price}}$. This is the bridge that lets the linear vol-space regression below inherit the price-space primacy of Axiom ?. (The vega factor is largest near the money and decays into the wings, which is exactly the desired emphasis: the liquid at-the-money quotes dominate the fit – but the wings, where this weight vanishes, are precisely where butterfly no-arbitrage must still be enforced (Section ??), so the $\mathbf{g} \geq 0$ scan cannot be left to the vega-weighted objective.)

4.3 Bayesian Calibration

4.3.1 Prior, likelihood, posterior

Calibration is Bayes' rule applied to attestations:

$$P(\vartheta \mid \text{attestations}) \propto P(\vartheta) \prod_i P(o_i \mid \vartheta),$$

where the prior $P(\vartheta)$ encodes yesterday's belief and admissibility, and the likelihood $P(o_i \mid \vartheta)$ is built from the pricing map $v_i^{\text{model}} = \pi_i(\vartheta)$ together with an observation-noise model for $\varepsilon_i = v_i - v_i^{\text{model}}$. Axiom ?? makes ϑ finite-dimensional; Axiom ?? restricts the support to Θ_{AF} ; Axiom ?? phrases the acceptance gate in price space; Axiom ?? constrains how successive posteriors may differ.

4.3.2 Worked example: a single listed option

Consider a single listed one-month at-the-money call on an equity index. The MCA fixes Black–Scholes with a named yield curve and dividend yield (illustratively, a flat 3% rate and 1% dividend – the MCA *records* such configuration; the pricing itself is written through the forward $F(T)$ and discount $D(T)$ per Remark ??). The spot S and strike K are known.

The unknown is one-dimensional, $\vartheta = (\sigma)$. Because the option is at the money, $m = \ln(K/F) = 0$, and by the mandated decomposition $\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ with the normalisation $\sigma_{\text{smile}}(0, T) = 0$, the single unknown is precisely the at-the-money component $\sigma_{\text{ATM}}(\ell_F, T)$. (Under the standing zero-rate collapse this call would be at the money in spot as well, $F = S$; here F stays general, flagged.)

Each attestation is a mid-price v_t ; a naive noise model sets $v_t = \pi(\sigma_t) + \varepsilon_t$ with ε_t Gaussian of standard deviation equal to half the bid–ask spread. A prior on the daily vol move makes the problem dynamic:

$$\sigma_t = \sigma_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, (0.01)^2),$$

a driftless random walk with a one-percentage-point (absolute) daily shock, optionally truncated to a bounded range. Before today’s price arrives this gives a predictive distribution for σ_t ; when v_t arrives, the likelihood $P(v_t | \sigma_t)$ and this predictive combine into the posterior $P(\sigma_t | I_t)$. This is the simplest non-trivial instance of the general machinery; every worked number here (0.01 shock, 3%, 1%) is a modelling illustration.

4.3.3 The linear–Gaussian state-space model

The one-option example generalises to a full surface once the state is a vector and the observation operator a matrix. Let $x_t \in \mathbb{R}^d$ be the latent state at knowledge time t – for a smile, $x_t = |\beta\rangle_t$, the parameter ket of the surface.

Dynamics. Consistency with Axiom ?? dictates the *driftless* transition (identity transition matrix $A = I_d$):

$$x_t = x_{t-1} + \zeta_t, \quad \zeta_t \sim \mathcal{N}(0, Q).$$

The process-noise covariance Q controls how fast the state may move between updates.

Remark 4.11 (The identity transition is a sticky-parameter convention – C2). Writing $A = I_d$ holds the whole state, including the ATM intercept σ_0 evaluated at the *current* ℓ_F , fixed in expectation between updates. But the mandated form $\sigma_{\text{ATM}}(\ell_F, T)$ and bridging identity C2 say that when the forward moves by $\Delta\ell_F$ between $t-1$ and t the ATM parameter drifts by $\text{vol.slope} \cdot \Delta\ell_F$ (plus a ∂_T roll as T shortens) – a routed, $\mathcal{F}$ -measurable, *predictable* motion, not news. With equity $\text{vol.slope} < 0$, an identity transition that stays silent about this would misattribute forward-move-induced ATM drift to process noise. The identity transition is therefore a *sticky-parameter convention*: ℓ_F and T are held fixed between updates, so the residual state is genuinely driftless. When the forward is allowed to move, carry the predictable drift explicitly,

$$\mu_{t|t-1} = \mu_{t-1|t-1} + \text{vol.slope} \cdot \Delta\ell_F e_{\sigma_0} + (T\text{-roll}),$$

where e_{σ_0} is the intercept coordinate. In the enlarged filtration $I_{t-1} \vee \mathcal{F}_t$ that has already observed $\Delta\ell_F$, this drift is nonzero and predictable, so the predicted mean is *not* a martingale there; Axiom ?? makes no claim in that filtration. Axiom ?? is stated in the base attestation filtration I_{t-1} , in which the forward move is not yet observed: for a martingale forward $\mathbb{E}[\Delta\ell_F | I_{t-1}] = 0$, so the predictable drift vanishes in expectation and $\mathbb{E}[\mu_{t|t-1} | I_{t-1}] = \mu_{t-1|t-1}$, preserving the martingale property in that filtration.

Observation. At time t the observation is $y_t \in \mathbb{R}^{J_t}$,

$$y_t = H_t x_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_t),$$

where H_t maps state components to the J_t instruments seen at t and R_t is their observation-noise covariance. For a smile fit the rows of H_t are the feature bras: if the strikes observed at t have log-moneynesses $\hat{m}_1, \dots, \hat{m}_{J_t}$, then

$$H_t = \begin{pmatrix} \langle \phi(\hat{m}_1) | \\ \vdots \\ \langle \phi(\hat{m}_{J_t}) | \end{pmatrix}, \quad (H_t x_t)_j = \langle \phi(\hat{m}_j) | \beta \rangle = \sigma^{\text{model}}(\hat{m}_j).$$

Here H_t is a slice of the design matrix $\Phi \in \mathbb{R}^{J \times (n+2)}$ of the calibration block: the observation operator of the filter is the design matrix of the regression.

Proposition 4.12 (Driftless dynamics give a martingale mean). *For the driftless model $x_t = x_{t-1} + \zeta_t$ with $\mathbb{E}[\zeta_t | I_{t-1}] = 0$, and under the correct (unbiased) observation model $\mathbb{E}[\varepsilon_t | I_{t-1}] = 0$, the posterior-mean sequence $\vartheta_n := \mu_{t|t}$ satisfies $\mathbb{E}[\vartheta_{n+1} | I_n] = \vartheta_n$, i.e. it is a martingale in the sense of Axiom ??.*

Proof. Two steps. First the *prediction* identity for the latent state: taking conditional expectations of the transition and using $A = I_d$ and $\mathbb{E}[\zeta_t | I_{t-1}] = 0$,

$$\mathbb{E}[x_t | I_{t-1}] = \mathbb{E}[x_{t-1} | I_{t-1}] + \mathbb{E}[\zeta_t | I_{t-1}] = \mu_{t-1|t-1}.$$

Second, the step Axiom ?? asks for, on the posterior *mean*. The update is $\mu_{t|t} = \mu_{t|t-1} + G_t e_t$ with $\mu_{t|t-1} = \mu_{t-1|t-1}$ and innovation $e_t = y_t - H_t \mu_{t|t-1} = H_t(x_t - \mu_{t|t-1}) + \varepsilon_t$. Under the correct model $\mathbb{E}[x_t | I_{t-1}] = \mu_{t-1|t-1}$ and $\mathbb{E}[\varepsilon_t | I_{t-1}] = 0$, so $\mathbb{E}[e_t | I_{t-1}] = 0$, and since G_t is I_{t-1} -measurable,

$$\mathbb{E}[\mu_{t|t} | I_{t-1}] = \mu_{t-1|t-1} + G_t \mathbb{E}[e_t | I_{t-1}] = \mu_{t-1|t-1}.$$

Identifying ϑ_n with $\mu_{t|t}$ gives $\mathbb{E}[\vartheta_{n+1} | I_n] = \vartheta_n$. □

Lemma 4.13 (Gating through $\mathcal{M}_t^2$ preserves the martingale). *Suppose the accept/inflate/reject decision of Section ?? is a function of the data only through $\mathcal{M}_t^2 = e_t^\top \Sigma_{e,t}^{-1} e_t$ (equivalently, through an acceptance region $\{\mathcal{M}_t^2 \leq u\}$ symmetric in e_t). Then the gated update still satisfies $\mathbb{E}[\mu_{t|t} | I_{t-1}] = \mu_{t-1|t-1}$ under the correct model, because $e_t \mapsto -e_t$ leaves the region fixed while flipping the sign of $G_t e_t$, so the region-conditioned mean of the increment vanishes. An asymmetric down-weighting (one that treats $+e_t$ and $-e_t$ differently) would break this and hence Axiom ??: only $\mathcal{M}_t^2$ -measurable gates are martingale-safe.*

This is the first sanity check of the whole scheme: the drift is zero *by construction under unbiased accepted attestations*, so the filter cannot manufacture parameter motion out of nothing. Any observed motion traces to an accepted attestation – with the caveat that biased data (stale or fat-finger, Axiom ??) does manufacture motion until the gate of Section ?? catches it; the martingale is a statement about correctly-labelled data, and it is Lemma ?? that keeps the gate itself from injecting drift.

4.3.4 The Kalman recursion

Write $\mu_{t|t-1}, \Sigma_{t|t-1}$ for the predicted mean and covariance and $\mu_{t|t}, \Sigma_{t|t}$ for the updated ones. The recursion is:

$$\begin{aligned} \text{Predict: } \mu_{t|t-1} &= \mu_{t-1|t-1}, & \Sigma_{t|t-1} &= \Sigma_{t-1|t-1} + Q, \\ \text{Innovation: } e_t &= y_t - H_t \mu_{t|t-1}, & \Sigma_{e,t} &= H_t \Sigma_{t|t-1} H_t^\top + R_t, \\ \text{Gain: } G_t &= \Sigma_{t|t-1} H_t^\top \Sigma_{e,t}^{-1}, \\ \text{Update: } \mu_{t|t} &= \mu_{t|t-1} + G_t e_t, & \Sigma_{t|t} &= (I_d - G_t H_t) \Sigma_{t|t-1}. \end{aligned}$$

The innovation e_t is the surprise in the new observations relative to the prediction, and its covariance $\Sigma_{e,t}$ measures how much surprise the model expects. Everything downstream – the update, the gate, the break detector – is a function of the pair $(e_t, \Sigma_{e,t})$.

Remark 4.14 (Standing regularity: $R_t \succ 0$). The gain $G_t = \Sigma_{t|t-1} H_t^\top \Sigma_{e,t}^{-1}$ and the whitening $\Sigma_{e,t}^{-1/2}$ of Proposition ?? both require $\Sigma_{e,t} \succ 0$. $R_t \succ 0$ is a standing condition of the recursion (with $\Sigma_{t|t-1} \succeq 0$ this suffices, and full row rank of H_t would suffice even for $R_t \succeq 0$). This is not automatic: duplicate venue quotes on the *same* instrument – the asynchrony of this chapter – make near-singular R_t a live case, and such rows must be deduplicated or their common noise modelled before inversion.

4.3.5 Batch update equals the recursion

A batch fit of a whole smile at one knowledge time is a single Kalman update in disguise, and writing it in information form makes the Bayesian content explicit.

Proposition 4.15 (Gaussian conjugate update / information form). *Let the prior on the state be $\mathcal{N}(\mu_0, \Sigma_0)$ and let a batch of observations satisfy $y = \Phi \beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, R)$ and Φ the design matrix. Then the posterior is Gaussian, $\mathcal{N}(\mu_\star, \Sigma_\star)$, with*

$$\Sigma_\star^{-1} = \Sigma_0^{-1} + \Phi^\top R^{-1} \Phi, \quad \mu_\star = \Sigma_\star (\Sigma_0^{-1} \mu_0 + \Phi^\top R^{-1} y),$$

and this coincides with one predict–update step of the Kalman recursion taken with $H = \Phi$ and prior covariance Σ_0 .

Proof. The log-posterior is, up to constants,

$$-\frac{1}{2}(\beta - \mu_0)^\top \Sigma_0^{-1}(\beta - \mu_0) - \frac{1}{2}(y - \Phi \beta)^\top R^{-1}(y - \Phi \beta),$$

a quadratic in β . Collecting the quadratic term gives the precision $\Sigma_\star^{-1} = \Sigma_0^{-1} + \Phi^\top R^{-1} \Phi$, and setting the gradient to zero gives $\Sigma_\star^{-1} \mu_\star = \Sigma_0^{-1} \mu_0 + \Phi^\top R^{-1} y$, i.e. the stated mean. For the equivalence with the recursion, Sherman–Morrison–Woodbury turns the information-form precision into the gain-form covariance $\Sigma_\star = (I - G\Phi)\Sigma_0$ with $G = \Sigma_0 \Phi^\top (\Phi \Sigma_0 \Phi^\top + R)^{-1}$. It remains to check the *mean*. Substituting $\Sigma_\star = (I - G\Phi)\Sigma_0$ into $\mu_\star = \Sigma_\star (\Sigma_0^{-1} \mu_0 + \Phi^\top R^{-1} y)$ gives

$$\mu_\star = (I - G\Phi)\mu_0 + (I - G\Phi)\Sigma_0 \Phi^\top R^{-1} y.$$

The Woodbury/push-through identity $(I - G\Phi)\Sigma_0 \Phi^\top R^{-1} = \Sigma_0 \Phi^\top (\Phi \Sigma_0 \Phi^\top + R)^{-1} = G$ (both equal $\Sigma_\star \Phi^\top R^{-1}$) collapses the second term to Gy , so

$$\mu_\star = \mu_0 - G\Phi\mu_0 + Gy = \mu_0 + G(y - \Phi\mu_0),$$

which is exactly one predict–update step with $H = \Phi$ and innovation $y - \Phi\mu_0$. $\square$

Two readings follow. First, when the observations are in vol space, the diagonal $R = \text{diag}(1/\omega_j^{\text{vol}})$ populated by the vol-space weights $\omega_j^{\text{vol}} = \nu_j^2 \omega_j^{\text{price}}$ makes the data term the weighted least-squares objective that reproduces the price-space primacy of Axiom ??; under the flagged homoscedastic-price collapse $\omega_j^{\text{vol}} \propto \nu_j^2$ this is the plain vega-weighted fit. Either way the posterior mean is the WRMSE-optimal fit and the prior term Σ_0^{-1} is a Tikhonov (ridge) regulariser – the fit degrades gracefully when strikes are sparse. Second, the update never rebuilds the posterior from scratch: it takes yesterday’s (μ_0, Σ_0) as the prior, in line with sequential consistency. The batch view and the recursive view are the same object.

4.4 Oracle Model and Asynchrony

The observation space is $\mathcal{O} = \{(i, v, t_{\text{event}}, t_{\text{seen}}, g)\}$, populated by multiple source streams differing in latency and reliability. Two clocks must be kept apart.

Definition 4.16 (Event time and knowledge time). The *event time* t_{event} is when the underlying market event occurred; the *knowledge time* t_{seen} is when the system first learns of it via the Oracle.

Calibration updates follow knowledge-time order, not event-time order: the posterior at each instant is a function of the information actually available then. Event time still enters *inside* the models – time-to-expiry T is computed from it – but the posterior evolution is driven by knowledge time. When two correlated legs are quoted on feeds with different latencies, their knowledge times diverge, and the discrepancy between what the system has stored and what its own model implies is the broken state of Section ??.

4.5 No-Arbitrage Constraints

The admissible region Θ_{AF} of Axiom ?? is a hard constraint: it is checked before any calibration is certified, and a violation is a rejection, not a warning.

4.5.1 Yield curves

Let $D(t)$ be the discount-factor curve for maturities $t \geq 0$. Admissibility requires

$$D(0) = 1, \quad D(t) > 0 \quad \forall t, \quad D(\cdot) \text{ non-increasing},$$

the last condition being equivalent to non-negative forward rates. Bounds on the curvature of forward rates may be added through the MCA.

Example 4.17. If a calibrated discount curve prices a five-year zero-coupon bond above a four-year bond of the same issuer and seniority, monotonicity is violated: the calibration has left Θ_{AF} and must be rejected.

4.5.2 Volatility surfaces

Let $w(m, T)$ denote the total implied variance as a function of strike log-moneyness $m = \ln(K/F)$ and expiry T . In terms of the mandated decomposition,

$$w(m, T) = \sigma(m, \ell_F, T)^2 T = (\sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T))^2 T,$$

and at the money $\sigma_{\text{smile}}(0, T) = 0$ gives the clean identity $w(0, T) = \sigma_{\text{ATM}}(\ell_F, T)^2 T$: the ATM total variance is exactly the ATM component, squared and scaled by the horizon. The forward-normalised moneyness $m = \ln(K/F(T))$ is what makes the following conditions the correct static no-arbitrage characterisation; they are stated for *deterministic rates and proportional (yield) dividends*, the regime in which the forward-normalised price $C/D(T)$ is a martingale in $K/F(T)$ (Gatheral, *The Volatility Surface*, 2006; Gatheral–Jacquier 2014). The discrete-cash-dividend case admitted in the dividend subsection below is treated separately (see the calendar bullet). Admissibility requires:

- **Positivity:** $w(m, T) > 0$ for all (m, T) .

- **No calendar arbitrage:** total implied variance is non-decreasing along the maturity axis at fixed forward-moneyness m . Because $\ell_F = \ln(F(T)/F_{\text{ref}})$ itself moves with T through the forward curve, the honest ATM comparison is a *total* derivative, not a partial:

$$\frac{d}{dT} w(0, T) = \sigma_{\text{ATM}}^2 + 2 \sigma_{\text{ATM}} T \left(\frac{\partial \sigma_{\text{ATM}}}{\partial T} + \text{vol.slope} \cdot \frac{d\ell_F}{dT} \right) \geq 0,$$

where the $\text{vol.slope} \cdot d\ell_F/dT$ term (equity $\text{vol.slope} < 0$, C2) is the routed forward-curve contribution that a bare $\partial w/\partial T$ silently drops. Under deterministic rates and proportional dividends this characterises calendar no-arbitrage; *with declared discrete cash dividends* the strikes $K = F(T)e^m$ jump across dividend dates and fixed- m monotonicity fails, so that regime instead uses a fixed-*strike* comparison against dividend-adjusted forwards, deferred to the MCA.

- **No butterfly arbitrage:** the risk-neutral density implied by the surface must be non-negative *and integrate to one*. With subscripts denoting partials in m , the Durrleman function (Durrleman 2004; Gatheral–Jacquier 2014, Lemma 2.2)

$$\mathbf{g}(m) = \left(1 - \frac{m w_m}{2w}\right)^2 - \frac{w_m^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w_{mm}}{2}$$

gives *local density non-negativity* exactly as $\mathbf{g}(m) \geq 0$ on $\mathbb{R}$. This is necessary but not by itself sufficient: full butterfly no-arbitrage additionally needs unit total mass, i.e. the wing condition $\lim_{m \rightarrow +\infty} d_+(m) = -\infty$ (equivalently call prices $\rightarrow 0$; otherwise the implied measure leaks mass to $+\infty$ and integrates to less than one). Density non-negativity *and* unit mass hold iff both conditions hold. The large-moneyness wing growth is further constrained by Lee’s moment bounds; the density-positivity scan must be run across the whole m -grid, including the wings where the vega weight of the vol-space fit vanishes.

- **Finite ATM skew:** $w_m(0, T)$ finite, controlling the smile skew $s(m) = \frac{d\sigma}{dm}$ at the money (the ratified table glyph; at fixed ℓ_F, T this equals $\partial \sigma_{\text{smile}}/\partial m$).

Parameterisation-specific sufficient conditions (SVI, SSVI, and relatives), including their own wing/mass constraints, are recorded in the MCA.

Example 4.18. If the calibrated smile at some maturity makes deep out-of-the-money calls cheaper than a vertical call spread covering the same region, $\mathbf{g}(m) < 0$ somewhere and a static butterfly arbitrage exists: the surface is outside Θ_{AF} .

4.5.3 Dividend and repo curves

For equity-like underlyings: continuous dividend yields satisfy $q(t) \geq 0$; the present value of declared discrete dividends does not exceed spot; and the forwards implied by (rates, q , repo) – equivalently by the discount curve $D(\cdot)$ of Remark ?? together with q and repo – agree with listed futures within tolerance. Repo curves are smooth, bounded below by modelled floors, and consistent with financing markets.

Example 4.19. If the curve implies that the present value of all future dividends exceeds the spot price, the model has shareholders paid more than the asset is worth – an economic impossibility revealing a calibration error.

4.5.4 Cross-asset constraints

Because Θ_{AF} is a hard set, each membership condition must be a *predicate*, not an approximation: “ $\approx$ ” is not checkable and would leave Θ_{AF} ill-defined for exactly the constraints this section calls

genuine, so INV-01 would be unverifiable. Each is therefore written as an explicit inequality carrying a named MCA tolerance, mirroring the $\tau_i, \tau_{\text{agg}}$ discipline of Axiom ?? . At minimum: FX triangular relationships,

$$|\text{FX}(A/C) - \text{FX}(A/B) \text{FX}(B/C)| \leq \tau_{\text{tri}};$$

interest-rate parity between FX forwards and discount curves; and put–call parity at every strike and maturity,

$$|C - P - D(T)(F(T) - K)| \leq \tau_{\text{pcp}}.$$

Put–call parity is an *exact* identity for European options; the slack τ_{pcp} therefore lives in the observation noise, not in the constraint itself, and is set per strike/maturity in the MCA. This is precisely where the general forward matters: with rates and dividends being calibrated, $D(T) \neq 1$ and $F(T) \neq S$, so put–call parity genuinely links the option surface to the rate and dividend curves. (Under the standing zero-rate collapse it would reduce to $|C - P - (S - K)| \leq \tau_{\text{pcp}}$; that collapse is *not* taken here, per Remark ??.)

Example 4.20. If the model prices calls and puts so as to systematically violate put–call parity for liquid EURUSD strikes, the calibration is wrong even when each option individually sits near its mid.

4.6 Robustness to Bad Data

Axiom ?? demands that bad attestations be caught before they touch the posterior. The tool is the innovation: an observation that is inconsistent with the model’s own prediction produces a large surprise, and surprise is measurable.

4.6.1 Innovation-based gating

For the batch y_t with predicted $(\mu_{t|t-1}, \Sigma_{t|t-1})$, form the innovation and its covariance

$$e_t = y_t - H_t \mu_{t|t-1}, \quad \Sigma_{e,t} = H_t \Sigma_{t|t-1} H_t^\top + R_t,$$

and the squared Mahalanobis distance

$$\mathcal{M}_t^2 = e_t^\top \Sigma_{e,t}^{-1} e_t.$$

Proposition 4.21 (Distribution of the gate statistic). *Assume $\Sigma_{e,t} \succ 0$ (the standing condition $R_t \succ 0$) and the correct linear–Gaussian model: x_t Gaussian, $\varepsilon_t \sim \mathcal{N}(0, R_t)$ with the true Q, R_t, H_t and exact feature-linearity of the surface. Then $e_t \sim \mathcal{N}(0, \Sigma_{e,t})$, the whitened innovation $\Sigma_{e,t}^{-1/2} e_t$ is a standard J_t -vector, and hence $\mathcal{M}_t^2 \sim \chi_{J_t}^2$.*

Proof. The innovation is a linear combination of the Gaussian x_t and ε_t , hence Gaussian; under the correct model it is mean-zero with covariance $\Sigma_{e,t}$ by construction, and $\Sigma_{e,t} \succ 0$ makes $\Sigma_{e,t}^{-1/2}$ well defined. Let $z = \Sigma_{e,t}^{-1/2} e_t$; then $\mathbb{E}[z] = 0$ and $\text{Cov}[z] = \Sigma_{e,t}^{-1/2} \Sigma_{e,t} \Sigma_{e,t}^{-1/2} = I_{J_t}$, so z is a standard Gaussian J_t -vector and $\mathcal{M}_t^2 = z^\top z = \|z\|^2 \sim \chi_{J_t}^2$. $\square$

This distribution is what makes the gate calibrated: a threshold u_{J_t} set at, say, the 95th percentile of $\chi_{J_t}^2$ has a known false-rejection rate. The gating rule is then (Figure ??):

- $\mathcal{M}_t^2 \leq u_{J_t}$: accept the batch and update normally.
- $\mathcal{M}_t^2$ moderately above u_{J_t} : down-weight by inflating R_t (larger R_t shrinks the gain G_t , so a suspect batch moves the state less).

- $\mathcal{M}_t^2$ far above u_{J_t} : reject; the posterior is left unchanged.

Component-wise, the standardised residuals

$$z_i = \frac{e_{t,i}}{\sqrt{(\Sigma_{e,t})_{ii}}}$$

pinpoint which individual observation is the outlier.

Remark 4.22 (The χ^2 calibration is nominal only under the correct model). Proposition ?? holds *only* for a correctly-specified Gaussian $\Sigma_{e,t}$ – yet Axiom ?? concedes heteroscedastic, fat-tailed, skewed, cross-correlated noise and Axiom ?? concedes outright wrong attestations, i.e. the null fails in precisely the states the gate exists to catch, and the defence is calibrated by the model it is meant to protect. Two consequences must be stated, not assumed away. (i) Under fat tails a χ^2 -based u_{J_t} *under-rejects* genuine extremes (real tail mass sits beyond the Gaussian quantile) while *over-rejecting* moderate innovations; the threshold should be recalibrated empirically on realised innovations, or a robust/heavy-tailed reference law used, rather than read off χ^2 . (ii) Model-*shape* misspecification (a smile that is not exactly feature-linear, stale Q/H_t during a regime shift) inflates innovations and makes the gate reject *good* data during genuine smile stress – a failure mode to monitor alongside the outlier rejection the gate is designed for.

Example 4.23. If a single SPX option quote arrives 20% away from the mid of every other source and pushes $\mathcal{M}_t^2$ past the gate, the filter rejects it rather than letting it distort the surface. A mis-keyed swap rate at 50% instead of 5% is excluded the same way, without blowing up the day’s yield curve.

Rejection leaves the state exactly where prediction placed it, $\Sigma_{t|t} = \Sigma_{t|t-1}$ and $\mu_{t|t} = \mu_{t|t-1}$, and the decision, with its diagnostics, is written to the audit record.

4.7 Temporal Coherence and Structural Breaks

4.7.1 Sequential updating and forgetting

Attestations are processed in knowledge-time order; each posterior is the previous one updated with the newly accepted data, never a fresh fit to all history. Old attestations fade not by deletion but through accumulated process noise: over k steps the prediction covariance has grown by kQ , so the influence of an observation k steps back is diluted by the intervening Q . Larger Q forgets faster. (Here k is a bare integer step count, the only role the Contract leaves to that letter.)

Example 4.24. If a yield-curve calibration still reacts strongly to a rate shock from a year ago while recent rates have been stable, Q is too small: the system is not forgetting old regimes fast enough.

4.7.2 Detecting structural breaks

By Proposition ?? the normalised innovations z_i should look like independent standard Gaussians. Structural breaks are detected as departures from that reference:

- rolling means of innovations drifting significantly from zero;
- empirical innovation variances persistently above or below the modelled $\Sigma_{e,t}$;
- non-negligible autocorrelation in the normalised innovations.

When a break is confirmed, the prior is widened (larger Q , or relaxed kernels), a regime-change record is produced, and optionally a batch re-fit on a recent window supplies a fresh starting posterior. After widening, the martingale property of Axiom ?? resumes in the new regime.

The covariance structure of the prior is central here. A term-structure model for volatility encodes the correlation between short-dated and long-dated vol factors through the off-diagonal of Q (and of $\Sigma_{t|t}$):

- In a stable regime short- and long-dated vols move together with high correlation; the prior reflects this and innovations stay well behaved.
- In a break where short-dated vol spikes while long-dated vol stays anchored (or vice versa), the empirical correlation changes. A prior that still assumes the old correlation generates persistent, structured innovations – biased means and non-zero autocorrelation.

Detecting the break is detecting that the innovation behaviour no longer matches what the prior covariance assumes; the remedy is to update Q and its correlation structure. Otherwise the filter either over-reacts (too much process noise in the wrong directions) or under-reacts and accumulates bias (too little).

4.8 Broken States and Cross-Asset Coherence

4.8.1 Broken states as inconsistent conditional expectations

Let $\vartheta_{\text{sys}}(T)$ be the parameter vector stored at knowledge time T and I_T the information set of accepted attestations up to T . The information-optimal parameter is the conditional expectation

$$\vartheta^*(T) := \mathbb{E}[\vartheta \mid I_T].$$

Definition 4.25 (Broken state). The system is in a broken state at time T if some component j satisfies

$$\vartheta_{\text{sys},j}(T) \neq \vartheta_j^*(T),$$

i.e. the stored parameter is not what the current information implies. The system has learned something it has not fully propagated into its own state.

Example 4.26. Let σ_{SPX} and σ_{SX5E} be vol levels with a joint model that makes them move together. If new SPX data pushes σ_{SPX} up by one vol point, the conditional expectation $\mathbb{E}[\sigma_{\text{SX5E}} \mid I_T]$ shifts too. If the system updates only σ_{SPX} and leaves σ_{SX5E} unchanged, the stored pair no longer equals the conditional expectation: the system is broken – it knows something about SX5E that its parameters do not reflect.

4.8.2 Spread options and asynchronous vol updates

The broken state becomes operationally dangerous for products that depend on *differences* of parameters. Consider an SPX–SX5E spread option, whose price depends on the spread between the two ATM vols and on their correlation. If the two surfaces update on different schedules – SPX immediately, SX5E only when its own (perhaps closed) session feeds – then new SPX information changes the conditional distribution of *both* legs, yet only one leg is written. The spread option then moves as if a single leg had reacted, becoming artificially volatile and inconsistent with the joint model.

Two features of the framework make this precise. First, the basket and spread constructions take their single-name inputs to be the ATM vols, $\sigma_i \equiv \sigma_{\text{ATM},i}(\ell_{F,i}, T)$ (bridging identity C1): the spread’s exposure runs through σ_{ATM} , so a stale σ_{ATM} on one leg is a mispriced spread. Second, the

two legs are not independent: under the joint prior a move in leg 1 shifts the conditional mean of leg 2's ATM vol, and the coherent size of that shift is set by the correlation between the two ATM-vol factors – the usual Gaussian conditional-mean coefficient $\Sigma_{21}\Sigma_{11}^{-1}$ of the prior covariance. How that vol move then feeds the *price* of leg 2 routes through its ATM slope $\text{vol.slope} = \frac{\partial \sigma_{\text{ATM}}}{\partial \ell_F}$. The precise cross-sensitivity $d\sigma_2/d\sigma_1$ – the multi-name shadow-Greek channel tying the joint correlation ρ to the two slopes – is derived in Volume II and is out of scope here; for this chapter it suffices that a coherent update propagates the move along these channels while a broken state ignores them.

The rule is therefore exhaustive: whenever new attestations change the conditional distribution of a subset of parameters, the system must either (i) update *all* affected parameters to their new conditional expectations, or (ii) explicitly flag that it has entered a broken state in which internal prices are not to be trusted.

4.9 Invariant Properties

The axioms imply a set of invariants that must hold on every certified calibration. Each is stated with a concrete failure it forbids.

- **INV-01 (Arbitrage-free).** Every certified ϑ lies in Θ_{AF} . *A surface implying negative local volatility $\sigma_{\text{loc}}(m, T)$ at some strike/maturity must never be labelled certified and served.*
- **INV-02 (Price-space residual bounds).** Certified calibrations satisfy the per-instrument and aggregate tolerances of Axiom ?? *If liquid one-year ATM SPX options are off by 2% of spot against a 0.5% tolerance, the calibration is rejected or re-weighted, not pushed to production.*
- **INV-03 (Sequential posterior consistency).** Each posterior is the previous one updated with the new accepted attestations; none is built independently of its predecessor. *The 10:01 surface is the 10:00 surface updated with the intervening minute, not a fresh fit to all history.*
- **INV-04 (Reproducibility).** Same ordered inputs, MCA, code version and numerical environment give the same posterior up to the declared tolerance. *Replaying last quarter's trades and quotes must reproduce that day's USD curve within rounding.*
- **INV-05 (Provenance completeness).** Every parameter set carries a provenance chain back to all inputs and the MCA. *A questioned CDS curve can be traced to the exact single-name and index quotes and config that produced it.*
- **INV-06 (Innovation gating).** No attestation touches the posterior without passing the checks of Section ??. *A mis-keyed 50% swap rate fails the gate and is dropped instead of wrecking the curve.*
- **INV-07 (Staleness monitoring).** When an object or its inputs age past the MCA thresholds, staleness is signalled. *A vol surface untouched for two hours in a fast market shows a stale-vol warning, not a pretence of currency.*
- **INV-08 (Dependency ordering).** A calibration depending on other calibrated objects uses only certified, internally consistent versions of them. *A swaption surface built on the USD discount curve is not fit against a half-updated curve snapshot.*
- **INV-09 (Joint coherence).** Jointly-modelled parameters stay consistent with both single-name calibrations and any multi-asset prices within tolerance. *The joint SPX/SX5E vol/correlation*

model keeps the implied spread-option level aligned with quoted spreads, and does not wander while the single-index surfaces look fine.

- **INV-10 (Smoothness bounds).** Parameter and price paths respect the smoothness implied by the priors and process noise; persistent violations are anomalies. *A forward-variance-swap term structure that flips by hundreds of basis points of variance day to day while ATM vols move smoothly is a model issue, not market information.*

4.10 Reference Products and Monitoring

The purpose of calibration is to extract signal and suppress noise, and the test of that is not the fit residual but the behaviour of prices for economically important products. The system maintains an explicit, versioned list of *reference products* – simple, liquid payoffs that depend directly on the calibrated objects and should be smooth in maturity and strike and stable over time except when the market genuinely moves.

The list should be small enough to understand yet rich enough to catch pathologies: forward rates across standard tenors; forward dividend yields; forward variance swaps, spot and forward-start; barrier options at standard levels; spread and simple basket payoffs sensitive to vol spreads and correlation (SPX–SX5E spread calls); call-versus-call structures; and local-vol slices $\sigma_{\text{loc}}(m, T)$ verifying that local vol changes smoothly across strikes. A useful scalar diagnostic is the variance-swap / ATM-vol ratio. To keep it *dimensionless*, quote the variance-swap fair strike $K_{\text{var}}(T)$ in variance units (annualised σ^2) and take the square root before dividing:

$$R_{\text{var}/\text{ATM}}(T) := \frac{\sqrt{K_{\text{var}}(T)}}{\sigma_{\text{ATM}}(\ell_F, T)},$$

a vol-quoted fair strike over the ATM vol, which is $O(1)$ (and ≥ 1 under a smile). It should be a smooth function of maturity and evolve smoothly in time; numerator and denominator now carry the same units, so a kink in $R_{\text{var}/\text{ATM}}$ signals contamination in the variance-swap leg or in σ_{ATM} (or both) rather than a units artefact.

For each reference product the system tracks levels and ratios: changes over time at fixed maturity and strike; cross-sectional smoothness across the surface at a fixed knowledge time; and smoothness of key ratios (variance-swap over ATM vol, barrier over vanilla, spread option over a combination of single-name options). Any calibration step that injects artificial jumps, oscillations, or excess volatility into these products or their ratios – beyond what the underlying attestations justify – is treated as the calibration amplifying noise rather than extracting signal, and is escalated as a model issue.

4.11 Scope and Extensions

The principles apply to yield curves and discount factors, volatility surfaces and smiles (equity, FX, rates, crypto), credit curves, correlation and covariance structures, and joint calibrations of related instruments. To bring a new parameter type into scope one specifies its parameter space Θ and admissible region Θ_{AF} ; the pricing map π ; the no-arbitrage constraints characterising Θ_{AF} ; the prior structure and smoothness properties; the price-space tolerances; the reference payoffs used as consistency checks; and the dependencies on existing calibrated objects. Each extension is recorded as an MCA together with an update to the invariant and monitoring definitions – so that the same audit, gating, and coherence guarantees that hold for a single option hold, unchanged in form, for the whole book.

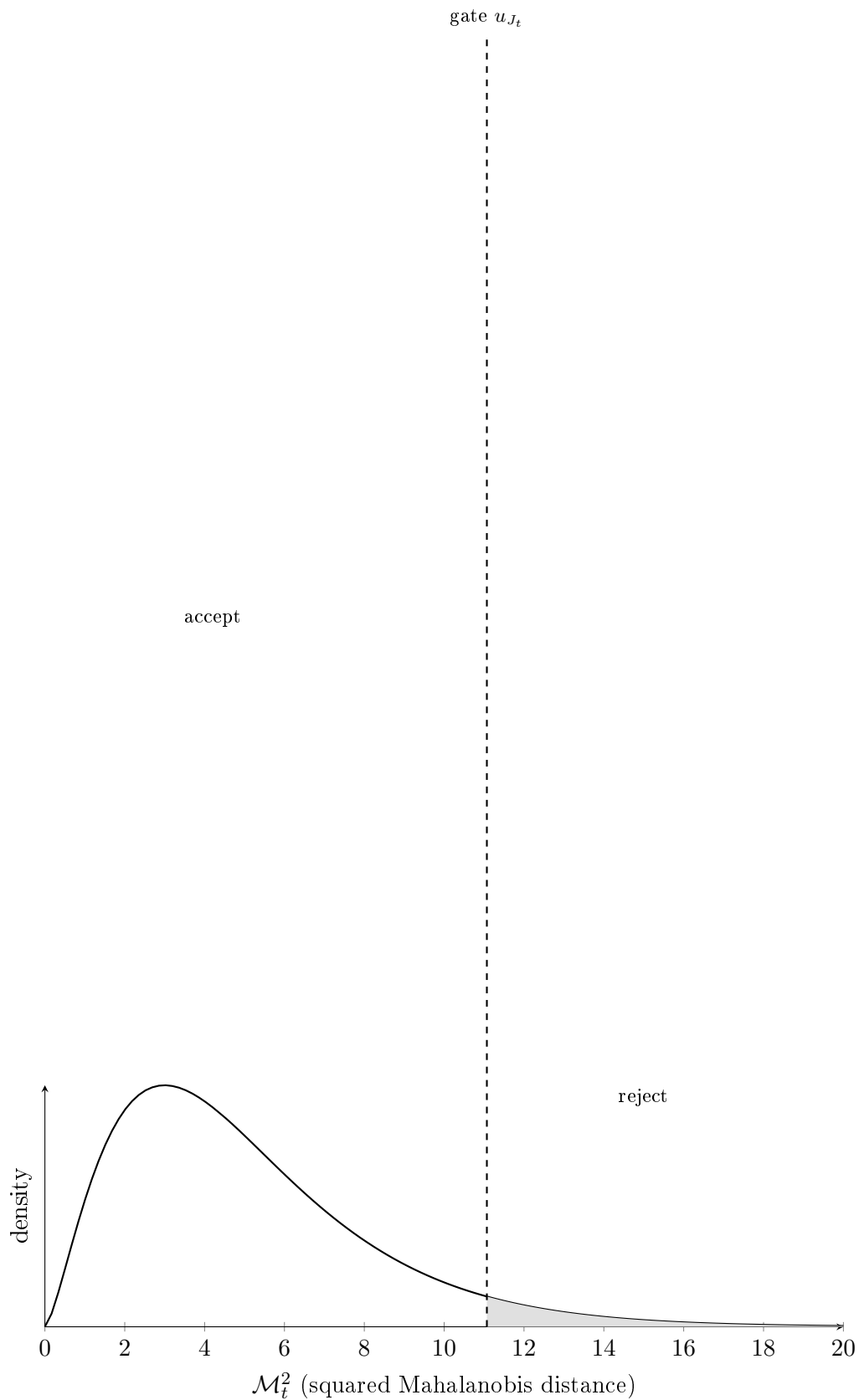


Figure 4.1: Innovation gating. The gate statistic $\mathcal{M}_t^2$ is $\chi_{J_t}^2$ -distributed under the correct model (Proposition ??, here illustrated for a modest number of observations). A threshold u_{J_t} at a fixed percentile splits the axis into an accept region and a reject/down-weight tail with a known false-rejection rate.